TOWARD ACCOUNTABLE AI SYSTEMS:

OBJECTIVE FAIRNESS, EVALUATION, AND ROBUSTNESS

By

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Abstract

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Artificial intelligence substantially influences decisions in domains where errors, latent bias, misunderstandings, and misplaced confidence can have serious ramifications. However, accountability in these systems remains difficult because it depends on more than model performance; it is also material that fairness is defined in a principled way, whether evaluation metrics are well-founded and remain reliable across groups and settings, and whether uncertainty is properly communicated. This dissertation examines accountability in artificial intelligence through four connected problems: how to define fairness, how model evaluation can become unreliable, when a metric is well-founded (measures the correct underlying thing and is axiomatically justified), and how to convey uncertainty.

Firstly, this dissertation develops the **Objective Fairness Index (OFI)**, a fairness metric grounded in the legal principles of objective testing. OFI measures disparity through differences in marginal benefit (a numerical estimate of “what happened” versus “what should have happened”), allowing fairness assessments to distinguish algorithmic discrimination from a disparity found in society itself. OFI is applied to binary-classification problems such as recidivism prediction, employment classification, and activity recognition in clinical datasets with synthetic data generation.

Next, this dissertation studies fairness metrics through the lens of **social choice theory**. By

treating fairness metrics as aggregation rules over groups, we develop a taxonomy of parity-gap metrics, adapt social-choice-style axioms to binary classification, and identify important pathologies and trade-offs, including aggregation reversals and an impossibility result showing that Consistency and Participation cannot be jointly guaranteed across the taxonomy.

The dissertation then studies **sample size-induced bias in classification metrics**. Here, we discover that many confusion-matrix-based metrics become unstable or jagged for small sample sizes, which can distort group comparisons. To address this problem, we introduce the **Metric Alignment Trial for Checking Homogeneity (MATCH)** test for assessing whether an observed subgroup score is probabilistically consistent with a reference distribution, and **Cross-Prior Smoothing (CPS)** for improving metric stability and reliability.

During **time-series similarity measure evaluation**, this dissertation rigorously checks if the literature's default, leave-one-out 1-nearest-neighbor assessed by accuracy, is appropriate. We find that it generalizes poorly to other objectives and that the Matthews Correlation Coefficient often provides sharper statistical discrimination than accuracy. Additionally, our objective-informed benchmarking findings suggest that the best time-series similarity measure is objective-dependent, and the statistical uncertainties often provide multiple "best" options.

Taken together, these contributions provide a rigorous addition toward accountable artificial intelligence. By introducing a new method for fairness assessment, formalizing desirable metric properties, improving the reliability of model evaluation, and revisiting time-series benchmarking approaches, this dissertation advances fairness formalizations, metric reliability, and interpretable uncertainty not only in artificial intelligence but also in its evaluations.

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INTRODUCTION

Artificial intelligence and machine learning systems now influence decisions in domains where mistakes can matter greatly, including healthcare, criminal justice, finance, and employment. In such settings, mere model performance estimates are insufficient. A system may achieve strong predictive scores and simultaneously yield unfair outcomes. Moreover, the evaluations of prediction or fairness may themselves be overconfident, violate desirable properties, or miss the necessary ontological bearing of our underlying concepts. When these issues are overlooked, evaluation becomes misleading and accountability becomes difficult to establish.

Accordingly, this dissertation studies accountability through four connected problems: how fairness should be defined, how model evaluation can become unreliable, when a metric is well-founded, and how to manage uncertainty. All together, the central claim is that objective assessments and critical evaluations are necessary not only for models, but also for the measurements, comparisons, and conclusions built around them.

1.1 Significance of Objective Fairness

Fairness in machine learning remains a contested concept. The literature contains many group-fairness metrics, each reflecting different normative commitments and statistical trade-offs (Verma and Rubin, 2018; Caton and Haas, 2024). This plurality indicates fundamental questions on what fairness means and creates a practical difficulty: in normal and high-stake settings alike, it is often unclear why one fairness metric should be preferred over another. Therefore, a fairness assessment should be created by a principled rationale with clear conceptual foundations and assumptions.

This dissertation approaches that problem using the backbone of case law and legal statutes in objective testing. In many legal settings, disparate outcomes alone do not hold one guilty of discrimination. It is material whether the assessment itself is rationally related to the task (an ontological grounding) and whether the resulting disparities reflect discriminatory treatment or legitimate differences in qualification. This distinction matters in machine learning because its repudiation risks conflating algorithmic discrimination with external disparities present in the surrounding social process.

Chapter 3 addresses this gap with the *Objective Fairness Index (OFI)*. OFI measures disparity through differences in marginal benefit, comparing what happened to a group with what should have happened under an objective assessment. This counterfactual comparison between observed and expected treatment provides a more principled way to distinguish algorithmic discrimination from disparity that is independent of the model. By grounding fairness in the legal principles of objective testing, OFI offers a more interpretable and defensible framework for assessing bias in machine learning systems.

The significance of this contribution is twofold. First, it provides a fairness metric grounded in a clearer normative rationale than many conventional choices. Second, it supports accountability by making fairness claims more interpretable in high-stakes contexts where legal, institutional, and human consequences matter. In this dissertation, objective fairness is not treated as the only fairness perspective, but as an especially important one when the goal is to evaluate whether a system is itself at fault.

1.2 Meta-Evaluation: Axiomatic Justification and Construct-Validity

A recurring theme of this dissertation is that evaluation tools should themselves be evaluated. In machine learning, metrics are often chosen because they are standard, convenient, or historically dominant. Yet popularity is not the same as justification. A metric should be supported by a credible argument that it measures the intended construct, and that its behavior satisfies desirable formal properties.

Chapter 4 addresses this issue for fairness metrics by connecting machine learning fairness to social choice theory. There, fairness metrics are considered as aggregation rules over groups. From this perspective, it is possible to study fairness metrics as structured decision rules with identifiable properties, trade-offs, and pathologies. The chapter develops a taxonomy of fairness metrics as *Parity-Gap Metrics*, adapts social-choice-style axioms to the binary-classification scenario, and shows that important tensions arise across the taxonomy, including aggregation reversals and an impossibility result showing that Consistency and Participation cannot be jointly guaranteed. In this way, the chapter provides an axiomatic basis for understanding when a fairness metric is justified and what kinds of guarantees it cannot provide simultaneously.

Another meta-evaluative work is presented in Chapter 6. This work questions whether a standard benchmarking convention in time-series similarity research actually measures what practitioners often assume it measures. The dominant default in that literature is leave-one-out 1-nearest-neighbor (1-NN) evaluated by accuracy. However, if the objective is to identify a generally strong similarity measure, then the objective should be realized and the performance evaluation carefully considered. Chapter 6 studies objective-informed benchmarking and shows that rankings based on 1-NN accuracy generalize poorly across other objectives, while the Matthews Correlation Coefficient often yields sharper discrimination than accuracy.

These chapters share a common message: evaluation is not neutral. Metric choice determines what becomes visible, what appears optimal, and what kinds of claims a researcher or practitioner is justified in making. For that reason, accountable AI requires evaluations to be derived from meta-evaluation analyses.

1.3 Challenges in Metric Reliability

Even when the correct construct has been identified, the measurement or metric may still behave unreliably. This is especially true for confusion-matrix-based metrics applied to groups with small and varying sample sizes. Group evaluations include any partitioning of the data into groups, including by protected attributes such as race, gender, or age. As such, group evaluation is pervasive in fairness analysis, auditing, and deployment review, yet the compared groups are often highly imbalanced. Under these conditions, an observed score intends to reflect model behavior, but the discrete and unstable behavior of the metric conflates the result.

Chapter 5 studies this problem as sample-size-induced bias in classification metrics. The chapter shows that many confusion-matrix-based metrics become unstable with jagged distributions and high variance. This is not a minor technical nuisance, but a realization that group comparisons can be distorted precisely in the settings where fairness assessments are often most important. We also count the number of undefined cases for common metrics, showing that these unstable estimates occur even when a metric is defined everywhere.

To address this problem, the dissertation introduces two tools. The Metric Alignment Trial for Checking Homogeneity (MATCH) tests whether an observed group score is probabilistically consistent with a reference distribution. Cross-Prior Smoothing (CPS) improves metric stability by borrowing information from a reference group to reduce the mean-squared error and mitigate small-sample pathologies. With these methods, we shift the discussion from simple estimates from metrics to asking whether the computed estimate is trustworthy enough to support a conclusion.

Our rigor in a metric's reliability is central to the dissertation's broader topic of accountability. If fairness is to be seriously assessed, then the metric for that assessment must be robust to common empirical conditions. Otherwise, the language of fairness can be undermined by unstable measurement.

1.4 Using Uncertainty

Uncertainty is often treated as secondary to prediction, yet in many applied settings it is one of the most important parts of a model's output. A deterministic prediction can create false confidence even when multiple plausible outcomes remain live. Similarly, an empirical comparison can exaggerate certainty when it reports a single winner without accounting for statistical indistinguishability or small-sample variation. In both cases, the core problem is the same: information is presented more decisively than the evidence warrants.

This dissertation treats uncertainty as essential to accountable assessment. In Chapter 5, uncertainty appears in the distributional behavior of classification metrics. MATCH explicitly evaluates whether an observed metric value is consistent with a reference distribution, and the chapter's analysis of jaggedness shows why group scores should not always be interpreted at face value. In Chapter 6, uncertainty appears in statistical testing and ranking. The results show that different diagnostics can support different winner groups, and that empirical conclusions should often acknowledge multiple statistically defensible choices rather than a single best measure.

Across these chapters, uncertainty is treated as information that should be modeled, communicated, and used. This is another component of the dissertation's central view of accountability: when evidence is uncertain, responsible systems and evaluations should say so.

1.5 Contributions

The overall objective of this dissertation is to strengthen accountability in artificial intelligence by addressing four connected problems: **1)** how fairness should be defined; **2)** how model evaluation can become unreliable; **3)** when a metric is well-founded; and **4)** how uncertainty should be quantified, interpreted, and communicated. The following chapters address these problems.

Chapter 2: Establish the technical foundations. Present the probability, confusion-matrix, and metric preliminaries used throughout the dissertation, providing the shared language needed for later chapters on fairness, evaluation, and uncertainty.

Supports all four problem focuses.

Chapter 3: Define objective fairness through OFI. Develop the Objective Fairness Index as a fairness metric grounded in objective testing, and show how it distinguishes algorithmic discrimination from broader disparity. Demonstrate its use in empirical fairness case studies and in evaluating bias mitigation with synthetic data generation.

Problem focus: defining fairness, well-founded metrics.

Chapter 4: Provide axiomatic justification for fairness metric choice. Reframe fairness metrics as aggregation rules over groups, develop a taxonomy of fairness metrics, adapt social-choice-style axioms to binary classification, and identify structural trade-offs and impossibility results that constrain metric selection.

Problem focus: well-founded metrics.

Chapter 5: Improve the reliability of group evaluation. Analyze sample-size-induced bias in classification metrics, characterize the instability and jaggedness that arise in small groups, and introduce MATCH and CPS as methods for more reliable group comparison and interpretation.

Problem focus: evaluation reliability, uncertainty.

Chapter 6: Reevaluate benchmarking through objective-informed diagnostics. Test whether the literature's standard 1-NN-with-accuracy benchmark for time-series similarity measures generalizes across other objectives, and show that metric rankings are objective-dependent, with MCC often providing stronger statistical discrimination than accuracy.

Problem focus: evaluation reliability, well-founded metrics, uncertainty.

Chapter 7: Synthesize the dissertation's argument. Bring these contributions together under the broader theme of accountable AI, showing how principled fairness, reliable evaluation, justified metric choice, and interpretable uncertainty support more defensible AI systems and conclusions.

Integrates all four problem focuses.

In Appendix B, we list the publications that have resulted out of the work in this dissertation. In summary, these objectives support the central claim: accountable artificial intelligence requires principled fairness definitions, evaluation procedures that remain reliable under realistic conditions, metrics whose uses are justified rather than assumed, and systems that properly handle and communicate uncertainty.

Chapter 2

BACKGROUND

2.1 Confusion Matrices and Classification

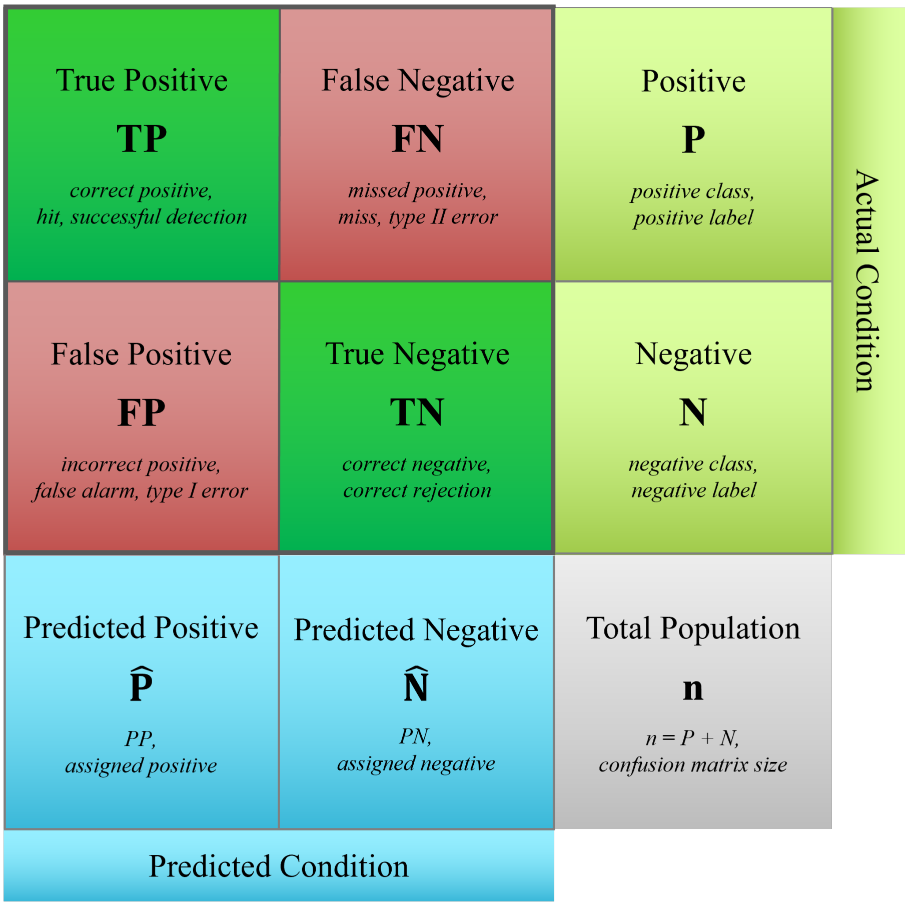


Figure 2.1: The binary confusion matrix illustrates the four primary outcomes of a binary classifier: True Positives, False Negatives, False Positives, and True Negatives. The additional cells summarize overall populations, providing context at a glance.

2.1.1 The Binary Confusion Matrix

The binary confusion matrix, CM, is defined in Definition 2.1. A detailed visualization is provided in Figure 2.1, which further illustrates the components and interactions within CM. This figure and accompanying discussion offer a comprehensive understanding of the matrix, highlighting the importance of each cell in evaluating classifier performance.

Definition 2.1 (Confusion Matrix)**.** *The binary confusion matrix,* CM*, summarizes the performance of a binary classifier and is defined as:*

Table 2.1: Confusion Matrix

	Predicted Positive	**Predicted Negative**
Actual Positive	TP	FN
Actual Negative	FP	TN

Overview: The confusion matrix above categorizes the outcomes of a binary classifier into four primary groups: **True Positives (TP)**, **False Positives (FP)**, **False Negatives (FN)**, and **True Negatives (TN)**. Each category helps evaluate the classifier's performance by describing its predictions compared to the actual labels.

Components of the Confusion Matrix:

- **True Positive (TP)**: Instances correctly predicted as positive.
  - *Alternative Names*: "Correct Positive", "Hit", "Successful Detection".
- **False Negative (FN)**: Instances that are actually positive but predicted as negative.
  - *Alternative Names*: "Missed Positive", "Miss", "Type II Error".
- **False Positive (FP)**: Instances that are actually negative but predicted as positive.
  - *Alternative Names*: "Incorrect Positive", "False Alarm", "Type I Error".
- **True Negative (TN)**: Instances correctly predicted as negative.
  - *Alternative Names*: "Correct Negative", "Correct Rejection".

Additional Notations:

- **$\hat{\text{P}}$ (Predicted Positive Total)**: The total number of instances that the model predicts as positive, regardless of their actual label, also referred to as **PP**.

- **$\hat{\mathbf{N}}$ (Predicted Negative Total)**: The total number of instances that the model predicts as negative, regardless of their actual label, also referred to as **PN**.
- **P (Actual Positive Total)**: The count of all actual positive instances in the dataset.
- **N (Actual Negative Total)**: The count of all actual negative instances in the dataset.
- **n (Total Population)**: The overall size of the dataset, equivalent to $\mathbf{n} = \mathbf{P} + \mathbf{N}$ and $\mathbf{n} = \hat{\mathbf{P}} + \hat{\mathbf{N}}$, representing the count of all instances.

Derived Metrics: Derived metrics, including accuracy, precision, recall, and specificity, serve as scalar summaries of model performance derived from the counts in the confusion matrix. These metrics reduce the complexity of a classifier's predictions into single-valued representations, making them more interpretable and facilitating comparisons across different models and datasets. However, their applicability is context-sensitive; the relevance and informativeness of each metric depend on the specific problem domain and the objectives of the analysis. Each metric emphasizes a distinct aspect of model performance, enabling a more nuanced evaluation of classifier behavior. For example:

- **Accuracy** measures overall correctness.
- **Precision** assesses the relevancy of positive predictions.
- **Recall** evaluates sensitivity to actual positives.
- **Specificity** captures the ability to identify negatives accurately.

These derived metrics offer an accessible and scalar representation of a model's effectiveness, which is particularly useful for comparative analysis and for tracking performance improvements over time. As discussed in greater detail in Chapter 5, we classify these performance metrics into three primary categories: Binomial Metrics, Joint Ratio Metrics (JRM), and specialized metrics, such as fairness measures. Each category provides unique insights that contribute to a comprehensive evaluation of the classifier, thereby refining our understanding of model performance across various dimensions.

2.1.2 Metric Definitions

In this section, we present a detailed overview of the classification metrics used throughout the dissertation. These metrics are categorized into three groups: Binomial Metrics, Joint Ratio Metrics (JRM), and Other Metrics, including fairness and specialized measures. Each table provides a formula, abbreviations, and brief description of the metric to clarify their definitions and usage. We use the binary confusion matrix from Definition 2.1.

Binomial Metrics

Binomial metrics, such as Accuracy and Prevalence, are fundamental to assessing the overall performance of a classification model. These metrics are based on binomial distributions and provide general insights into model behavior across various datasets. Refer to Table 2.2 for a complete breakdown of these metrics.

Joint Ratio Metrics (JRM)

Joint ratio metrics (JRM) are metrics derived from the confusion matrix, providing insight into the relationships between different types of classification error. These metrics include widely used measures such as True Positive Rate (TPR), False Positive Rate (FPR), and Positive Predictive Value (PPV). A comprehensive list of JRMs, their formulas, and descriptions can be found in Table 2.3.

Other Metrics

In addition to the JRMs and Binomial Metrics, we also include specialized metrics such as our Objective Fairness Index (discussed in Chapter 3), Matthews Correlation Coefficient (Matthews, 1975), and Prevalence Threshold. These metrics extend beyond basic performance evaluation and focus on more complex aspects, including fairness and correlation in imbalanced datasets. See Table 2.4 for further details.

Table 2.2: Binomial Metrics: General performance metrics using a binomial distribution of outcomes, commonly used for basic model evaluation.

Metric	Abbreviation	Formula	Description
Binomial Metric	N/A	$\frac{c_i + c_j}{n}$	A ratio of cells c_i and c_j to the total count n.
Accuracy	ACC	$\frac{\text{TP} + \text{TN}}{n}$	Proportion of correct predictions to all predictions.
Prevalence	PREV	$\frac{\text{TP} + \text{FN}}{n}$	Proportion of actual positive instances in the population. (Synonym: P)
Predicted Positive Rate	PPR	$\frac{\text{TP} + \text{FP}}{n}$	Fraction of instances predicted as positive by the classifier. (Synonyms: $\hat{P}$, PP)
Inaccuracy	INACC	$\frac{\text{FP} + \text{FN}}{n}$	Fraction of incorrect predictions out of total instances. (Synonyms: Error Rate, Misclassification Rate)
Negative Prevalence	NPREV	$\frac{\text{TN} + \text{FP}}{n}$	Proportion of actual negative instances in the population. (Synonym: Complement of Prevalence, N)
Predicted Negative Rate	PNR	$\frac{\text{TN} + \text{FN}}{n}$	Proportion of predicted negative instances in the population. (Synonyms: $\hat{N}$, PN)

Table 2.3: Joint Ratio Metrics (JRM): Metrics that describe the relationships between confusion matrix components as ratios, commonly used to assess various types of classification performance errors.

Metric	**Abbrev.**	**Formula**	**Description**
Joint Ratio Metric	JRM	$\frac{c_i}{c_i + c_j}$	A ratio between a cell c_i and the sum of c_i and another cell c_j.
True Positive Rate	TPR	$\frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$	Proportion of actual positives correctly identified. (Synonyms: Sensitivity, Recall, Hit Rate)
False Positive Rate	FPR	$\frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}}$	Proportion of actual negatives incorrectly classified as positive. (Synonym: Fall-Out)
True Negative Rate	TNR	$\frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}}$	Proportion of actual negatives correctly identified. (Synonym: Specificity)
False Negative Rate	FNR	$\frac{\mathrm{FN}}{\mathrm{FN} + \mathrm{TP}}$	Proportion of actual positives incorrectly classified as negative. (Synonym: Miss Rate)
Positive Predictive Value	PPV	$\frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$	Proportion of predicted positives that are actual positives. (Synonym: Precision)
Negative Predictive Value	NPV	$\frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FN}}$	Proportion of predicted negatives that are actual negatives.
False Discovery Rate	FDR	$\frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TP}}$	Proportion of predicted positives that are false positives.
False Omission Rate	FOR	$\frac{\mathrm{FN}}{\mathrm{FN} + \mathrm{TN}}$	Proportion of predicted negatives that are false negatives. (Synonym: False Reassurance Rate)

Table 2.4: Other Metrics: Specialized metrics focused on fairness, correlation, and balancing different forms of classification errors.

Metric	Abbrev.	Formula	Description
Original F_1 Score	F_1	$2 \div \left(\frac{TP + FP}{TP} + \frac{TP + FN}{TP}\right)$	Harmonic mean of precision and recall (simplified). (Synonyms: F_1 Measure, Dice-Sørensen Coefficient)
Simplified F_1 Score	F_1	$\frac{2 \cdot TP}{2 \cdot TP + FP + FN}$	Simplified version of the Original F_1 Score.
Matthews Correlation Coefficient	MCC	$\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$	Balanced metric accounting for all confusion matrix values, robust for imbalanced data. (Synonyms: Phi Coefficient, ϕ, r_ϕ)
Prevalence Threshold	PT	$\frac{\sqrt{TPR \cdot FPR} - FPR}{TPR - FPR}$	Threshold at which the positive prediction rate balances misclassification rates.
Marginal Benefit	$\mathcal{B}$	$\frac{FP - FN}{n}$	Considers objective testing principles in law; represents the benefit gained or lost (cost) for a group.
Objective Fairness Index	OFI	$\mathcal{B}_1 - \mathcal{B}_2$	The disparity between two groups' marginal benefits.
Treatment Equality	TE	$\frac{FN_1}{FP_1} - \frac{FN_2}{FP_2}$	Compares false negatives and false positives between two groups.

2.2 Probability Distributions

Here, we give necessary preliminaries on probability distributions.

2.2.1 Common Terms

In probability and statistics, a **distribution** describes how the values of a random variable are distributed. This section defines key terms and concepts related to distributions.

A probability distribution is classified as a **discrete distribution** if a random variable X assumes a countable set of distinct values. Conversely, a probability distribution is referred to as a **continuous distribution** if the random variable X can take on infinitely many values within a continuous range.

Key notations associated with distributions are as follows:

- X represents the random variable.
- $\mathcal{X}$ is the *support* of the distribution, defining the set of all possible values that X can assume.
- x denotes a specific realization or observed value of X.
- $X \sim D$ means that X is sampled from the distribution D.

The **probability mass function (PMF)** is used for discrete random variables. It gives the probability that X takes on a specific value x:

$$P(X = x) = f_X(x), \quad \text{where } \sum_x f_X(x) = 1. \tag{2.1}$$

For example, the PMF of a fair six-sided die is:

$$f_X(x) = \begin{cases} \frac{1}{6}, & x \in \{1, 2, 3, 4, 5, 6\}, \\ 0, & \text{otherwise.} \end{cases}$$

The **probability density function (PDF)** is used for continuous random variables. It describes the relative likelihood of a random variable X taking on a value near x. For example, assume $f_X(a) = 0.1$ and $f_X(b) = 0.2$ for $(a, b) \in \mathbb{R}^2$. Then, the following calculates how likely a is compared to b.

$$\text{Relative likelihood of a to b:} \quad \frac{f_X(a)}{f_X(b)} = \frac{1}{2}. \tag{2.2}$$

So, a value X being near a is half as likely to occur as X being near b.

To find the probability that X lies within a range $[a, b]$, we use:

$$P(a \leq X \leq b) = \int_a^b f_X(x)\, dx.$$

The **cumulative distribution function (CDF)** applies to both discrete and continuous random variables. It gives the probability that a random variable X is less than or equal to a value x:

$$F_X(x) = P(X \leq x) = \begin{cases} \sum_{k \leq x} f_X(k), & \text{if } \mathcal{X} \text{ is discrete,} \\ \int_{-\infty}^{x} f_X(t)\, dt, & \text{if } \mathcal{X} \text{ is continuous.} \end{cases} \tag{2.3}$$

The CDF satisfies:

$$F_X(-\infty) = 0, \quad F_X(\infty) = 1.$$

Key Properties

- **Mean** (μ): The mean represents the expected value of the random variable. It provides a measure of central tendency, indicating where the distribution is centered. For a random variable X, the mean is defined as:

$$\mu = \mathbb{E}[X] = \begin{cases} \sum_{x \in \mathcal{X}} xP(X = x) & \text{for discrete distributions,} \\ \int_{-\infty}^{\infty} x f_X(x)\, dx & \text{for continuous distributions.} \end{cases} \tag{2.4}$$

- **Variance** (σ^2): The variance quantifies the spread of the distribution around the mean. It is useful for understanding the variability of outcomes, which is critical for risk assessment and uncertainty quantification. Its defined as:

$$\sigma^2 = \mathbb{E}[(X - \mu)^2]. \tag{2.5}$$

- **Skewness** (γ_1): Skewness measures the asymmetry of the probability distribution. Positive skewness indicates a longer tail on the right, while negative skewness reflects a longer tail on the left. This helps in identifying directional biases in the distribution. It is calculated as:
$$\gamma_1 = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3}. \tag{2.6}$$
- **Kurtosis** (β_2): Kurtosis measures the "tailedness" of the distribution, capturing the sharpness of its peak and the heaviness of its tails. This is to identify bias toward the center or extremes. It is given by:
$$\beta_2 = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4}. \tag{2.7}$$
- **Excess Kurtosis** (γ_2): Excess kurtosis refines the concept of kurtosis by quantifying the deviation of the distribution's tail heaviness and peak sharpness relative to the normal distribution, which has an excess kurtosis of 0. It helps identify whether the data exhibits fatter tails (positive excess kurtosis) or thinner tails (negative excess kurtosis) compared to the normal distribution. We subtract three since the normal distribution's kurtosis is 3.
$$\gamma_2 = \beta_2 - 3 \tag{2.8}$$

These properties are crucial in distribution analyses. They provide insights into the distribution's shape, central tendency, and variability, enabling a comprehensive understanding of the underlying data.

2.2.2 The Gaussian Distribution

The Gaussian distribution, also known as the **normal distribution**, is one of the most fundamental probability distributions in statistics and machine learning. It is characterized by its bell-shaped curve and is mathematically defined in Definition 2.2.

Definition 2.2 (Gaussian Distribution)**.** *The probability density function (PDF) of the Gaussian* *distribution is given by:*

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}, \tag{2.9}$$

where:

- $\mu \in \mathbb{R}$ *is the mean of the distribution,*
- $\sigma^2 > 0$ *is the variance of the distribution.*

Using the PDF, the CDF of a Gaussian random variable X is:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\, dt. \tag{2.10}$$

The Gaussian distribution possesses several key properties that make it widely applicable. Firstly, the distribution is symmetric around the mean, with the mean, median, and mode all coinciding at the center of the distribution. Additionally, the Gaussian distribution is solely characterized by two parameters: the mean μ (location) and variance σ^2 (spread). The standard normal distribution is a common, special case of the Gaussian distribution with $\mu = 0$ and $\sigma^2 = 1$, formally $\mathcal{N}(0, 1)$. The PDF and CDF for $\mathcal{N}(0, 1)$ are publicly available in precomputed standard normal tables.

Additionally, the normal approximation is widely used in practice due to the Central Limit Theorem, which states that the sum of a large number of independent and identically distributed random variables tends to a normal distribution. Furthermore, the Gaussian distribution is linear for independent variables, offering great conveniences. For example, if $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ are independent, then:

$$aX_1 + bX_2 \sim \mathcal{N}(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2), \tag{2.11}$$

where $(a, b) \in \mathbb{R}^2$.

We list the first four moments and excess kurtosis of the Gaussian distribution in Table 2.5.

Table 2.5: Moments and Excess Kurtosis of the Normal Distribution

Property	Symbol	Alternative Notation	Mathematical Expression
Mean	μ	$\mathbb{E}[X]$	μ
Variance	σ^2	$\mathrm{Var}(X)$	σ^2
Skewness	γ_1	$\mathrm{Skew}(X)$	0
Kurtosis	β_2	$\mathrm{Kurt}(X)$	3
Excess Kurtosis	γ_2	—	0

2.2.3 The Binomial Distribution

The binomial distribution is a discrete probability distribution that describes the number of successes in a fixed number of independent Bernoulli trials, each with the same probability of success. A Bernoulli trial is a random experiment with exactly two possible outcomes: "success" (with probability p) and "failure" (with probability $q = 1 - p$).

Definition 2.3 (The Binomial Distribution)**.** *Mathematically, the binomial distribution can be expressed with its PMF:*

$$P(X = k) = \binom{n}{k} p^k q^{n-k}, \quad \textit{for } k = 0, 1, 2, \ldots, n,$$

where:

- *n is the total number of trials,*
- *k is the number of successes,*
- *p is the probability of success on a single trial,*
- *$q = 1 - p$ is the probability of failure on a single trial,*
- *$\binom{n}{k}$ is the binomial coefficient, given by:*

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \tag{2.12}$$

which represents the number of ways to choose k successes from n trials. The binomial coefficient is said as "n choose k", and is also expressed as: $C(n, k)$ or $n\mathrm{C}k$.

Example Suppose a coin is flipped 10 times, and the probability of getting heads (success) in each flip is 0.5. Let X denote the number of heads obtained. Then $X \sim \text{Binomial}(10, 0.5)$. The probability of getting exactly 4 heads is:

$$P(X = 4) = \binom{10}{4}(0.5)^4(1 - 0.5)^6 = \frac{10!}{4! \cdot 6!} \cdot (0.5)^4 \cdot (0.5)^6 \approx 0.205. \tag{2.13}$$

2.2.4 Key Properties

The binomial distribution is characterized by several fundamental properties and assumptions, which form the basis for its application in probabilistic modeling. These include:

- **Support:** The random variable X takes discrete values in the set $\{0, 1, \ldots, n\}$, representing the possible number of successes in n trials.
- **Independence:** Each trial is independent of all others, ensuring that the outcome of one trial does not influence the outcomes of others.
- **Summation Property:** The probabilities of all possible outcomes sum to 1:

$$\sum_{k=0}^{n} P(X = k) = 1, \tag{2.14}$$

  satisfying the requirements of a probability distribution.

The moments and excess kurtosis of the binomial distribution, as outlined in Table 2.6, provide critical insights into the behavior of binomial random variables. These properties serve as essential tools for analyzing and interpreting outcomes in scenarios involving binary experiments.

In Chapter 5, we use these properties to analyze binomial metrics (Table 2.2) and their biases.

Table 2.6: Moments and Excess Kurtosis of the Binomial Distribution

Property	Symbol	Alternative Notation	Mathematical Expression
Mean	μ	$\mathbb{E}[X]$	np
Variance	σ^2	$\mathrm{Var}(X)$	npq
Skewness	γ_1	$\mathrm{Skew}(X)$	$\frac{q-p}{\sqrt{npq}}$
Kurtosis	β_2	$\mathrm{Kurt}(X)$	$\frac{1-6pq}{npq}+3$
Excess Kurtosis	γ_2	—	$\frac{1-6pq}{npq}$

Chapter 3

DEFINING AND MEASURING FAIRNESS IN AI: OBJECTIVE FAIRNESS INDEX (OFI)

Leveraging current legal standards, we define bias through the lens of marginal benefits and objective testing with the novel metric "Objective Fairness Index". This index combines the contextual nuances of objective testing with metric stability, providing a legally consistent and reliable measure. Utilizing the Objective Fairness Index, we provide fresh insights into sensitive machine learning applications, such as COMPAS (recidivism prediction), and bias mitigation analyses, highlighting the metric's practical and theoretical significance. The Objective Fairness Index allows one to differentiate between discriminatory tests and systemic disparities.

3.1 Introduction

In the 2010s, machine learning took major leaps forward in everyday life. However, we soon realized that bias is rampant in machine learning models. One of the most prominent examples regarding COMPAS, which deals with Americans' civil rights in the United States justice system (Brennan et al., 2009; Dieterich et al., 2016). Researchers are actively working to mitigate bias in general in a myriad of ways (Dai et al., 2022; Damak et al., 2022; Wang et al., 2022; Mehrabi et al., 2021; Caton and Haas, 2024; Corbett-Davies and Goel, 2018; Zhang et al., 2020; Hong et al., 2021; Kwon et al., 2023; Russo and Tonia, 2023; Zhang et al., 2023; Cornacchia et al., 2023; Gao et al., 2023; Le and Deng, 2023; Briscoe et al., 2021). Despite being the cornerstone of many proposals, definitions of bias remain inconsistent (Caton and Haas, 2024; Verma and Rubin, 2018). Due to various philosophies and situations, there is no consensus on a formal definition. This work focuses on the legal context, namely the disparate impact metric (DI). DI is useful as a flag or reparative measure, however it is insufficient in objective-testing laws.

To address this gap, we introduce the Objective Fairness Index (OFI), extending our work in Briscoe et al. (2024a,b). We formalize OFI by defining bias as the difference between marginal benefits, derived via the lens of objective testing. In law, objective testing is the philosophy that correct assessments found via rational, related tests are not discriminatory. Additionally, we provide applications to two critical bias scenarios: COMPAS and Folktables' adult employment dataset.

The principle of objective testing in non-discrimination laws is recognized not only in the US but worldwide. According to Chopin and Germaine (2017), 34 out of 35 surveyed countries (including Spain, Germany, Switzerland, Turkey, and Romania) have such laws in their national legislation. Going beyond the precedents found in the U.S. and the 34 other countries, we find that South Africa (Employment Equity Act), Canada (Public Service Employee Relations Commission v British Columbia Government Service Employees' Union), and Australia (Fair Work Act) uphold similar legal standards. Our work with the Objective Fairness Index aims to establish an agreeable and naturally interpretable definition of bias, grounded in these comprehensive legal frameworks and precedents.

We make three key contributions in this chapter:

- We introduce the Objective Fairness Index (OFI) and substantially support it with insights from the legal literature, highlighting the gap of bias metrics in the literature.
- We bring novel insights into two popular ML fairness applications: COMPAS (predicts recidivism) and Folktables' Adult Employment (predicts employment).
- Using a different dataset, (CASAS' smart home data) we demonstrate how OFI is used to assess bias and bias mitigation. Bias mitigation is achieved via a custom GAN: Conditional HydraGAN.

3.2 Limitations of Disparate Impact

This section gives context to the founding of the Objective Fairness Index by reviewing the Supreme Court of the United States' decision that spurred disparate impact and a following precedent. Under objective-testing laws, we show that OFI is better than DI in identifying discriminatory tests.

3.2.1 Legal Foundations of Disparate Impact

Disparate impact's significance comes from United States labor laws regarding the biased treatment of protected groups. In Griggs v. Duke Power Co., 401 U.S. 424 (1971), the Supreme Court unanimously agreed that employers must prove that any test or assessment used must be related to job performance (objective). Written by Chief Justice Burger, the decision says that a lower admittance rate of a protected class warrants further investigation—even if such discrimination may be unintentional. Since this precedent was passed from a lawsuit under Title VII of the Civil Rights Act of 1964, the Court's ruling extends to decision-making whenever protected classes are concerned.

To summarize this ruling, employers must have objective and fair tests (neutral in form) and should not inadvertently advantage a protected class (neutral in impact). Should there be a disparate impact, the employer is subject to scrutiny and bears the burden of proof that the test is objective and "consistent with business necessity". The disparate impact bias metric is defined in Definition 3.1. Where $P = \text{FN} + \text{TP}$ denotes the amount of positive labels, $\hat{P} = \text{FP} + \text{TP}$ denotes the amount of positive predictions, $N = \text{FP} + \text{TN}$ denotes the amount of negative labels, and $\hat{N} = \text{FN} + \text{TN}$ denotes the amount of negative predictions.

Definition 3.1. *Disparate impact compares the rates of positive outcomes between two groups using ratios.*

$$DI \triangleq \hat{P}_i / n_i \div \hat{P}_j / n_j \tag{3.1}$$

The implication of Equation 3.1 is that $DI = 1$ suggests no bias. However, the real world can have small and non-representative sampling, causing unbiased decisions to be $DI \neq 1$. As such, a Four-Fifths Rule (80% Rule) has been established in law as an initial screening tool to identify potential discrimination (Biddle, 2017). This rule is codified in 29 CFR § 1607.4 - Information on impact by the U.S. Equal Employment Opportunity Commission (1978). This rule suggests that if $DI > 5/4$ then group i has positive bias, if $DI < 4/5$ then there is positive bias for j, otherwise DI finds no bias.

In Griggs v. Duke Power Co., the Court goes on to note that Congress does not intend to require businesses to give jobs to unqualified candidates, and that context should still be considered, emphasizing that truly objective assessments are not discriminatory. However, because disparate impact lacks situational context, it cannot capture this legal nuance.

Disparate impact has been influential enough for employers to discard assessments that fail the heuristic to prevent any federal investigation or social uprising, regardless of the assessments' objectivity. Ricci v. DeStefano, 557 U.S. 557 (2009) is the first of these cases to be argued in the Supreme Court when New Haven, Connecticut dismissed objective assessments that benefited the white and Hispanic firefighters over other protected classes. The Court ruled in the firefighters' favor—that omitting an objective performance test solely due to differing acceptance rates (disparate impact) is a violation of the rights of those who correctly benefit from such a test. This provides us with a use case where equal positive prediction rates (which became zero after the test's omission) are discriminatory.

3.2.2 Defining Benefit and Bias

We define bias as a function of what happened to the group (benefit) and what should have happened to the group (expected benefit). An objective test will show that benefit equals expected benefit. Consistent with common understanding and legal precedents, we define benefit as an advantage that is gained from a decision or action, such as hiring from a machine-learning prediction. We achieve formal definitions of these terms using binary confusion matrices, allowing generalization to situations where information such as specific features cannot be used due to privacy, legal protection, or information loss.

We define an individual benefit in Definition 3.2 to help describe a group's benefit in Definition 3.3. Let us assume the positive prediction is the beneficial prediction in all definitions.

Definition 3.2. *An individual q's benefit b_q is the individual's prediction. $b_q = 1$ if predicted positive, otherwise 0.*

Definition 3.3. *A group's benefit is the arithmetic mean of all individual benefits within the*

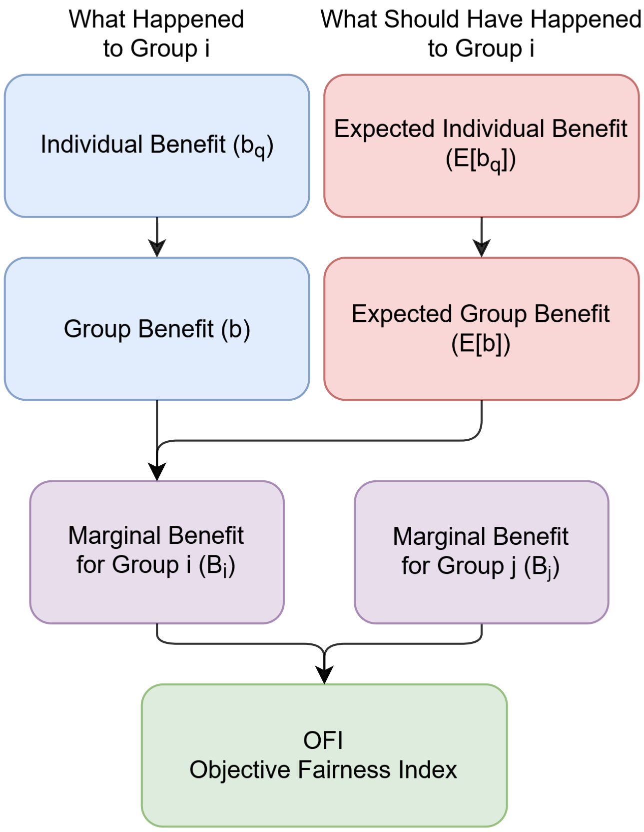


Figure 3.1: Objective Fairness Index's conceptual framework.

group.

$$b \triangleq \frac{1}{n} \sum_{\forall b_q} b_q = \frac{1 \cdot \hat{P} + 0 \cdot \hat{N}}{n} = \frac{\mathrm{TP} + \mathrm{FP}}{n} \tag{3.2}$$

The Objective Fairness Index also integrates the expected benefit of an individual (Definition 3.4) and the group's expected benefit (Definition 3.5) by using the actual labels.

Definition 3.4. *An individual's expected benefit* $\mathbb{E}[b_q]$ *is their actual label.* $\mathbb{E}[b_q] = 1$ *if labeled positive, otherwise 0.*

Definition 3.5. *A group's expected benefit is the arithmetic mean of all individual expected benefits within the group.*

$$\mathbb{E}[b] \triangleq \frac{1}{n} \sum_{\forall b_q} \mathbb{E}[b_q] = \frac{1 \cdot P - 0 \cdot N}{n} = \frac{\mathrm{TP} + \mathrm{FN}}{n} \tag{3.3}$$

Formally defining benefit and expected benefit allows a counterfactual test of "what happened" versus "what should have happened". As mentioned, this is of paramount importance in law. In Ricci v. DeStefano (2009), the city of New Haven omitted their test solely based on the ratio of benefits to be obtained by each protected class of firefighters. However, this discriminates against the firefighters qualified by this objective test. Pursuant to the Supreme Court's ruling, We present a solution by scoring this group's treatment as the "marginal benefit" ($\mathcal{B}$ in Definition 3.6).

Definition 3.6. *Marginal benefit ($\mathcal{B}$) is the difference between a group's benefit and expected benefit.*

$$\mathcal{B} \triangleq b - \mathbb{E}[b] = (\mathrm{FP} - \mathrm{FN})/n \tag{3.4}$$

This marginal benefit score suggests: $\mathcal{B} < 0$ an unjust lack of benefit; $\mathcal{B} = 0$ an appropriate amount of benefit; $\mathcal{B} > 0$ an unjust surplus of benefit. With marginal benefit defined, We now formalize the definition of the Objective Fairness Index in Definition 3.7. We provide a graphical representation in Figure 3.1.

Definition 3.7. *The Objective Fairness Index is the difference between two groups' marginal benefits.*

$$\mathrm{OFI} \triangleq \mathcal{B}_i - \mathcal{B}_j = \frac{\mathrm{FP}_i - \mathrm{FN}_i}{n_i} - \frac{\mathrm{FP}_j - \mathrm{FN}_j}{n_j} \in [-2, 2] \tag{3.5}$$

3.2.3 Improvement over Disparate Impact

To illustrate the Objective Fairness Index's significance, consider the instances in Table 3.1, similar to the case of New Haven and its firefighters. The same objective test is given in all scenarios. In Scenario A, the employer fulfills their promise and promotes (OFI = −0.06, DI = 0.38). In Scenario B, the test is discarded (OFI = 0.22, DI = 1). Note Scenario B's DI score is undefined. However, since the philosophy of DI suggests equal positive prediction rates indicate no bias, we use DI = 1.

With OFI's absolute value being greater in Scenario B than in Scenario A, OFI is consistent with the ruling of Ricci v. DeStefano (2009). However, DI disagrees with this precedent, suggesting for strong bias in Scenario B and none in Scenario A.

In Scenario α, we conduct a thought experiment with a smoothing technique. Scenario α adds one qualified individual to each group that benefits (e.g., is correctly promoted) to Scenario B.

Table 3.1: Comparison of OFI, DI, and Law (Ricci v. DeStefano) Across Different Scenarios

Group	Scenario	TP	FN	FP	TN	n	b	$\mathbb{E}[b]$	$\mathcal{B}$	OFI: $\mathcal{B}_i - \mathcal{B}_j$	DI: b_i/b_j	Ricci v. DeStefano
i	A	1	0	0	5	6	1/6	1/6	0	−0.06	0.38	Likely no bias
j	A	7	0	1	10	18	8/18	7/18	1/18			
i	B	0	1	0	5	6	0/6	1/6	−1/6	0.22	NaN or 1	Bias for i
j	B	0	7	0	11	18	0/18	7/18	−7/18			
i	α	1	1	0	5	7	1/7	2/7	−1/7	0.23	2.71	Bias for i
j	α	1	7	0	11	19	1/19	8/19	−7/19			

Adding this single correct prediction changes DI's score from 1 to 2.71, suggesting a transition from no bias to heavy bias for group i. However, OFI adjusts from 0.22 to 0.23 in the same Scenario α, illustrating its robustness.

Unlike DI, the Objective Fairness Index indicates the correct directional bias in both scenarios. Additionally, OFI's defined-everywhere property and robustness against small changes showcase its stability. However, note that DI is complementary to OFI in systemic disparity detection. With OFI, we can identify if an algorithm is at fault (OFI $\not\approx 0$) or if a systematic disparity exists outside of the algorithm (OFI ≈ 0 and DI $\not\approx 1$).

3.3 Baseline Desiderata for Bias Metrics

Ideally, metrics should be explainable and have compelling reasons for their existence such as legal precedents. We present and define five desirable traits for all metrics: real-valued, directed, symmetric, bounded, and defined everywhere. Next, we define how a metric satisfies objective testing. Finally, we present a table comparing many popular bias metrics in Table 3.2. We introduce two new symbols: s for the result of a bias metric M, and p to represent a scalar pivot. A pivot only occurs in directional metrics and is the pinpoint of bias direction. The subscript ϕ is a reminder that the symbol represents a property.

3.3.1 Real-Valued

Real-valued metrics provide a continuous scale to quantify the bias difference between groups, offering a nuanced understanding of bias magnitude. The advantage of real-valued metrics

Table 3.2: Popular confusion-matrix-based bias metrics and checks for satisfying desirable properties. Each metric contrasts group i with group j. For all metrics, a greater score indicates more bias toward group i. Abbreviations: Satisfies Objective Testing (OBJ), Real-Valued ($\mathbb{R}$), Directed (DIR), Symmetric (SYM), Bounded in Range (BND), Defined Everywhere (ALL).

Metric	Formula	OBJ	$\mathbb{R}$	DIR	SYM	BND	ALL
Accuracy Difference	$ACCD \triangleq \frac{\mathrm{TP}_i+\mathrm{TN}_i}{n_i} - \frac{\mathrm{TP}_j+\mathrm{TN}_j}{n_j}$	×	✓	✓	✓	✓	✓
MCC Difference	$MCCD \triangleq MCC_i - MCC_j$	×	✓	✓	✓	✓	×
Disparate Impact	$DI \triangleq \frac{\hat{P}_i}{n_i} \div \frac{\hat{P}_j}{n_j}$	×	✓	✓	×	×	×
Predictive Parity	$PP \triangleq \frac{\mathrm{TP}_i}{P_i} - \frac{\mathrm{TP}_j}{P_j}$	×	✓	✓	✓	✓	×
Treatment Equality	$TE \triangleq \frac{\mathrm{FN}_i}{\mathrm{FP}_i} - \frac{\mathrm{FN}_j}{\mathrm{FP}_j}$	×	✓	✓	✓	×	×
Equalized Odds	$\mathrm{EO} \triangleq (\mathrm{FPR}_i = \mathrm{FPR}_j) \wedge (\mathrm{TPR}_i = \mathrm{TPR}_j)$	×	×	×	✓	✓	×
Statistical Parity	$SP \triangleq \hat{P}_i = \hat{P}_j$	×	×	×	✓	✓	✓
Average Absolute Odds Difference	$AAOD \triangleq \frac{1}{2}\left(\left\|\frac{\mathrm{FP}_i}{N_i} - \frac{\mathrm{FP}_j}{N_j}\right\| + \left\|\frac{\mathrm{TP}_i}{P_i} - \frac{\mathrm{TP}_j}{P_j}\right\|\right)$	×	✓	×	✓	✓	×
Difference in Conditional Acceptance	$DCA \triangleq \frac{P_i}{\hat{P}_i} - \frac{P_j}{\hat{P}_j}$	×	✓	✓	✓	×	×
Difference in Conditional Rejection	$DCR \triangleq \frac{N_j}{\hat{N}_j} - \frac{N_i}{\hat{N}_i}$	×	✓	✓	✓	×	×
Difference in Positive Proportion & Labels	$DPPL \triangleq \frac{\hat{P}_i}{n_i} - \frac{\hat{P}_j}{n_j}$	×	✓	✓	✓	✓	✓
Objective Fairness Index	$OFI \triangleq \frac{\mathrm{FP}_i - \mathrm{FN}_i}{n_i} - \frac{\mathrm{FP}_j - \mathrm{FN}_j}{n_j}$	✓	✓	✓	✓	✓	✓

over Boolean or categorical metrics is their ability to capture a wide range of outcomes and degrees of bias, rather than merely indicating the presence or absence of bias. This property is crucial for fine-grained analysis and comparison across different models or systems. For instance, Equalized Odds is introduced as being binary in its basic form, yet adapting the metric into a real-valued measure by considering the differences in false positive rates (FPR) and true positive rates (TPR) between groups highlights degrees of bias and direction, allowing for more detailed analyses.

Real-valued metrics are essential for creating a detailed and actionable understanding of bias, allowing law enforcement and other stakeholders to prioritize issues based on the magnitude of disparity. However, the interpretation of these values requires careful consideration, as the scale and range can vary significantly between different metrics, potentially leading to confusion or misinterpretation. We formally define real-valued properties in Definition 3.8.

Definition 3.8. *The real-valued property R_ϕ is satisfied when a metric can give any real value within some non-empty range given by (a, b) where a and b are defined by the metric's range and are in $(-\infty, \infty)$.*

$$\mathbb{R}_\phi \triangleq s \in \mathbb{R} \cap (a, b) \tag{3.6}$$

3.3.2 Direction

Direction is necessary to understand which group is favored or disadvantaged by a model's predictions, providing essential insights beyond the mere presence or amount of bias. This property is vital for diagnosing the nature of bias and guiding corrective actions. By identifying the direction of bias, practitioners can tailor interventions to support the disadvantaged group, whether through re-balancing training data, adjusting model parameters, or applying post-processing fairness techniques.

However, the utility of directional information depends on the clarity and consistency of its definition across metrics. Ambiguities in how direction is calculated or interpreted can undermine its effectiveness in fairness assessments. Therefore, a standardized approach to determining and reporting direction can enhance the utility of this property in bias metrics.

In the literature, we find that this approach often comes with a pivot where the pivot indicates the absence of bias. If the bias score is less than the pivot, group j holds the advantage. On the other hand, if the score is greater than the pivot, group i holds the advantage.

Upon all reviewed directional bias metrics, we find that they are split into two categories: ratios and Manhattan distance. For ratios such as disparate impact, one is the pivot. Otherwise, Manhattan-distance directional metrics (including OFI) have zero as the pivot. We formally define the directional property in Definition 3.9.

Definition 3.9. *The directional property D_ϕ is satisfied when a metric indicates which group* *holds the advantage using a pivot p and bias score s.*

$$D_\phi \triangleq s \begin{cases} \textit{suggests bias for group i and against group j} & \textit{if } s > p; \\ \textit{suggests no bias} & \textit{if } s = p; \\ \textit{suggests bias against group j and for group i} & \textit{if } s < p. \end{cases} \tag{3.7}$$

3.3.3 Symmetry

Symmetry ensures that the metric yields the same magnitude of bias, regardless of which group is denoted as “group i” or “group j”. This property simplifies the analysis by eliminating the need to consider group order in the calculation, thereby enhancing the interpretability and comparability of results.

The challenge with ensuring symmetry lies in the design of the metric itself. Metrics that naturally lend themselves to symmetry, such as differences or ratios, are preferable. Disparate impact is a ratio of ratios and is therefore asymmetric. However, OFI and others are symmetric. We define symmetry in Definition 3.10.

Definition 3.10. *The symmetric property S_ϕ of a metric M yields equal magnitudes of bias that are suitable for direct comparison.*

$$S_\phi \triangleq |M(i, j)| = |M(j, i)| \tag{3.8}$$

3.3.4 Bounded

Bounded metrics have a fixed range, making them easier to interpret and compare across different contexts. A bounded range, such as $[0, 1]$ or $[-1, 1]$, provides a clear reference point for evaluating the severity of bias, with the bounds representing the absence or extremity of bias. This property is particularly useful for non-experts or stakeholders who need to understand bias metrics without delving into complex statistical details. Disparate impact is unbounded while OFI is bounded.

The primary challenge with bounded metrics lies in their score distribution and ensuring that it accurately reflects the nuances of bias across the entire range. Metrics that are too coarse (e.g., binary scores) or that saturate easily (e.g., sigmoidal function) may not capture subtle variations in bias, while those that are too sensitive might fluctuate wildly for minor changes in the underlying data or model behavior.

We satisfy checks for coarseness with the real-value property. Additionally, none of the metrics we review inherently saturate easily. Nonetheless, some situations may cause metrics to saturate. Thus, checking for context-dependent saturation requires more information such as the model's hypothesis distribution and its data. We define the bounded properly in Definition 3.11.

Definition 3.11. *The bounded property B_ϕ is satisfied if the metric M yields scores bounded by finite scalars a and b, or if the set of possible scores is finite.*

$$B_\phi \triangleq s \begin{cases} s \in [a, b] & \text{if } M \models R_\phi; \\ s \text{ is an element of a finite set} & \text{otherwise.} \end{cases} \tag{3.9}$$

3.3.5 Universal Applicability

Metrics defined for any possible confusion matrix (CM) and population size ($n > 0$) are robust and versatile, capable of handling a wide variety of scenarios without becoming undefined or losing meaning. The universal-applicability property ensures that the metric can be consistently applied across different datasets and models, regardless of the size or distribution of the groups being compared. This is especially important when considering groups with a low sample size.

Achieving this property requires careful mathematical formulation to avoid divisions by zero, undefined logarithms, or other issues that can arise in edge cases. Metrics that are not universally applicable may require additional handling or assumptions, complicating their use and interpretation. We formally define the universal-applicability property in Definition 3.12.

Definition 3.12. *The universal-applicability property U_ϕ is satisfied for any metric being defined everywhere for any non-empty group ($n > 0$). $\mathbb{S}$ describes the set of all valid scores from M.*

$$U_\phi \triangleq M(\text{CM}) \mapsto \mathbb{S}, \forall \text{CM} \,|_{n>0} \tag{3.10}$$

3.3.6 Objective Testing

Objective testing, a concept rooted deeply in legal precedents such as Griggs v. Duke Power Co. (1971) et seq., evaluates bias through a lens that compares what happened (actual benefit) against what should have happened (expected benefit) under a non-discriminatory system. This approach is encapsulated in the Objective Fairness Index (OFI) metric, which uniquely satisfies this property by employing a counterfactual analysis that incorporates both actual and potential outcomes.

An objective-bias metric performs a counterfactual test against benefit b with expected benefit $\mathbb{E}[b]$, like $\mathcal{B}$ does. We define the objective-testing property in Definition 3.13. This is the only property highlighting the intra-group case.

Definition 3.13. *The intra-objective-testing property ($\mathcal{B}_\phi$) is satisfied when some metric M increases or decreases as b does, and does the opposite for $\mathbb{E}[b]$. Furthermore, it recognizes context by having correct predictions (TP and TN) be evidence for objectivity, lowering the bias score respectively. We informally summarize this in Equation 3.11 below. We assume that only one sample is added or removed in each scenario.*

$$\mathcal{B}_\phi \triangleq \begin{cases} M(b, \mathbb{E}[b]) \uparrow & \textit{as } b \uparrow \textit{ or } \mathbb{E}[b] \downarrow \\ M(b, \mathbb{E}[b]) \downarrow & \textit{as } b \downarrow \textit{ or } \mathbb{E}[b] \uparrow \\ M(b, \mathbb{E}[b]) \textit{ goes toward } 0 & \textit{as correct values} \uparrow \\ M(b, \mathbb{E}[b]) \textit{ moves away from } 0 & \textit{as correct values} \downarrow \end{cases} \tag{3.11}$$

OFI compares two expressions satisfying $\mathcal{B}_\phi$, finding the disparity of marginal benefit. As such, OFI satisfies the inter-objective-testing property as defined in Definition 3.14. The inter-objective-testing property may be shortened to "objective-testing property" or O_ϕ.

Definition 3.14. *The objective-testing property (O_ϕ) is satisfied when the metric $M(B_i, B_j)$ takes one parameter for each group, both satisfying $\mathcal{B}_\phi$. Additionally, O_ϕ finds bias for group i (as* *decided from threshold p) iff B_i is greater than B_j.*

$$O_\phi \triangleq M(B_i, B_j) > p \iff B_i > B_j \text{ s.t. } B \models \mathcal{B}_\phi \tag{3.12}$$

From Table 3.2, we see that OFI is the only objective-testing metric and the only metric to satisfy all desirable properties and is the only one to satisfy the objective-testing property.

Nuances

In this section, we further illustrate how OFI complies with O_ϕ unlike two deceptively similar metrics: treatment equality (TE) and difference in conditional acceptance (DCA). In Table 3.3, we give Scenarios C and D for groups i and j, the scores obtained by each metric, and the threshold for when that metric determines bias against group i. We define confusion matrices in the form [TP, FN, FP, TN].

Table 3.3: Comparative Analysis of Deceptively Similar Bias Metrics, Highlighting OFI's Unique Consideration of Objective Testing

Description	Scenario C	Scenario D	Threshold
Group i's CM	$[1, 2, 1, 1]$	$[1, 2, 1, 1]$	-
Group j's CM	$[1, 1, 2, 2]$	$[1, 1, 2, 4]$	-
TE = $\mathrm{FN}_i/\mathrm{FP}_i - \mathrm{FN}_j/\mathrm{FP}_j$	1.5	1.5	TE > 0
DCA = $P_i/\hat{P}_i - P_j/\hat{P}_j$	0.83	0.83	DCA > 0
OFI = $\mathcal{B}_i - \mathcal{B}_j$	**-0.37**	**-0.33**	OFI < 0

Table 3.3 compares a scenario where Group i's CM is stagnant and Group j gains three more TNs in Scenario D. This highlights how neither TE nor DCA recognizes the information of three more correct predictions. However, OFI captures the subtlety by calculating the marginal benefit for each group and scenario. We only claim that OFI uniquely satisfies the objective-testing property, and stress that reparative metrics such as TE or DI be considered as necessary.

OFI stands out by not only quantifying the discrepancy between groups' marginal benefits but also providing a means to objectively test for bias across different groups, making it invaluable for legal consistency. Importantly, OFI is not designed to be reparative but rather to identify and quantify bias, aligning closely with the objective testing sought in legal contexts. When appropriate, we recommend using a reparative metric such as DI. For example, South Africa's Employment Equity Act (Section 6.2) requires considering affirmative action and objective testing.

3.4 Applying OFI: Empirical Case Studies

We explore the application of the Objective Fairness Index in practical scenarios: COMPAS (predicts recidivism), and Folktables' Adult Employment dataset (predicts employment).

For COMPAS, we obtain the confusion matrix from actual COMPAS predictions and cases of recidivism occurring within two years (Angwin et al., 2016a; Gursoy and Kakadiaris, 2022). Since this work assumes that the positive label is beneficial, we flip the COMPAS positive label to be "not a recidivist".

Figure 3.2 shows bias scores for each ethnicity pair in the COMPAS dataset. Here, we see that OFI affirms DI's findings. This is an important revelation, as some argue that the disparate prediction rates are correct. In return, some argue that the labeled dataset suffers from systematic bias. However, this case illustrates that OFI can sidestep the need to prove incorrect ground labels, and shows that even if one assumes correct ground labels, there is algorithmic bias against certain races in COMPAS (OFI $\not\approx 0$). We believe this revelation will encourage more people to take corrective measures.

For the Folktables dataset, we analyze data from the Georgia census (393,236 observations) to predict employment status using a Random Forest classifier (RF) and Naïve Bayes (NB). In RF, OFI corroborates the findings from DI. However, OFI reveals differences to DI in the NB experiment (see Figure 3.3). Here, DI suggests a positive bias for Pacific Islanders in 7/8 cases. In contrast, OFI shows positive bias in 2/8 cases.

With the lens of objective testing, we find that Pacific Islanders suffer from algorithmic bias, despite having high disparate impact scores. These studies demonstrate OFI confirming suspicions of algorithmic bias (COMPAS), and when protected classes suffer from algorithmic bias, despite a high positive prediction rate.

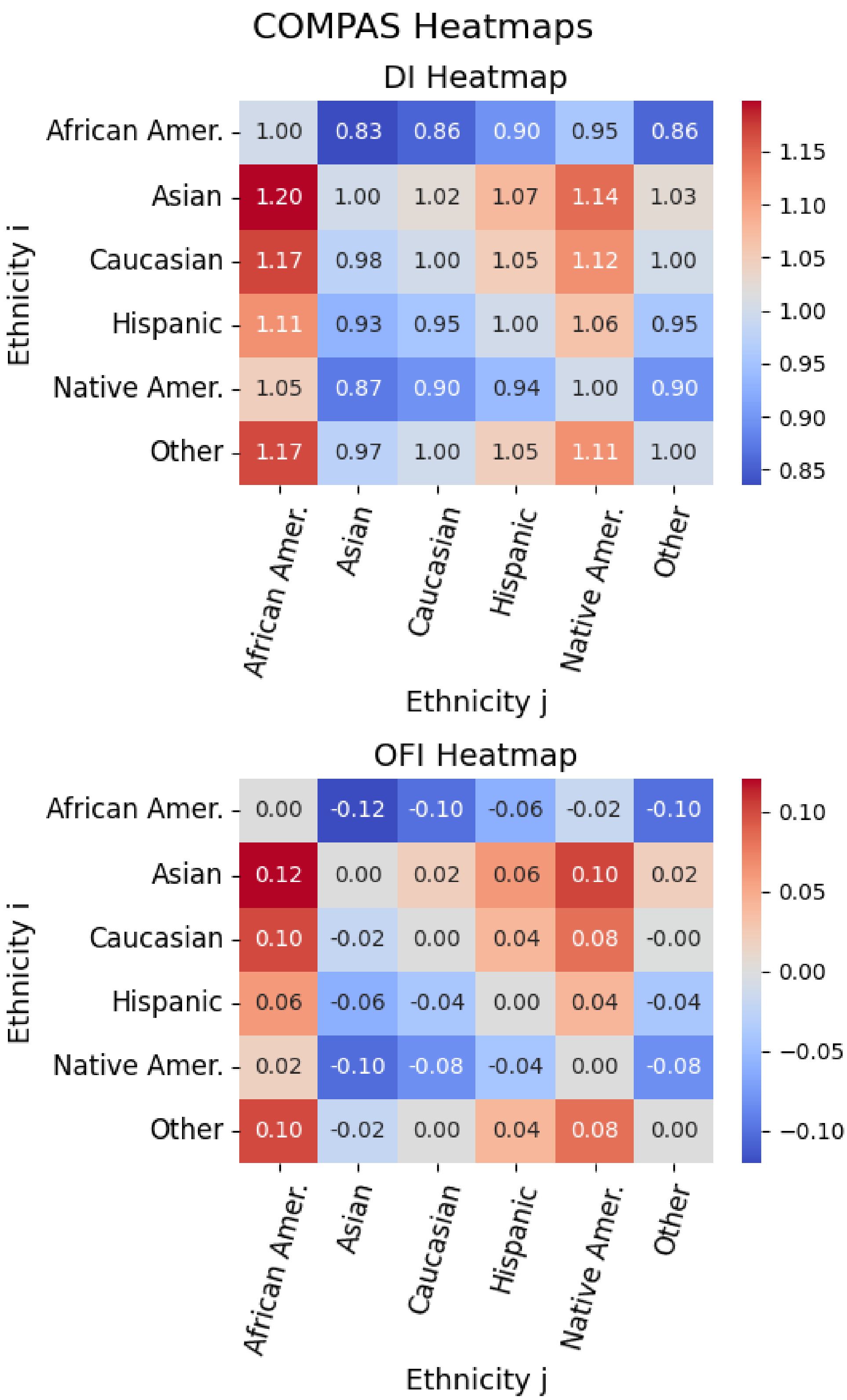


Figure 3.2: OFI shows that COMPAS manifests algorithmic bias in addition to manifesting disparate impact (DI). We use the predictions and labels published in Angwin et al. (2016a).

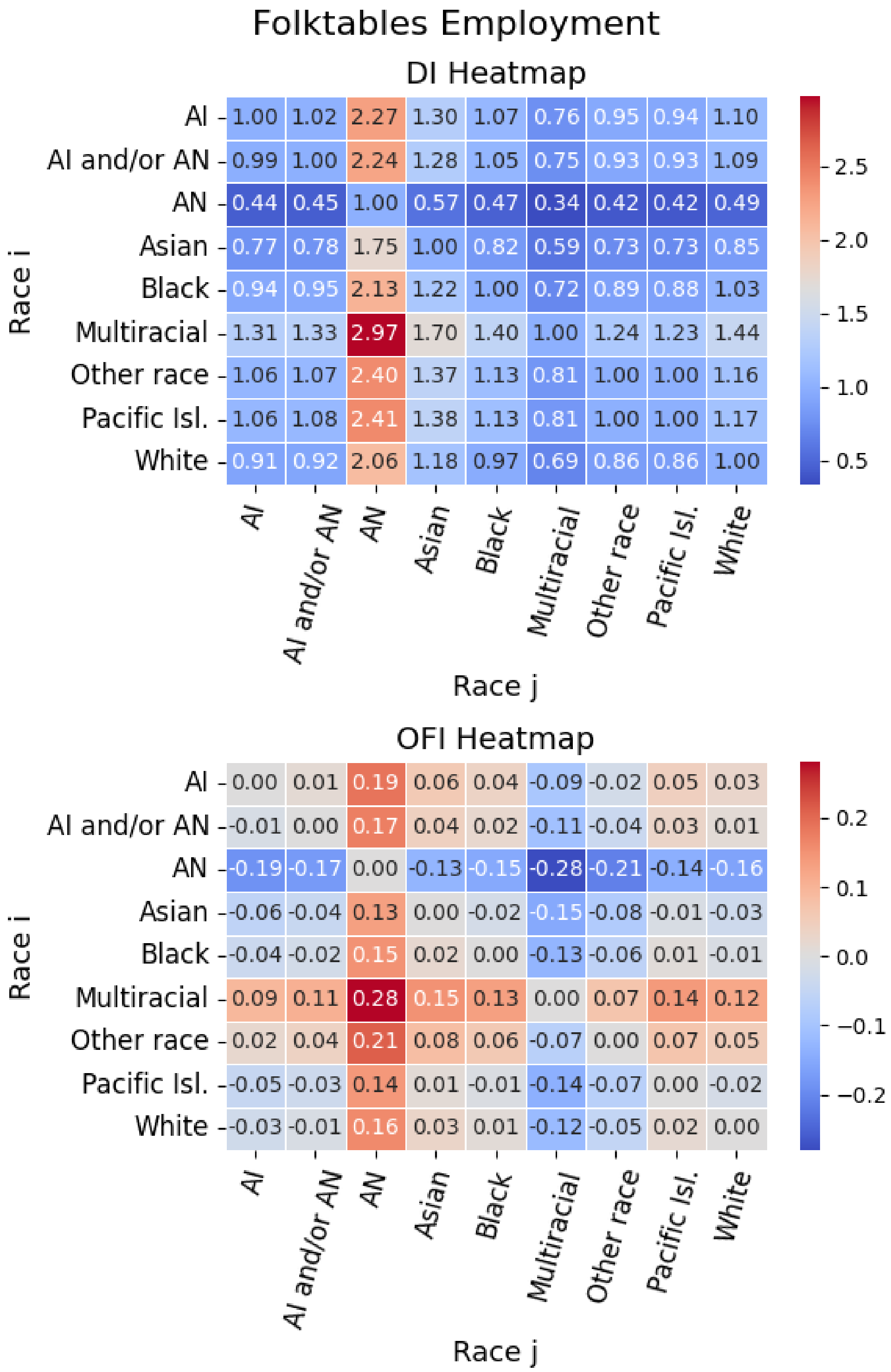


Figure 3.3: DI conveys that Pacific Islanders have positive bias for 7/8 comparisons. However, OFI shows that Pacific Islanders suffer from algorithmic bias in 6/8 cases. **Key:** AI is American Indian, AN is Alaska Native, Pacific Isl. is Pacific Islander.

3.5 Evaluating Conditional HydraGAN for Bias Mitigation with the Objective Fairness Index

In this section, we evaluate Conditional HydraGAN's effectiveness in mitigating bias in the CASAS dataset using the Objective Fairness Index (OFI) and other bias metrics. Our analysis demonstrates the utility of OFI in evaluating bias mitigation strategies and promoting fairness in healthcare outcomes.

3.5.1 Background

In 2014, Goodfellow et al. introduced the Generative Adversarial Network (GAN), which became highly influential for its capability to produce realistic synthetic data. A GAN consists of two neural networks, a generator and a discriminator, which engage in a competitive, two-player, zero-sum game. The generator takes a random latent vector and generates synthetic data, aiming to deceive the discriminator into classifying this data as real. The discriminator, trained using binary cross-entropy, evaluates both real and synthetic data, attempting to distinguish between them.

Shortly after, Mirza and Osindero (2014) introduced the Conditional GAN (CGAN), which extends GANs by allowing users to control the output distribution. CGAN accomplishes this by concatenating a conditional vector to the generator's latent vector, enabling users to specify attributes for the generated output.

In subsequent years, Arjovsky et al. (2017) refined the GAN architecture by introducing the Wasserstein GAN (WGAN), which reframes the generator/discriminator setup as an actor/critic model. In WGAN, the critic employs the Wasserstein loss function to score the generator's output, aiming for improved stability and gradient flow. However, Arjovsky et al. found that large neural weights could destabilize training, leading them to implement weight clipping as a control measure. Later, Gulrajani et al. (2017) addressed the limitations of weight clipping by introducing a gradient penalty (WGAN-GP), which imposes a softer constraint on weight sizes and enhances stability by penalizing large weights in the loss function.

We integrate the principles of CGAN and WGAN-GP into a Conditional Wasserstein GAN (CWGAN). Building on this, we develop Conditional HydraGAN, an extension that incorporates DeSmet and Cook's (2024) HydraGAN—a multi-objective GAN with multiple critics/discriminators, each specialized for different aspects of data quality and diversity.

3.5.2 Introduction to Conditional HydraGAN

In the pursuit of equitable healthcare outcomes, particularly within geriatric populations, addressing algorithmic bias is essential. Machine learning models deployed in clinical settings face the challenge of biases that, if uncorrected, can propagate healthcare disparities. This section presents *Conditional HydraGAN*, a multi-objective generative adversarial network (GAN) developed specifically for mitigating biases in geriatric clinical datasets. Unlike DeSmet and Cook's original HydraGAN model (2024), which functions as a general-purpose data augmenter, the Conditional HydraGAN introduces conditional attributes to generate synthetic data, enforcing an arbitrary output distribution of sensitive features, such as age and gender. By enabling customized data generation, Conditional HydraGAN enhances the representativeness of training data, promoting fairness in health outcome predictions.

We use the CASAS' dataset to illustrate OFI's effectiveness outside of traditional bias datasets. Our conditional HydraGAN serves as a valuable instance where OFI can objectively evaluate bias and its mitigation across subgroups. Utilizing OFI, we analyze the effectiveness of Conditional HydraGAN by measuring bias both pre- and post-synthetic augmentation. This approach ensures that any reduction in bias through Conditional HydraGAN is rigorously and transparently evaluated.

3.5.3 Methodology: Conditional HydraGAN Design and Implementation

The Conditional HydraGAN employs a multi-agent architecture where a generator is balanced by multiple critics, each responsible for different objectives, such as data realism and fairness. To achieve targeted bias mitigation, we introduce conditional vectors representing sensitive attributes into the generator's input space, as done in Briscoe et al. (2022). This design allows the network to learn variations specific to underrepresented subgroups within the geriatric dataset. During training, each critic provides feedback not only on the data's realism but also on the extent to which bias is minimized with respect to specific protected attributes.

For the CASAS dataset, Conditional HydraGAN is guided by three core objectives: individual diversity, activity diversity, and data realism. In the original HydraGAN model, these goals would typically be addressed through three separate critics. However, by incorporating conditional vectors, Conditional HydraGAN inherently fulfills the individual diversity requirement, eliminating the need for a dedicated critic on this aspect.

Activity diversity, while essential, is given a more flexible role by being treated through a loss function rather than as a strict conditional constraint. This design choice allows the generator to prioritize individual diversity while permitting more natural variability in the types of activities represented. By relaxing activity diversity into a loss function, the model gains the flexibility to balance this objective against the overarching goal of data realism, ensuring that synthetic data better reflect the authentic, often unpredictable patterns of geriatric health behaviors. Treating activity diversity as a loss function minimizes the risk of producing overly rigid activity distributions that may lack representational richness, while still guiding the model toward generating a broad range of activity types. The activity diversity critic minimizes the divergence-based diversity loss. The divergence-based diversity loss measures the MSE between the actual proportion of activities and the desired proportion for each activity label.

Data realism remains independent of the conditional attributes, as it relies on the generator's ability to produce samples that reflect the broader distribution of the geriatric dataset without being tightly constrained to specific activity patterns. By decoupling data realism from conditional constraints, Conditional HydraGAN focuses on generating data that is both demographically aligned and realistic, capturing the complex and naturally varied patterns found within geriatric populations. This configuration allows Conditional HydraGAN to strike a balance between maintaining essential diversity criteria and achieving realistic, nuanced data that can better support bias mitigation in healthcare applications. The data realism critic minimizes the Wasserstein + Gradient Penalty (WGAN+GP) loss (Gulrajani et al., 2017).

3.5.4 Results: Evaluating Bias Mitigation with OFI

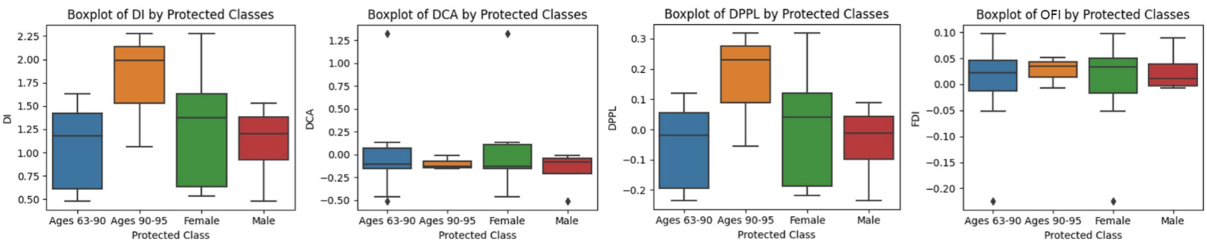
We focus on two sensitive attributes: age and gender. In the case of gender, we assess bias for the traditional male and female groups. In the case of age, we assess bias for the older 25% of the sample in comparison with the group containing the younger 75% of the participants. Rather than evaluate bias for all of the metrics listed in Table 3.2, we select a subset including Disparate Impact (DI), Difference in Conditional Acceptance (DCA), Difference in Proportionate Positives and Labels (DPPI), and Fairness Disparity Index (OFI). This is a representative set: the remaining metrics yield very similar results to these.

The selected bias metrics focus on a task, in this case activity recognition. To simplify analyses, we aggregate activity categories into two classes: *active* behavior (bed-toilet transition, cook, eat, enter home, leave home, hygiene, wash dishes) and *sedentary* behavior (relax, sleep, work, other). For DCA, DPPL, and OFI, a no-bias score is zero. For DI, the no-bias score is one. A closer value to the no-bias score indicates less bias. A positive value indicates a bias toward the sedentary categories, while a negative value indicates a bias toward the active categories.

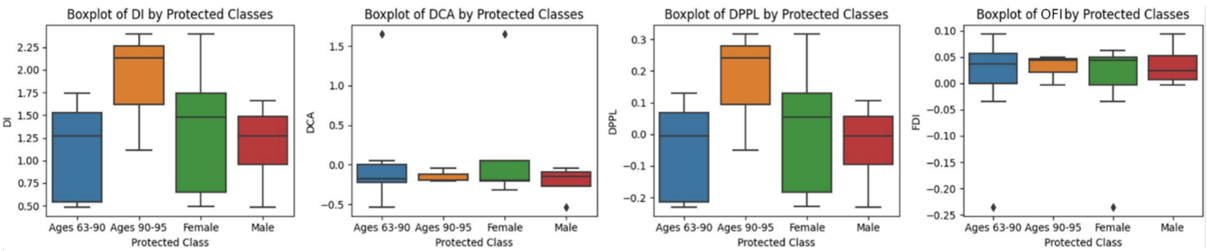
While Aminikhanghahi and Cook (2019) employ a random forest for CASAS human activity recognition, this work uses predictions from a multilayer perceptron (MLP) classifier. We find that the MLP is better able to infer classifications of the real data using the synthetic data. This is likely because the synthetic data's sensors and activities are processed by a softmax activation function while the real data are one-hot encoded. We do not use argmax to one-hot encode the generated data because some information is lost in the process. Empirically, the MLP better transfers knowledge between the synthetic softmax and real one-hot encoded probability density functions.

3.5.5 Algorithmic Bias

We compare the algorithmic bias between the original and expanded datasets in Figure 3.4. We see that Conditional HydraGAN yields improvements in OFI, DI, and DCA. DPPL shows about no difference, however this metric assesses the proportion of positive labels (is sedentary),



(a) Evaluating the original dataset with DI, DCA, DPPL, and OFI.



(b) Evaluating DI, DCA, DPPL, and OFI on the synthetic dataset generated by Conditional HydraGAN.

Figure 3.4: Comparison of algorithmic bias using original and synthetic datasets, illustrating bias metrics before and after synthetic data generation using the Conditional HydraGAN approach.

Using the Original Dataset

Figure 3.4a depicts boxplots of quantified bias on the original dataset. Since the Ages 90-95 class receives high scores in DI and DPPL, we see they are more likely to be predicted as sedentary. Referencing OFI, which gives context for correct predictions, we see that these positive predictions are largely correct as the Ages 90-95 OFI score is near zero. This makes sense as the data reflects that these older patients tend to be sedentary more often than the younger class. However, since OFI is positive, we see that there is still a slight bias for predicting the older age group as sedentary more often than they should be.

Using the Expanded Dataset

We train the model using synthetic data generated by HydraGAN then quantify the bias by testing on real data. We summarize the bias results in Figure 3.4b. Here, individual diversity is enforced by querying the generator with each individual's label. This individual diversity also improves the protected class's diversities. In total, 3,072,000 synthetic points are created to be realistic and improve diversity for the underrepresented groups.

We see in Table 3.4 that the synthetic data mitigates algorithmic bias overall (the bold cells indicate improvement). For OFI, DPPL, and DCA, a score of zero indicates no bias. For DI, the ideal score is 1. Over many trials, these scores have a negligible standard deviation. Some improvement results seem off due to rounding.

Table 3.4: Comparing synthetic and real data for each protected class using bias metrics.

Metric	Data	Ages 63-90	Ages 90-95	Female	Male
OFI	Synthetic	0.002	0.026	-0.001	0.026
	Real	0.004	0.030	-0.001	0.033
	Improves	**0.002**	**0.005**	-0.000	**0.007**
DPPL	Synthetic	-0.056	0.165	0.012	-0.043
	Real	-0.055	0.170	0.012	-0.036
	Improves	-0.002	**0.005**	**0.000**	-0.007
DI	Synthetic	1.060	1.778	1.280	1.103
	Real	1.109	1.891	1.340	1.174
	Improves	**0.049**	**0.113**	**0.060**	**0.071**
DCA	Synthetic	0.003	-0.100	0.046	-0.172
	Real	0.012	-0.151	0.061	-0.219
	Improves	**0.009**	**0.051**	**0.014**	**0.048**

3.5.6 Data Bias

We see that our infinitely strong conditional drastically improves representation for all individuals in Table 3.5. The softer constraint (done by a critic) improves the diversity of activities between 4% and 20%, depending on the metric considered.

Table 3.5: Comparison of synthetic data and real data to the target uniform distribution.

	Activities			**Individuals**		
Metric	**Synthetic**	**Real**	**Improvement**	**Synthetic**	**Real**	**Improvement**
KS statistic	0.863	0.900	**4.11%**	0.000	0.733	**100%**
KL Divergence	1.822	2.302	**20.9%**	0.000	0.252	**100%**
JS Distance	0.652	0.725	**10.1%**	0.000	0.250	**100%**

3.5.7 Conclusion

The Conditional HydraGAN demonstrates that targeted synthetic data generation can effectively address specific biases in clinical data, particularly in the context of geriatric health monitoring. By applying OFI, we not only ensure that Conditional HydraGAN's bias mitigation is quantitatively assessed but also highlight the broader applicability of OFI in evaluating the fairness of diverse machine learning interventions. Conditional HydraGAN thus represents a practical tool for advancing bias mitigation strategies, with the OFI providing an objective lens for assessing fairness improvements in real-world clinical applications.

3.6 Conclusions and Future Directions

In this work, we introduce the Objective Fairness Index (OFI) to evaluate bias in a legally grounded and context-aware perspective. By leveraging legal frameworks and precedents, particularly those stemming from objective testing and disparate impact theory, OFI captures the nuanced understanding of bias that encompasses both what occurred and what ought to have occurred.

In Section 3.2, we illustrate that the Objective Fairness Index is legally grounded, even where the disparate impact metric fails. Specifically, OFI finds no bias in objective testing while DI may or may not, and OFI can correctly detect bias where DI does not. The analyses include legal-case-driven situations and popular bias problems, including COMPAS (risk of recidivism).

The Objective Fairness Index addresses a significant gap in the machine learning and AI fairness literature by offering a metric that not only aligns with legal standards but also provides novel insights into binary classification problems. Our findings highlight the limitations of existing metrics, such as disparate impact, which lack context and may lead to misinterpretation or oversight of critical bias factors. OFI offers a solution by integrating objective testing directly into the evaluation of bias.

For future work, we envision OFI in more complex scenarios, including multi-class classification problems, regression tasks, and recommender systems. Another promising avenue is the exploration of OFI's role in the development of debiasing techniques.

Additionally, corrective procedures should be investigated. As an objective measure, OFI identifies discriminatory selections, whereas reparative measures such as DI are for identifying social disparities. Thus, if the OFI fails, correction at the algorithmic or employer level is appropriate. However, if OFI shows no bias but reparative measures do, this suggests systematic disparity and can rationalize social programs. We recommend that all measures and corrective procedures are routinely scrutinized regarding ethics and law.

There are some inherent limitations to applying concepts in law to binary classification. Examples include the "good faith" of the test-maker (Canada's Meiorin test), proportional responses (UK's Equality Act 2010, suggesting a non-binary decision), and expectations of "reasonable accommodations" for certain situations (Americans with Disabilities Act). We hope to explore solutions for these in more complex scenarios where they are addressable.

This work underscores the importance of interdisciplinary approaches in tackling bias in AI. By grounding OFI in well-established legal precedents and demonstrating its applicability to contemporary machine learning challenges, we contribute a tool for researchers striving for fairness and transparency in AI systems.

Chapter 4

FAIRNESS IN CLASSIFICATION: BRIDGING SOCIAL CHOICE THEORY AND MACHINE LEARNING FAIRNESS

Fairness in binary classification is commonly assessed with metrics such as Statistical Parity or Disparate Impact. However, choosing among them remains a difficult problem. These metrics contain different assumptions, react differently to aggregation and edge cases, and can lead practitioners to inconsistent conclusions about which groups are disadvantaged. We address this problem by transforming ideas from Social Choice Theory (SCT) to confusion-matrix-based fairness metrics. We define a taxonomy of *Parity-Gap Metrics* that unifies common group-fairness metrics and organizes them into *Single-Estimate Parities* or *Multi-Estimate Parities*. In this framework, we adapt SCT-style properties to the binary-classification setting and characterize their behavior for certain metric classes. Our results identify important pathologies and trade-offs, including dictatorship, Simpson's Paradox-style reversals under aggregation, and an impossibility to jointly satisfy "Consistency" and "Participation". The resulting taxonomy, property summaries, and decision diagram provide a more principled basis for selecting fairness metrics in practice.

Some evaluative properties relevant to fairness metrics were already introduced in Chapter 3, where they were used to assess OFI as a bias metric. Those properties serve here as baseline desiderata. This present chapter extends that earlier discussion by introducing social-choice-theoretic axioms that are native to aggregation problems, such as participation, consistency, and dictatorship-style pathologies. Accordingly, Chapter 3 asks whether a metric is well behaved in isolation, whereas this chapter asks how fairness metrics behave as aggregation rules over groups.

4.1 Introduction

Machine–learning (ML) systems now often inform decisions about credit, hiring, healthcare access, and public policy. In these settings, choosing a fairness metric determines which disparities are visible, which protected groups appear disadvantaged, and what interventions a practitioner may be justified in taking. Groups in these audits are often defined by legally or socially sensitive attributes (e.g., race, sex, age, disability status) so metric choice can directly shape if a deployment is viewed as acceptable, questionable, or harmful. A large literature has responded with a plethora of group–fairness metrics—e.g., Statistical Parity, Equalized Odds, and Disparate Impact—each capturing different normative and statistical desiderata (Verma and Rubin, 2018; Castelnovo et al., 2022; Caton and Haas, 2024; Feng et al., 2025; Shaham et al., 2025). Yet practitioners still face three unresolved problems: (i) *selection*—which metric is appropriate for a given deployment; (ii) *combinations*—whether multiple metrics can be satisfied or optimized simultaneously; and (iii) *robustness*—whether a metric behaves sensibly under natural perturbations of the data and model.

These questions mirror those addressed by the classic discipline of *Social Choice Theory* (SCT), which studies how to aggregate individual preferences into collective decisions under transparent axioms such as direction, anonymity, participation, non-dictatorship, and Condorcet consistency (Arrow, 1951; Gibbard, 1973; Brandt et al., 2016). SCT's central lesson is that desirable decision rules often come with unavoidable trade-offs. Analogous tensions have been proven in ML fairness; for example, Kleinberg et al. (2016) show that calibration and multiple error–parity constraints cannot hold simultaneously except in trivial cases.

We bring SCT's mature axiomatic toolkit to the evaluation of fairness metrics for *binary classification* at the group level. Our core idea is simple: treat each fairness metric as a *rule* that orders groups by (dis)advantage, then examine that rule under SCT-style axioms. This matters because fairness metrics are ultimately comparative judgments, and we should know if this judgment is stable under aggregation or overturned by small perturbations. Doing so allows us to replace rules-of-thumb or ad-hoc metric comparison with a principled checklist: which axioms does a metric (or metric class) satisfy, which undesirable properties can occur, and what trade-offs are unavoidable?

4.1.1 Motivation and Scope

Why SCT for fairness metrics? Metric choice today is often justified by convention, domain preference, or legal guidance. Our axiomatic analysis supplies a *testable* basis for selection because each property can be verified directly from a metric's functional form or finite confusion-matrix instances, and is *design-agnostic* in that it applies to any classifier output, regardless of preprocessing techniques, architecture, training procedure, or any other part of the machine-learning pipeline. For example, if a setting demands skew-symmetry and Condorcet consistency, then any metric that violates either axiom is simply ineligible—no tuning can fix it. Conversely, when axioms are mutually incompatible, the SCT lens immediately makes those conflicts explicit, avoiding wasted effort chasing unattainable goals.

Scope. We focus on *group* fairness in *binary classification* where each group is summarized by a confusion matrix. In practice, these groups are not arbitrary partitions, but are often sensitive, salient groups such as racial groups in recidivism prediction or sex-based groups in loan-approval systems. Specifically:

- We analyze metrics that are functions of confusion-matrices and their rates (e.g., true-positive rate, false-positive rate); we call this super-class *Parity-Gap Metrics* (PGM).
- We assume standard i.i.d. sampling within each group and examine robustness to sampling/replication operations relevant to practice (e.g., perturbations, data augmentation that preserves empirical rates).
- We do not study individual fairness, counterfactual/causal notions, ranking/regression settings, or allocation problems beyond what is needed to interpret axioms in the domain of binary classification.

This scope covers common fairness audits that rely on confusion-matrices and aligns with both regulatory practice and model-development workflows.

What this yields in practice. The SCT perspective covers concrete, implementation-level consequences: some metric classes may suggest bias for one group over two independent tests, but flip the bias for the other group upon aggregation (violating Consistency); some are undefined on common edge cases (breaching Universal Domain); and some allow a single observation to flip the verdict at arbitrarily large sample sizes (a “dictatorship” pathology). Our taxonomy and proofs isolate these behaviors class-by-class, enabling principled metric selection tied to deployment requirements.

4.1.2 Contributions

Our main result from this paper is the impossibility result summarized in Theorem 4.1.

Theorem 4.1. *Across the Parity-Gap Metric (PGM) taxonomy, Consistency and Participation cannot be jointly guaranteed.*

The remainder of our contributions build the framework necessary to state, interpret, and prove this result, and invites further research into inherent limitations of metric selection. Our contributions are summarized as follows:

- **A unifying taxonomy for confusion-matrix fairness metrics** (Sections 4.3 and 4.4). We formalize *Parity-Gap Metrics* (PGM) as fairness rules (fairness metrics) built by combining one or more *intra-group* metrics through a coordinate-wise combiner and an aggregator function. As visualized in Figure 4.1, the taxonomy cleanly splits into *Single-Estimate Parities* (SEP) and *Multi-Estimate Parities* (MEP), with natural leaves such as equality, difference, ratio (SEP) and vector-equality, max-gap, norm-threshold, compensatory linear, and additive-monotone families (MEP). By construction, this covers all fairness metrics commonly used in practice. Our “intra-group” framework builds from Briscoe et al.’s (Briscoe et al., 2025) categorization of confusion-matrix metrics.
- **SCT axioms for classification metrics** (Section 4.5). We adapt a set of foundational SCT properties to the binary-classification setting for both intra-group and inter-group metrics, including direction, participation, consistency, independence of irrelevant alternatives (IIA), Condorcet consistency, and universal domain. Each axiom is formally stated and supplemented with a plain-language motivation.
- **Detection of fairness pathologies and their causes** (Sections 4.5 and 4.6). Using the taxonomy and our introduced axioms, we identify when metric classes can reverse bias

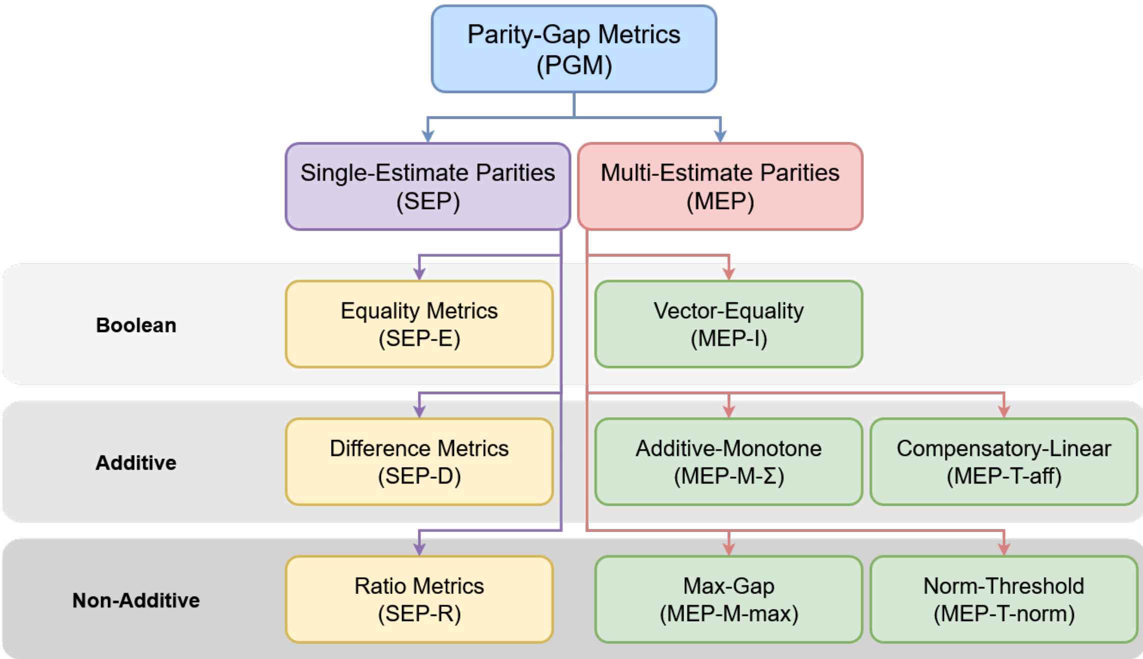


Figure 4.1: The taxonomy of fairness metrics is rooted in Parity-Gap Metrics before splitting off into two major parts: metrics that use a single intra-group estimate (SEP) and metrics that use multiple intra-group estimates (MEP). For example, a metric in MEP-I uses the Indicator function to check for perfect parity over all its intra-group estimates.

indication upon aggregation (violating consistency), be overturned by a single observation at arbitrary n (dictatorship), and more.

- **Guidance for metric selection** (Section 4.6). From our class characterizations, we derive a concise decision chart that maps deployment desiderata to admissible metric classes, such as clarifying when max-gap (risk-averse), compensatory-linear (considering trade-offs), or norm-threshold (non-linear gap penalties) formulations are appropriate.

4.2 Background

We begin with an overview of Social Choice Theory (SCT) before discussing related work.

4.2.1 Overview of Social Choice Theory

SCT studies how to combine many local judgments into one collective decision. In machine learning and AI, we often question how to combine model outputs, losses, and/or benchmarks into a system-level verdict. Similarly, SCT philosophizes how preferences, votes, or judgments of individuals may be mapped into a collective ranking or choice, and what axiomatic properties/guarantees this aggregation should satisfy. SCT goes beyond declaring a winner and defines important, desirable properties such as non-dictatorship, consistency, and independence of irrelevant attributes (Arrow, 1950; May, 1952; Sen and Pattanaik, 1969).

A notable result in SCT is Arrow's impossibility theorem. For any ranked-ballot voting system, Arrow proved that when there are at least three candidates, no rule nor procedure can simultaneously satisfy four desirable and natural conditions: *unrestricted domain*, meaning that an individual may rank candidates however they desire; *Pareto efficiency*, meaning that if every person prefers x to y, then the aggregate must also prefer x to y; *Independence of Irrelevant Alternatives*, meaning that the final ranking between x and y should not depend on relative preferences for z, which is often used as a way to rule out the possibility of spoiler candidates, and *non-dictatorship*, meaning that no single individual determines the collective decision. This theorem is impactful since all properties are desirable, yet their joint satisfaction is impossible. More recently, Milgrom (2024) formalizes "Kenneth Arrow's Last Theorem", a dictatorship-style result for environmental policies over time. Motivated by these results, we seek and find inherent tradeoffs in our work of fairness metric selection.

In binary classification, our fairness metrics are inherently a set of rules mapping individual predictions to a scalar value—it aggregates lower-level evidence into a high-level comparative judgment. Our inputs are group-partitioned confusion matrices and the output is a verdict about parity, disparity, or directional advantage between groups (see examples in Table 4.1). SCT has discovered several fundamental incompatibilities in voting theory and other aggregation settings (Young, 1975; Fishburn, 1977; Moulin, 1988; List and Pettit, 2002). Our research works toward importing this rich background in the ML fairness literature.

4.2.2 Related Work

Prior work has invoked SCT notions in allocation or preference-aggregation contexts and impossibility theorems have mapped high-level tensions in ML fairness (Kleinberg et al., 2016). Our contribution is different in kind. We introduce a *metric-class*-level treatment

Table 4.1: Confusion–matrix-based fairness metrics grouped by functional category. Each metric compares a protected group i with a reference group j.

Difference-of-Scores Metrics
Format: Difference of intra-group metrics.
Accuracy Difference (ACCD)
$\frac{TP_i + TN_i}{n_i} - \frac{TP_j + TN_j}{n_j}$
Predictive Parity (PP)
$\frac{TP_i}{\hat{P}_i} - \frac{TP_j}{\hat{P}_j}$
Difference in Positive Proportion (DPPL)
$\frac{\hat{P}_i}{n_i} - \frac{\hat{P}_j}{n_j}$
Ratio-Based Metrics
Format: Ratio of intra-group metrics.
Disparate Impact (DI)
$\frac{\hat{P}_i/n_i}{\hat{P}_j/n_j}$
Other Metrics
Equalized Odds (EO)
$(FPR_i = FPR_j) \wedge (TPR_i = TPR_j)$
Statistical Parity (SP)
$\frac{\hat{P}_i}{n_i} = \frac{\hat{P}_j}{n_j}$
Average Absolute Odds Difference (AAOD)
$\frac{1}{2}\left(\left\lvert\frac{FP_i}{N_i} - \frac{FP_j}{N_j}\right\rvert + \left\lvert\frac{TP_i}{P_i} - \frac{TP_j}{P_j}\right\rvert\right)$

that (i) unifies most deployed metrics under a single construction (PGM), (ii) carries a full axiomatic analysis from intra-group rates to inter-group gaps, and (iii) translates the results into actionable guidance.

Recent scholarship has begun to import the axiomatic vocabulary of SCT into algorithmic-fairness research, but existing efforts are still fragmented and often focus on a single axiom or application domain. Conitzer et al. (2019) introduce group fairness for the allocation of indivisible goods, strengthening envy-freeness to protect arbitrary coalitions, yet their analysis and applications stops short of a few terms specific to "cake-cutting" problems (proportionality and envy-freeness derivatives).

Shah (2023) defines a preference-based agenda for algorithmic fairness, importing classical cooperative notions such as proportionality, the core, envy-freeness, and the stronger "group fairness" axiom from social choice into modern decision systems. Crucially, however, the analysis remains allocation-centric and sidesteps the statistical-classification "zoo" that motivates our study.

In 2023, Zanger-Tishler et al. (2023) exposes how data with proxy labels may become less accurate and less fair when provided with more features. This work ends with considering feature selection under measurement error.

Liu et al. (2025) emphasizes that fairness should not be merely a descriptive measure after training, but a design objective that must be translated into concrete choices about data, metrics, stakeholders, and deployment constraints. While considering vehicle license allocations, Rong et al. (2019) take a similar posture and argue that allocation mechanisms should not be solely judged by efficiency and revenue, but also an explicit notion of equality.

Watson-Daniels et al. (2023) examine *predictive multiplicity*—the phenomenon where multiple empirically similar classifiers yield divergent predictions for the same individual. They introduce *ambiguity* and *discrepancy* to formally quantify individual-level uncertainty. While their framework highlights the fragility of deterministic predictions under model multiplicity, it stops short of addressing inter-group fairness considerations.

In the bias-metric literature, the *Objective Fairness Index* (Briscoe and Gebremedhin, 2024) offers a legally grounded alternative to Disparate Impact, but its axiomatic profile remains

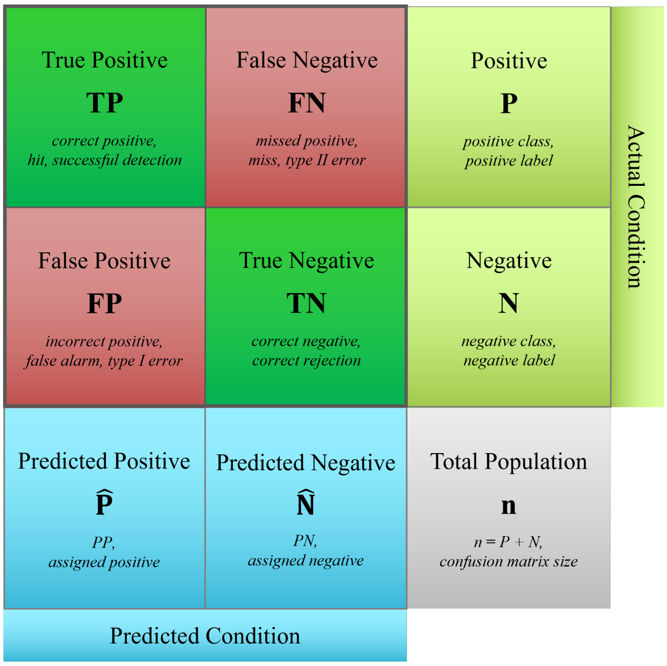


Figure 4.2: The binary confusion matrix illustrates the four primary outcomes of a binary classifier: True Positives, False Negatives, False Positives, and True Negatives. The additional cells summarize overall populations, providing context at a glance.

unexplored.

In summary, while each of the above strands addresses an isolated concern, our unified intra-group taxonomy, inter-group taxonomy, and social-choice theory framework synthesizes a single, principled checklist for evaluating metrics, especially fairness metrics.

4.3 Notations and Intra-Group Metrics

In this section, we provide notations and a formal taxonomy for intra-group metrics. These are necessary to introduce our main taxonomy of fairness metrics in Section 4.4.

4.3.1 Notations of Operators and Groups

In this work, we use "$\triangleq$" to denote that the left-hand side (LHS) is *always* defined by the right-hand side (RHS); ":=" denotes that the LHS is assigned to the RHS temporarily (such as for an example); and "=" denotes an equivalence relationship.

Let $\perp \triangleq \text{undef}$ and $\mathbb{R}_\perp \triangleq \mathbb{R} \cup \{\perp\}$. For analytical purposes, we generalize each inter-group metric M as a function of intra-group metrics g and non-empty groups $\mathcal{G} = \{A, B, C, \ldots\}$, where each group $I \in \mathcal{G}$ is represented by a binary confusion matrix (as illustrated in Figure 4.2). Using codomains $(\mathcal{V}_g, \mathcal{V}_M)$,

Intra-Group:

$$g : \mathcal{G} \longrightarrow \mathcal{V}_g \subseteq \mathbb{R}_{\perp}, \quad \forall I \in \mathcal{G} : I \mapsto {\mathbb{N}_{\geq 0}}^{2\times 2}; \tag{4.1}$$

Inter-Group:

$$M : \mathcal{G} \times \mathcal{G} \longrightarrow \mathcal{V}_M \subseteq \mathbb{R}_{\perp}, \quad \forall I \in \mathcal{G} : I \mapsto {\mathbb{N}_{\geq 0}}^{2\times 2}. \tag{4.2}$$

For simplicity, $I \in \mathcal{G}$ sometimes flattens to be a quadruple $I \mapsto {\mathbb{N}_{\geq 0}}^{4}$. This uses either $\mathrm{CM} \triangleq [\mathrm{TP}, \mathrm{FN}, \mathrm{FP}, \mathrm{TN}]$ or some shuffle of CM's cells, defined as $\vec{c} \triangleq [c_i, c_j, c_k, c_l]$. There is no loss in meaning during these conveniences or generalizations.

4.3.2 Intra-Group Metrics

We partition intra-group metrics into two classes: Group Proportions and Group Counts. These classes have members as we explain below and visualize in Figure 4.3.

Group Proportions is the class of normalized intra-group metrics: Cell-Pair Density Metrics (CPD), Joint-Ratio Metrics (JRM), and Cell-Pair Trade-Off Metrics (CPT). These are the most common intra-group metrics, and all consist of finding some proportionality within a group.

CPD contains six unique metrics (Table 4.2). Metrics in this class are useful to find the likelihood of one row, column, or diagonal of a confusion matrix occurring given the total count n.

$$\mathrm{CPD} \triangleq \left\{ (c_i + c_j)/n \;\mid\; i \neq j \right\}. \tag{4.3}$$

JRM contains twelve unique metrics (see Table 4.3). Metrics in this class are useful to find the likelihood of one cell c_i given a row, column, or diagonal of a confusion matrix that includes c_i.

$$\mathrm{JRM} \triangleq \left\{ c_i/(c_i + c_j) \;\mid\; i \neq j \right\}. \tag{4.4}$$

CPT is a class of metrics that is underexplored in the literature. Only marginal benefit

$\mathcal{B} = (\mathrm{FP} - \mathrm{FN})/n$ is currently defined (Briscoe and Gebremedhin, 2024). In this class, metrics are useful when one needs to calculate a normalized trade-off between two cells, such as for objective testing.

$$\mathrm{CPT} \triangleq \left\{ (c_i - c_j)/n \;\middle|\; i \neq j \right\}. \tag{4.5}$$

Group Counts is the class of metrics that resolve some weighted summation of cells. We split this class into Monotonic Counts (MC) and Non-Monotonic Counts (NMC). These are uncommon metrics compared to Group Proportions.

GC is defined by

$$\mathrm{GC} \triangleq \left\{ \vec{c} \cdot \vec{w} \;\middle|\; \vec{w} \in \mathbb{R}^4 \right\}. \tag{4.6}$$

GC covers the trivial journalling of cell counts such as the count of Type I errors (FP) or the count of correct predictions (TP + TN). However, it also has a more interesting use case of calculating expected benefits or costs. This version has been applied in business contexts as described in (Larner, 2024) using the definition provided in equation 4.10 below.

MC is defined by

$$\mathrm{MC} \triangleq \left\{ \vec{c} \cdot \vec{w} \;\middle|\; \vec{w} \in {\mathbb{R}_{\geq 0}}^4 \right\}. \tag{4.7}$$

An example of MC is calculating misclassification costs:

$$\mathrm{MC} = w_{\mathrm{FP}}\mathrm{FP} + w_{\mathrm{FN}}\mathrm{FN}. \tag{4.8}$$

NMC is the complement of MC:

$$\mathrm{NMC} \triangleq \mathrm{GC} \setminus \mathrm{MC}. \tag{4.9}$$

NMC allows one to consider trade-offs or deficits, like calculating net profit from business churning:

$$\text{Net Profit} \triangleq w_{\mathrm{P}}\mathrm{P} - w_{\hat{P}}\hat{P} = \left(w_{\mathrm{P}} - w_{\hat{P}}\right)\mathrm{TP} + w_{\mathrm{P}}\,\mathrm{FN} - w_{\hat{P}}\,\mathrm{FP}. \tag{4.10}$$

In this example, we gain w_{P} per positive client and lose $w_{\hat{P}}$ advertising to clients we think are positive.

Table 4.2: Cell-Pair Density Metrics (CPD). Metrics that are a proportion equal to the sum of two cells to the total count.

Metric	Abbrev.	Formula	Description
Cell-Pair Density Metric	CPD	$\frac{c_i + c_j}{n}$	A ratio of cells c_i and c_j to the total count n.
Accuracy	ACC	$\frac{\text{TP} + \text{TN}}{n}$	Proportion of correct predictions to all predictions.
Prevalence	PREV	$\frac{\text{TP} + \text{FN}}{n}$	Proportion of actual positive instances in the population. (Synonym: P)
Predicted Positive Rate	PPR	$\frac{\text{TP} + \text{FP}}{n}$	Fraction of instances predicted as positive by the classifier. (Synonyms: $\hat{P}$, PP)
Inaccuracy	INACC	$\frac{\text{FP} + \text{FN}}{n}$	Fraction of incorrect predictions out of total instances. (Synonyms: Error Rate, Misclassification Rate)
Negative Prevalence	NPREV	$\frac{\text{TN} + \text{FP}}{n}$	Proportion of actual negative instances in the population. (Synonym: Complement of Prevalence, N)
Predicted Negative Rate	PNR	$\frac{\text{TN} + \text{FN}}{n}$	Proportion of predicted negative instances in the population. (Synonyms: $\hat{N}$, PN)

Table 4.3: Joint Ratio Metrics (JRM): Metrics that describe the relationships between confusion matrix components as ratios, commonly used to assess various types of classification performance errors. The final four are uncommon and do not mix correct and incorrect cells.

Metric	**Abbrev.**	**Formula**	**Description**
Joint Ratio Metric	JRM	$\frac{c_i}{c_i + c_j}$	A ratio between a cell c_i and the sum of c_i and another cell c_j.
True Positive Rate	TPR	$\frac{\text{TP}}{\text{TP} + \text{FN}}$	Proportion of actual positives correctly identified. (Synonyms: Sensitivity, Recall, Hit Rate)
False Positive Rate	FPR	$\frac{\text{FP}}{\text{FP} + \text{TN}}$	Proportion of actual negatives incorrectly classified as positive. (Synonym: Fall-Out)
True Negative Rate	TNR	$\frac{\text{TN}}{\text{TN} + \text{FP}}$	Proportion of actual negatives correctly identified. (Synonym: Specificity)
False Negative Rate	FNR	$\frac{\text{FN}}{\text{FN} + \text{TP}}$	Proportion of actual positives incorrectly classified as negative. (Synonym: Miss Rate)
Positive Predictive Value	PPV	$\frac{\text{TP}}{\text{TP} + \text{FP}}$	Proportion of predicted positives that are actual positives. (Synonym: Precision)
Negative Predictive Value	NPV	$\frac{\text{TN}}{\text{TN} + \text{FN}}$	Proportion of predicted negatives that are actual negatives.
False Discovery Rate	FDR	$\frac{\text{FP}}{\text{FP} + \text{TP}}$	Proportion of predicted positives that are false positives.
False Omission Rate	FOR	$\frac{\text{FN}}{\text{FN} + \text{TN}}$	Proportion of predicted negatives that are false negatives. (Synonym: False Reassurance Rate)
True-Positive Share of Correct Predictions	—	$\frac{\text{TP}}{\text{TP} + \text{TN}}$	Proportion of correct predictions that are true positives.
True-Negative Share of Correct Predictions	—	$\frac{\text{TN}}{\text{TN} + \text{TP}}$	Proportion of correct predictions that are true negatives.
False-Positive Share of Errors	—	$\frac{\text{FP}}{\text{FP} + \text{FN}}$	Proportion of misclassifications that are false positives.
False-Negative Share of Errors	—	$\frac{\text{FN}}{\text{FN} + \text{FP}}$	Proportion of misclassifications that are false negatives.

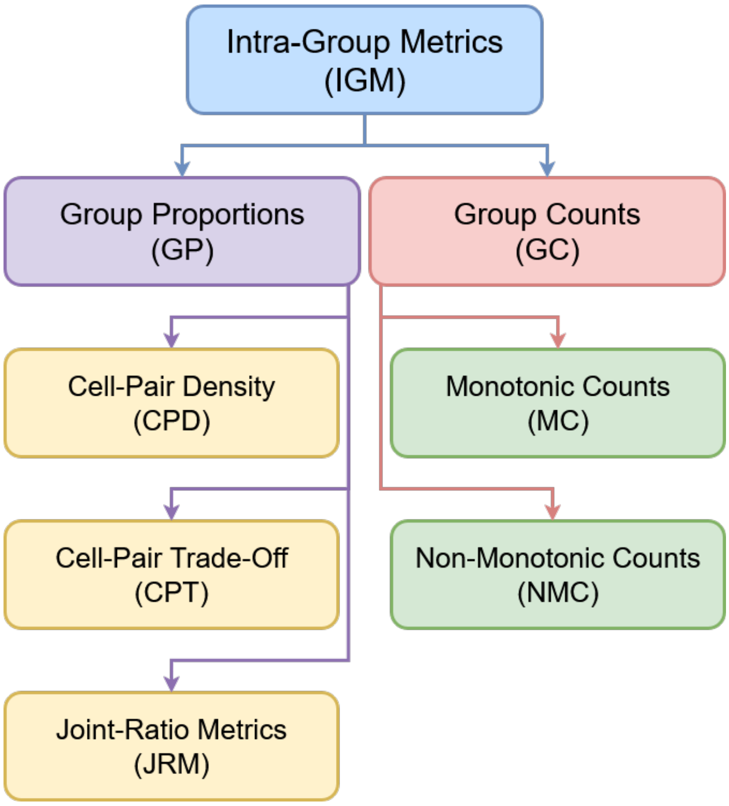


Figure 4.3: The intra-group metric taxonomy.

4.3.3 The No-Bias Constant

For every inter-group metric we define perfect parity p as the "no-bias" constant. Perfect parity p is significant as it indicates a successful lack of bias according to the metric estimate—all fairness metrics have one and only one $p \in \mathbb{R}$. The concrete value of p is commonly known for each individual fairness metric (under PGM), but we do not know of any other work that abstracts this no-bias constant as we do here.

To help explain bias assessment with p, we demonstrate metrics as *directed* (indicates direction of bias) or *undirected* (indicates presence or magnitude, but not direction of bias). In this example, we assume that a higher g is favorable and adapt the nomenclature of Amazon (2021): *positive bias* for A indicates beneficial bias for A and *negative bias* for A means detrimental bias against A.

- For **directed metrics**:
  - if $M(A, B) < p$, then there is positive bias for Group A compared to Group B;

- if $M(A, B) = p$, then there is no bias between groups A and B;
- if $M(A, B) > p$, then there is negative bias for Group A compared to Group B;
- if $M(A, B) = \perp$, then no conclusion is drawn.

- For **undirected metrics**:
  - if $M(A, B) \neq p$, then there is positive bias for either Group A or Group B;
  - if $M(A, B) = p$, then there is no bias between groups A and B;
  - if $M(A, B) = \perp$, then no conclusion is drawn.

For analytical purposes, we define p as the score achieved when all cells are equivalent:

$$p = M(\{n/4\}^{2\times 2}, \{n/4\}^{2\times 2}). \tag{4.11}$$

Perfect parity p in Equation 4.11 holds as equivalent to the desired (no-bias) value for all fairness metrics that we survey.

Note: In many instances, a threshold is allowed around p to signal no bias. We endorse this, as not only is there almost always a probabilistic nature in classification assessment (Briscoe et al., 2025), but the combinatorial nature of confusion matrices may not allow strict equivalence of group-specific estimates. For example, consider two groups whose sizes are two different prime numbers; their accuracies can only be equivalent at 0 and 1.

4.4 Taxonomy of Fairness Metrics in Classification

We introduce the taxonomy of fairness metrics as a tree structure in Figure 4.1, rooted in the PGM class which contains most common fairness-classification metrics. We give concrete examples of bias metrics in Table 4.1. In this section, we give formal definitions and details for each class of metrics.

The first split in our taxonomy is based on how many intra-group estimates are used for each group. Some fairness metrics compare groups with only one scalar summary, such as predicted positive rate. Other fairness metrics compare groups with multiple scalar summaries. We categorize these into two broad families: Single-Estimate Parities (SEP) and Multi-Estimate

Parities (MEP).

4.4.1 Parity Gap Metrics

The highest-level class of inter-group metrics that we consider is Parity Gap Metrics (PGM). This class encompasses most common fairness metrics in classification and hence it is exactly the scope of our analysis. All metrics in PGM are either directed or undirected metrics with a reachable, no-bias constant p.

We define PGM as the set of all metrics with an aggregator function h (e.g., an average) over a non-constant, coordinate-wise "combiner" φ (e.g., a difference) of d non-constant, intra-group metrics $\vec{g}$ (e.g., TPR and FPR).

$$\vec{g} = (g_1, g_2, \ldots, g_k, \ldots, g_d), \quad \forall g_k : \mathcal{G} \to \mathcal{V}_k \subseteq \mathbb{R}_\perp. \tag{4.12}$$

The resolved, coordinate-wise combiner over $\vec{g}$ is written as

$$\vec{\varphi}(A, B; \vec{g}) = \Big(\varphi\big(g_1(A), g_1(B)\big), \ \ldots, \ \varphi\big(g_d(A), g_d(B)\big)\Big).$$

Then the general form of a metric M in PGM is

$$M(A, B) \triangleq h(\vec{\varphi}(A, B; \vec{g})). \tag{4.13}$$

Accordingly, its codomain is

$$\mathcal{V}_M = h\Big(\varphi\big(\mathcal{V}_{g_1} \times \mathcal{V}_{g_1}\big), \ldots, \varphi\big(\mathcal{V}_{g_d} \times \mathcal{V}_{g_d}\big)\Big) \subseteq \mathbb{R}_\perp. \tag{4.14}$$

Concisely, the PGM class is defined by

$$PGM \triangleq \{M(A, B) \mid M(A, B) \triangleq h(A, B; \vec{\varphi}), \quad h : \mathcal{G} \times \mathcal{G} \longrightarrow \mathbb{R}_\perp, \quad A, B \in \mathcal{G}\}. \tag{4.15}$$

For example, we construct the Average Absolute Odds Difference:

$$d = 2, \quad \vec{g} \coloneqq (\mathrm{FPR}, \mathrm{TPR}), \quad \varphi(A, B; g_k) \coloneqq |g_k(A) - g_k(B)|, \quad h(\vec{\varphi}) \coloneqq \tfrac{1}{d} \sum_{\varphi_k \in \vec{\varphi}} \varphi_k,$$

$$\begin{aligned} \mathrm{AAOD}(A, B) &\triangleq h\left(|\,\mathrm{FPR}(A) - \mathrm{FPR}(B)|,\ |\,\mathrm{TPR}(A) - \mathrm{TPR}(B)|\right) \\ &= \tfrac{1}{2}\big(|\,\mathrm{FPR}(A) - \mathrm{FPR}(B)| + |\,\mathrm{TPR}(A) - \mathrm{TPR}(B)|\big). \end{aligned} \tag{4.16}$$

4.4.2 Single-Estimate Parities (SEP)

Metrics in the SEP class compare groups using exactly one intra-group estimate. These metrics are often the most interpretable since the fairness judgment is traceable to one familiar quantity, such as accuracy or recall. Metrics in SEP are also easier to optimize and visualize than metrics with multiple intra-group estimates.

Single-Estimate Parities (SEP). $M \in \mathrm{SEP}$ if and only if its construction uses a *single* scalar estimate per group ($d = 1$). For any $g : \mathcal{G} \to \mathbb{R}_{\perp}$,

$$\mathrm{SEP} \triangleq \{M \in \mathrm{PGM} \mid d = 1\}. \tag{4.17}$$

SEP contains three classes of metrics: one for equalities (SEP-E), one for differences (SEP-D), and one for ratios (SEP-R).

Equality Metrics (SEP-E) SEP-E is the class of metrics with the form

$$\mathrm{SEP{-}E} \triangleq \Big\{ M \in \mathrm{SEP} \ \Big| \ M(A, B) = \begin{cases} \perp & \text{if } \perp \in \{g(A),\, g(B)\}, \\ 1 & \text{if } g(A) = g(B), \\ 0 & \text{otherwise.} \end{cases} \Big\}. \tag{4.18}$$

Simply, metrics in SEP-E check for equality between two intra-group estimates. These metrics are rarely met in practice and are useful if strict adherence is required by some policy or law.

Difference Metrics (SEP-D) SEP-D is the class of metrics with the form

$$\mathrm{SEP\!-\!D} \triangleq \left\{ M \in \mathrm{SEP} \;\middle|\; \varphi(A, B; g) = g(A) - g(B) \right\}. \tag{4.19}$$

In words, metrics in SEP-D calculates the Manhattan distance between two groups' estimates. These metrics are useful when one needs a magnitude for bias.

Ratio Metrics (SEP-R) SEP-R is the class of metrics with the form

$$\mathrm{SEP\!-\!R} \triangleq \left\{ M \in \mathrm{SEP} \;\middle|\; \varphi(A, B; g) = g(A)/g(B) \right\}. \tag{4.20}$$

These metrics are the ratio of two groups' intra-group estimate. Metrics in SEP-R are useful when one needs to think of one group estimate as a multiple of another, or in policy matters (Briscoe and Gebremedhin, 2024).

4.4.3 Multi-Estimate Parities (MEP)

Some fairness notions require multiple estimates from each group. For example, one may want to consider both true-positive rates and false-positive rates simultaneously, or to combine several terms into a single fairness score. These metrics belong to the Multi-Estimate Parities (MEP) class.

Multi-Estimate Parities (MEP). A metric M is in MEP if and only if it relies on *two or more* estimates for each group:

$$\mathrm{MEP} \triangleq \{M \in \mathrm{PGM} \mid d \geq 2\}. \tag{4.21}$$

Metrics in MEP are useful when a single summary would hide important trade-offs. MEP metrics evaluate fairness over a profile of single-group metrics, capturing richer desiderata such as "equal opportunity *and* equal misclassification cost."

MEP contains five classes of metrics: MEP-I (indicator or vector-equality metrics), MEP-M-Σ (additive-monotone metrics), MEP-M-max (max-gap metrics), MEP-T-aff (compensatory-linear metrics), and MEP-T-norm (norm-threshold metrics).

Vector-Equality Metrics (MEP-I) MEP-I is the class of metrics with the form

$$\text{MEP-I} \triangleq \{ M \in \text{MEP} \mid \varphi(A, B; g_k) = \mathbf{1}[g_k(A) = g_k(B)], \quad h = \bigwedge_{\varphi_i \in \vec{\varphi}} \varphi_i \}. \tag{4.22}$$

In words, $M(A, B) = 1$ iff $g_k(A) = g_k(B)$ for every coordinate k; otherwise 0 (or $\perp$ if any coordinate is undefined). Metrics in MEP-Indicator are effectively a "pass/fail" rule that only accepts equality on all tracked group estimates. MEP-I is useful when one has a statute or policy that demands strict equality across multiple dimensions.

Additive-Monotone Metrics (MEP-M-Σ) Metrics in MEP-M-Σ use non-negative weights $\vec{w}$ and have the form

$$\text{MEP-M-}\Sigma \triangleq \{ M \in \text{MEP} \mid \varphi : \mathbb{R}_\perp \times \mathbb{R}_\perp \to \mathbb{R}_{\geq 0, \perp}, \quad h = \vec{w} \cdot \vec{\varphi} \}. \tag{4.23}$$

Intuitively, φ is a non-negative disparity such as $|x - y|$ or $|x/y|$, and h is a (non-negative) weighted summation over $\vec{\varphi}$. MEP-M-Σ is useful when one wants every unfairness "point" to be a debit—debits only add up. MEP-M-Σ makes an ideal objective for optimization, model ranking, or compliance dashboards where every disparity must count and none may offset another.

Max-Gap Metrics (MEP-M-max) Metrics in MEP-M-max have the form

$$\text{MEP-M-Max} \triangleq \{ M \in \text{MEP} \mid h(A, B; \vec{\varphi}) = \max\left(\varphi_k \in \vec{\varphi}(A, B; \vec{g})\right) \}. \tag{4.24}$$

Metrics in this class report the *worst* individual gap between the two groups. This is suitable for risk-averse contexts where the largest unfairness dimension dominates (e.g., safety-critical systems).

Compensatory-Linear Metrics (MEP-T-aff) Metrics in MEP-T-aff use weights $\vec{w} \in \mathbb{R}^d$ and have the form

$$\text{MEP-T-aff} \triangleq \{ M \in \text{MEP} \mid h(A, B; \vec{\varphi}) = \vec{w} \cdot \vec{\varphi}(A, B) \}. \tag{4.25}$$

Hence $M(A, B)$ has perfect parity along the hyper-plane $\vec{w} \cdot \vec{\varphi}(A, B) = p$.

In this class, positive and negative coefficients let gains in one coordinate offset losses in another. This allows one to capture explicit trade-offs when stakeholders can price different error types (e.g. balancing FPR vs. FNR).

Norm-Threshold Metrics (MEP-T-norm) The metrics in MEP-T-norm fix a norm $\|\cdot\|_{p-\text{norm}}$ on $\mathbb{R}^d$ where $p \geq 2$ of p-norm is a finite integer. Then, MEP-T-norm is defined by

$$\text{MEP-T-norm} \triangleq \{ M \in \text{MEP} \mid h(A, B; \vec{\varphi}) = \left\|\vec{\varphi}(A, B; \vec{g})\right\|_{p-\text{norm}} \}. \tag{4.26}$$

We restrict p-norm to avoid coinciding with MEP-M-Σ and MEP-M-max. A p-norm measures the distance of the profile of gaps with increasing sensitivity to outliers as the p-norm increases. This is useful when gap disparities are non-linear, analogous to "sum-of-square errors".

4.5 SCT Properties of Classification Metrics

Having mapped the ever-growing "zoo" of fairness metrics into a concise taxonomy, we now confront a deeper question: which leaves of that tree behave sensibly under foundational principles of fairness? Social Choice Theory (SCT) offers a rigorous starting point in the form of axioms such as non-dictatorship, participation, and consistency. However, bringing these ideas into binary classification is more than renaming terms from voting theory. In SCT, axioms constrain rules that aggregate preferences over candidates. Here, we compare protected groups summarized by confusion matrices, and the "social outcome" is the result induced by a fairness metric relative to the "no-bias" baseline p.

This translation is one of our central theoretical contributions. We reform SCT axioms to be native in binary classification fairness. For example, *dictatorship* becomes the pathology which a single additional observation flips the direction of bias even at arbitrarily large sample sizes;

participation becomes invariance under rate-preserving replication; and *consistency* becomes preservation of a bias verdict under aggregation of independent and agreeing datasets.

Accordingly, this section adapts or extends SCT axioms into thirteen axioms for the binary-classification setting. In Section 4.5.1 we consider intra-group metrics, and Section 4.5.2 discusses inter-group metrics.

4.5.1 Intra-Group Metric Properties

We introduce properties derived by social-choice-theoretic principles for intra-group metrics. Recall that an intra-group metric $g : \mathcal{G} \to \mathbb{R}_{\perp}$ and $A \in \mathbb{R}^{2\times2}_{\geq 0}$ is a confusion matrix representing a group (data subset) in $\mathcal{G}$. Furthermore, in our analyses, we define a midpoint $m = g(\{n/4\}^{2\times2})$, and "polarity reversal" of x as a reflection across the midpoint, $\mathrm{polrev}(x)$.

$$\mathrm{polrev}(x) \;=\; \begin{cases} -x, & \text{if } x \in \mathbb{R} \text{ is a signed metric,} \\ 1-x, & \text{if } x \in [0,1] \text{ is a normalized metric,} \\ 1/\mathrm{x}, & \text{if } x > 0 \text{ is a ratio-based metric,} \\ \neg x, & \text{if } x \in \{0,1\} \text{ is Boolean.} \end{cases} \tag{4.27}$$

This polarity reversal can be considered as an "opposite" of bias under the given metric's semantics.

Definition 4.1 (π-Involution Symmetry)**.** *With $A = (TP, FN, FP, TN)$, we define three self–inverse maps:*

$$\begin{aligned} \pi_{corr}(A) &= (TN, FP, FN, TP) && \textit{(flip correct structure),} \\ \pi_{pred}(A) &= (FN, TP, TN, FP) && \textit{(flip prediction),} \\ \pi_{label}(A) &= (FP, TN, TP, FN) && \textit{(flip ground–truth label).} \end{aligned}$$

*Then, a metric g is **π-Involution Symmetric** iff $g(\pi(A)) = \mathrm{polrev}(g(A))$ for any self-inverse mapping.*

π-Involution Symmetry is useful when a practitioner needs to identify invariance to correctness, predictions, or ground truth.

For the next property, let $A^+ = A + e_c$ for the matrix obtained by adding one instance to the indicated cell c. Additionally, let X be any pair of cells. Exhaustively, these pairs are: positive labels (P), negative labels (N), positive predictions ($\hat{P}$), negative predictions ($\hat{N}$), correct predictions (C), and incorrect predictions (I).

Definition 4.2 (Intra-Group Non-Dictatorship)**.** *The intra-group metric g is a dictatorship if some A exists such that $g(A) \neq m$ and an increment in some cell can shift $g(A^+)$ past the midpoint. Formally, metric g satisfies **Non-Dictatorship** iff*

$$\big\{\forall A \;\mid\; g(A) \notin \{m, \perp\}\big\} \;:\; \begin{cases} g(A) > m \implies g(A^+) \geq m, \\ g(A) < m \implies g(A^+) \leq m. \end{cases} \tag{4.28}$$

This ensures that for an arbitrarily large n, no individual can flip the indication of bias, nor can any conclusion be derived from a single individual if no conclusion could previously be made.

Definition 4.3 (Intra-Group Participation)**.** ***Intra-Group Participation*** *is satisfied if for any augmentation that leaves the relevant group empirical rates unchanged (even if cardinalities change), $g(A)$ is invariant.*

Let $\bar{c}$ be a normalized confusion matrix (all values sum to one). Then, Participation is satisfied iff

$$\forall (p_1, p_2) \in \mathbb{R}^2_{>0} : \quad g(p_1 \cdot \bar{c}_A) = g(p_2 \cdot \bar{c}_A). \tag{4.29}$$

This requires metrics to be independent of scale occurring because of uniform replication.

Definition 4.4 (Universal Domain (UD))**.** *Metric g has a **Universal Domain** if there exist finite bounds $L < U$ such that*

$$L \;\leq\; g(A) \;\leq\; U.$$

In words, UD requires the result to be finite for every admissible dataset. UD bans "division-by-zero" or other gaps in the metric's domain.

4.5.2 Inter-Group Metric Properties

Recall that an inter-group metric is $M : \mathcal{G} \times \mathcal{G} \to \mathbb{R}_{\perp}$.

Definition 4.5 (Consistency)**.** *To standardize discussion for the deviation from parity p, we define*

$$\operatorname{sign}_p(x) \triangleq \operatorname{sign}(x - p).$$

Then, for two disjoint datasets D_1, D_2 drawn from the same population:

*Metric M is **Consistent** if $\operatorname{sign}_p(M_{D_1}(i, j)) = \operatorname{sign}_p(M_{D_2}(i, j))$ then, on the union $D_{1,2} \triangleq D_1 \cup D_2$,*

$$\operatorname{sign}_p(M_{D_1 \cup D_2}(i, j)) = \operatorname{sign}_p(M_{D_1}(i, j))$$

for everywhere M_{D_1}, M_{D_2}, and $M_{D_{1,2}}$ is defined.

A Consistent metric ensures that agreement across independent samples is respected after aggregation.

Definition 4.6 (Direction)**.** *Metric M is **directed** if there is a constant p (independent of $\varphi \in \vec{\varphi}$) such that $M(A, B) \gtrless p$ is informative of which group has positive or negative bias and $M(A, B) = p$ suggests no bias.*

Directed metrics allow a practitioner to identify the group that is advantaged or disadvantaged for the given evidence.

Definition 4.7 (Skew Symmetry)**.** *Metric M is **Skew Symmetric** if $M(A, B) = -M(B, A)$ for all distinct A, B.*

Skew Symmetry pairs naturally with the Directed property. When Skew Symmetry is satisfied, we know that the total bias observed is exactly the absolute value. This allows practitioners to easily compare two models where the scores are similar in absolute value. For example, if the scores are -1 and 1 then the total bias is equal and the models favor different groups.

Definition 4.8 (Inter-Group Participation)**.** *Extending Intra-Group Participation to multiple groups,* ***Inter-Group Participation*** *is satisfied if for any augmentation that leaves the relevant group empirical rates unchanged (even if cardinalities change), $M(A, B)$ is invariant.*

Let $\bar{c}$ be a normalized confusion matrix (all values sum to one). Then, Participation is satisfied iff

$$M(p_1 \cdot \bar{c}_A, q_1 \cdot \bar{c}_B) = M(p_2 \cdot \bar{c}_A, q_2 \cdot \bar{c}_B) \tag{4.30}$$

for all $[p_1, p_2, q_1, q_2] \in \mathbb{R}^4_{>0}$.

This requires metrics to be independent of scale occurring because of uniform replication.

Definition 4.9 (Non-Imposition)**.** *Metric M satisfies* ***Non-Imposition*** *if, with any chosen reference group $i \in \mathcal{G}$, there exists some dataset such that the metric can produce* any desired ranking *of all the other groups.*

Non-Imposition ensures the evaluation of bias itself is reasonable—meaning that it is theoretically possible for any given group to have more bias than any other group.

Definition 4.10 (Inter-Group Non-Dictatorship)**.** *Metric M is a dictatorship if some (A, B) exists such that $M(A, B) \neq p$ and an increment in some cell in A can shift $M(A^+, B)$ past the midpoint for any arbitrarily large n. Formally, M satisfies* ***Non-Dictatorship*** *iff*

$$\forall (A, B) \;\mid\; M(A, B) \neq m \;:\; \begin{cases} M(A, B) > p \quad \Longrightarrow \quad M(A^+, B) \geq p, \\ M(A, B) < p \quad \Longrightarrow \quad M(A^+, B) \leq p. \end{cases} \tag{4.31}$$

Non-Dictatorship ensures that for any arbitrarily large n, no individual can flip the indication of bias (but may weaken the magnitude of bias even to zero).

Definition 4.11 (Condorcet Consistency)**.** *Metric M is* ***Condorcet-Consistent*** *if for every dataset and all mutually distinct groups $A, B, C \in \mathcal{G}$,*

$$M(A, B) > p \;\wedge\; M(B, C) > p \quad \Longrightarrow \quad M(A, C) > p,$$

whenever the three comparisons are defined.

Condorcet consistency forbids "rock–paper–scissors" cycles of bias. If group A is measured as positive bias over B and B over C, the metric must also favor A over C.

Definition 4.12 (Independence of Irrelevant Alternatives (IIA))**.** *Metric M satisfies **IIA** if for every distinct $A, B, C \in \mathcal{G}$ and for all datasets (D, D') that coincide on the confusion-matrix rows of A and B but may differ on one or more rows of C,*

$$A^D = A^{D'} \;\wedge\; B^D = B^{D'} \implies M(A^D, B^D) = M(A^{D'}, B^{D'}). \tag{4.32}$$

That is, the comparison between A and B is immune to changes affecting any third ("irrelevant") group C. IIA allows practitioners to perform straightforward, pairwise comparisons, simplifying the debugging process.

Definition 4.13 (Anonymity)**.** *For every metric M, **Anonymity** requires that permuting (i) the ordering of the individuals within each group and (ii) their identifiers across groups—but leaving the aggregate contingency-table counts unchanged—must leave $M(A, B)$ unchanged for all $A, B \in \mathcal{G}$.*

Anonymity ensures that (i) ordering of individuals does not affect the group score, and (ii) that individual identity does not affect group score. For example, swapping individual Alice from group A and Bob from group B, where Alice and Bob are otherwise identical, should not change the aggregate score for either group.

4.6 Results

Here, we provide results for property satisfaction, the fairness-metric decision diagram, and our main theorem. We relegate most proofs to the appendix, highlighting only the inter-group properties "Consistency" and "Participation" in this section since we also prove that they are mutually exclusive.

Table 4.4: Legend of property symbols.

Symbol	Meaning
✓	Guaranteed to hold
×	Not guaranteed to hold
∅	Undefined or not applicable
MC	Holds only for $g \in$ MC
GP	Holds only for $g \in$ GP

Table 4.5: Summary table of properties held by intra-group metric classes.

Property	JRM	CPD	CPT	MC	NMC
≥ 1 π-Involution Symmetries	✓	✓	✓	∅	∅
Participation	✓	✓	✓	×	×
Universal Domain	×	✓	✓	✓	✓
Non-Dictatorship	✓	✓	✓	×	×

Table 4.6: Summary table of properties held by SEP.

Property	SEP-E	SEP-D	SEP-R
Direction	×	✓	✓
Skew-Symmetry	×	✓	×
Consistency	×	MC	MC
Participation	GP	GP	GP
Non-Imposition	×	✓	✓
Non-Dictatorship	∅	×	×
Condorcet Consistent	∅	✓	✓
IIA	✓	✓	✓
Anonymity	✓	✓	✓

4.6.1 Summaries of Property Satisfaction

Using the legend in Table 4.4, we summarize property satisfaction for all intra-group metrics in Table 4.5 and for SEP in Table 4.6.

For intra-group metrics (IGMs), we prove property satisfaction for all IGMs and their social-theoretic properties ($5 \times 4 = 20$ use cases). We see that CPD and CPT satisfy all relevant properties.

For inter-group metrics, we have eight classes of PGMs, five classes of IGMs, and nine social-theoretic properties. This causes an explosion of possible use cases ($8 \times 5 \times 9 = 360$). To

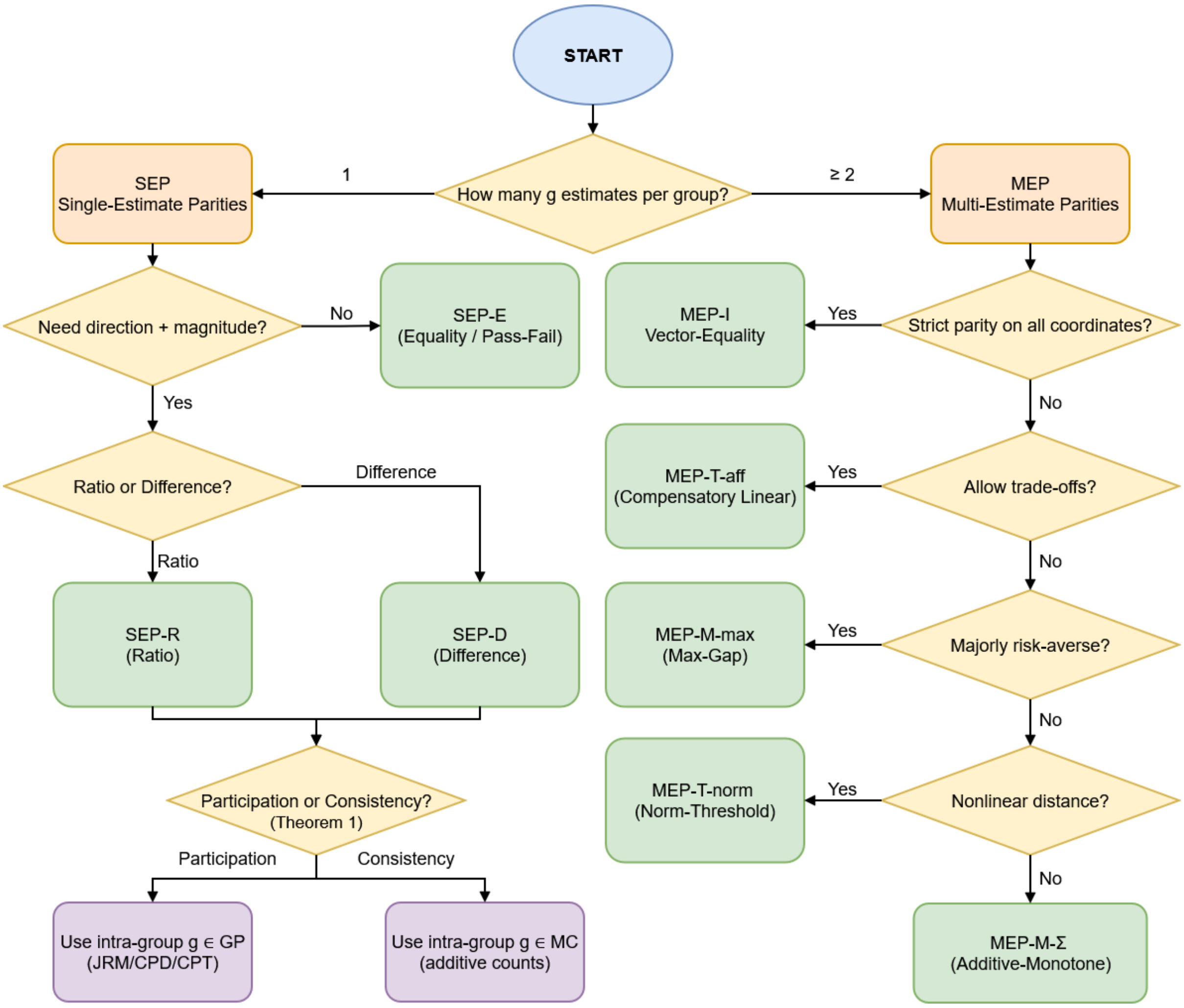


Figure 4.4: Decision diagram for selecting which metric class in the taxonomy, with some specialized property checking for intra-group metrics.

keep the analysis tractable, we prove property states for the three most common PGMs (SEP-E, SEP-D, and SEP-R) for four IGMs (CPD, CPT, JRM, and MC) over all nine properties. This is still $3 \times 4 \times 9 = 108$ use cases; we aggregate use cases into single proofs when possible for conciseness. Table 4.6 illustrates concrete examples of the mutual exclusivity of Participation and Consistency.

Additionally, we provide a decision diagram for fairness metric class in Figure 4.4. This figure assists practitioners in fairness metric selection with the branching of class properties in our taxonomy. This figure complements our property satisfaction table as it illustrates a higher level review of the metric selection process.

4.6.2 Consistency versus Participation

Here, we give proofs and results for Consistency and Participation over SEP classes before concluding with our main theorem of their mutual exclusivity.

Consistency

Consistency fails for Group Proportion intra-group classes due to Simpson's paradox.

Recall that D_1 and D_2 are disjoint datasets and their union is $D_{1,2} \triangleq D_1 \cup D_2$. Similarly, the corresponding confusion matrices are combined additively. To standardize discussion for the deviation from parity p, we define

$$\mathrm{sign}_p(x) \triangleq \mathrm{sign}(x - p). \tag{4.33}$$

Thus Consistency must satisfy

$$\mathrm{sign}_p(M_{D_1}(A, B)) = \mathrm{sign}_p(M_{D_2}(A, B)) \implies \mathrm{sign}_p(M_{D_1 \cup D_2}(A, B)) = \mathrm{sign}_p(M_{D_1}(A, B)). \tag{4.34}$$

Lemma 4.1. *SEP-D and SEP-R with the Group Proportions intra-group metric class can violate Consistency.*

Proof. This is related to Simpson's paradox. For example:
Dataset D_1:

$$A = (\mathrm{TP} = 6, \mathrm{FN} = 4, \mathrm{FP} = 0, \mathrm{TN} = 0),$$

$$B = (\mathrm{TP} = 2, \mathrm{FN} = 8, \mathrm{FP} = 0, \mathrm{TN} = 0).$$

Then $\mathrm{ACC}_{D_1}(A) = 6/10 = 0.6$ and $\mathrm{ACC}_{D_1}(B) = 2/10 = 0.2$, so $M_{D_1}(A, B) = 0.6 - 0.2 = 0.4 > 0$.

Dataset D_2:

$$
\begin{aligned}
A =& (\mathrm{TP} = 9, \mathrm{FN} = 1, \mathrm{FP} = 0, \mathrm{TN} = 0), \\
B =& (\mathrm{TP} = 44, \mathrm{FN} = 6, \mathrm{FP} = 0, \mathrm{TN} = 0).
\end{aligned}
$$

Then $\mathrm{ACC}_{D_2}(A) = 9/10 = 0.9$ and $\mathrm{ACC}_{D_2}(B) = 44/50 = 0.88$, so $M_{D_2}(A, B) = 0.9 - 0.88 = 0.02 > 0$.

On the union $D_{1,2}$, add the counts. Dataset $D_{1,2}$:

$$
\begin{aligned}
A =& (\mathrm{TP} = 15, \mathrm{FN} = 5, \mathrm{FP} = 0, \mathrm{TN} = 0), \\
B =& (\mathrm{TP} = 46, \mathrm{FN} = 14, \mathrm{FP} = 0, \mathrm{TN} = 0).
\end{aligned}
$$

Hence $\mathrm{ACC}_{D_{1,2}}(A) = 15/20 = 0.75$ while $\mathrm{ACC}_{D_{1,2}}(B) = 46/60 \approx 0.7667$, so $M_{D_{1,2}}(A, B) \approx 0.75 - 0.7667 = -0.0167 < 0$.

This counterexample is for CPD. Similar counterexamples exist for CPT and JRM. □

Lemma 4.2. *SEP-D satisfies Consistency for the intra-group class GC.*

Proof. If $\mathrm{sign}_p(M_{D_1}(A, B) = \mathrm{sign}_p(M_{D_2}(A, B)) = +1$ then $g_{D_1}(A) > g_{D_1}(B)$ and $g_{D_2}(A) > g_{D_2}(B)$. Then by additivity:

$$
g_{D_{1,2}}(A) = g_{D_1}(A) + g_{D_2}(A) > g_{D_1}(B) + g_{D_2}(B) = g_{D_{1,2}}(B). \tag{4.35}
$$

The case for both groups having sign of -1 follows the same logic. For both groups having sign of 0, it is the equality case.

Similarly, the NMC case. □

Lemma 4.3. *SEP-R satisfies consistency wherever defined. (MC version).*

Proof. This proof is similar to Lemma 4.2. Recall that for SEP-R,

$$
M_D(A, B) = \frac{g_D(A)}{g_D(B)}, \qquad p = 1,
$$

Since $g \in \mathrm{MC}$, $g_D \geq 0$ for any dataset D. Now consider this case:

$$\frac{g_{D_1}(A)}{g_{D_1}(B)} > 1 \qquad \text{and} \qquad \frac{g_{D_2}(A)}{g_{D_2}(B)} > 1.$$

Because all $g_D \geq 0$ and we only consider cases where M is defined ($g_D(B) > 0$), this is equivalent to

$$g_{D_1}(A) > g_{D_1}(B) \qquad \text{and} \qquad g_{D_2}(A) > g_{D_2}(B).$$

Now we add the inequalities:

$$g_{D_1}(A) + g_{D_2}(A) \;>\; g_{D_1}(B) + g_{D_2}(B),$$

All g in MC are linear combinations, hence they are additive ($g_{D_1} + g_{D_2} = g_{D_{1,2}}$). So,

$$g_{D_{1,2}}(A) > g_{D_{1,2}}(B).$$

Therefore,

$$\frac{g_{D_{1,2}}(A)}{g_{D_{1,2}}(B)} > 1.$$

By flipping the inequality sign, the case for $M_D(A, B) < 1$ follows, and $M_D(A, B) = 1$ is the equality case. □

Lemma 4.4. *SEP-R fails consistency. (NMC version).*

Proof. Let $g = \mathrm{TP} - \mathrm{FP} \in \mathrm{NMC}$. Then consider these two datasets:

Dataset D_1:

$$A_1 = (\mathrm{TP} = 0, \mathrm{FN} = 0, \mathrm{FP} = 4, \mathrm{TN} = 0), \qquad B_1 = (\mathrm{TP} = 0, \mathrm{FN} = 0, \mathrm{FP} = 2, \mathrm{TN} = 0).$$

Here, $M_{D_1}(\cdot) = -4/-2 = 2 > 1$.

Dataset D_2:

$$A_2 = (\mathrm{TP} = 6, \mathrm{FN} = 0, \mathrm{FP} = 0, \mathrm{TN} = 0), \qquad B_2 = (\mathrm{TP} = 5, \mathrm{FN} = 0, \mathrm{FP} = 0, \mathrm{TN} = 0).$$

Here, $M_{D_2}(\cdot) = 6/5 > 1$.

On their union:

$$g_{D_{1,2}}(A) = 6 - 4 = 2, \qquad g_{D_{1,2}}(B) = 5 - 2 = 3,$$

and $M_{D_{1,2}} = 2/3 < 1$. □

Lemma 4.5. *SEP-E fails consistency. (GP version).*

Proof. For our counterexample, we choose g as accuracy. Recall that $p = 1$ for SEP-E. Dataset D_1 (both groups have accuracy 0 but different sizes):

$$A_1 = (\mathrm{TP} = 0, \mathrm{FN} = 100, \mathrm{FP} = 0, \mathrm{TN} = 0),$$
$$B_1 = (\mathrm{TP} = 0, \mathrm{FN} = 1, \mathrm{FP} = 0, \mathrm{TN} = 0).$$

Then $g_{D_1}(A) = 0 = g_{D_1}(B)$, so $M_{D_1}(A, B) = 1$.

Dataset D_2 (both groups have accuracy 1 but sizes swapped):

$$A_2 = (\mathrm{TP} = 1, \mathrm{FN} = 0, \mathrm{FP} = 0, \mathrm{TN} = 0),$$
$$B_2 = (\mathrm{TP} = 100, \mathrm{FN} = 0, \mathrm{FP} = 0, \mathrm{TN} = 0).$$

Then $g_{D_2}(A) = 1 = g_{D_2}(B)$, so $M_{D_2}(A, B) = 1$.

Union $D_{1,2}$:

$$A = (\mathrm{TP} = 1, \mathrm{FN} = 100, \mathrm{FP} = 0, \mathrm{TN} = 0) \Rightarrow g(A) = \tfrac{1}{101},$$
$$B = (\mathrm{TP} = 100, \mathrm{FN} = 1, \mathrm{FP} = 0, \mathrm{TN} = 0) \Rightarrow g(B) = \tfrac{100}{101}.$$

With $g(A) \neq g(B)$, $M_{D_{1,2}}(A, B) = 0$. □

Lemma 4.6. *SEP-E fails consistency. (GC version).*

Proof. We construct a counterexample with $g \in \mathrm{MC}$. Let D_1 and D_2 be datasets so each separately has equality under SEP-E, but the union $D_{1,2}$ does not. Let $g = \mathrm{TP}$, $M = \mathbf{1}[g(A) =$

$g(B)]$.

Dataset 1:

$$A = [\mathrm{TP} = 2, 0, 0, 0], B = [\mathrm{TP} = 1, 0, 0, 0]$$

Here, $g(A) = 2 \neq g(B) = 1$. Hence, $M_{D_1} = 0$.

Dataset 2:

$$A = [\mathrm{TP} = 1, 0, 0, 0], B = [\mathrm{TP} = 2, 0, 0, 0]$$

Here, $g(A) = 1 \neq g(B) = 2$. Hence, $M_{D_2} = 0$.

Dataset Union:

$$A = [\mathrm{TP} = 3, 0, 0, 0], B = [\mathrm{TP} = 3, 0, 0, 0]$$

Here, $g(A) = g(B) = 3$. Hence, $M_{D_{1,2}} = 1$. This violates consistency.

For NMC, use the same example with each TP multiplied by -1. □

We have proven that SEP-E violates Consistency for any intra-group metric; SEP-D only satisfies Consistency if all $g \in$ GC; and SEP-R only satisfies Consistency if $g \in$ MC.

Participation

We now move to prove which classes satisfy Participation.

Lemma 4.7. *All $M \in PGM$ satisfy Participation if every $g \in \vec{g}$ is a Group Proportion.*

Proof. Participation requires that data augmentation with equivalent empirical rates does not change the result. For all $g \in$ GP, any such augmentation results in the same proportions. As such, every Group Proportion g is unchanged, hence M receives the same values in its arguments. Succinctly,

$$\left(\forall g \in \vec{g} : g \in \mathrm{GP}\right) \implies M \models \text{Participation}.$$

□

Lemma 4.8. *No $M \in PGM$ is guaranteed to satisfy Participation if any $g \in \vec{g}$ is in GC.*

Proof. Metrics in GC do not guarantee identical outputs when cardinality changes since the increase for both group cells may be different despite having the same empirical rates. For example, Group A may have $[1, 1, 1, 1]$ and Group B has $[1, 1, 1, 1]$. With a uniform rate, we can add $[2, 2, 2, 2]$ to Group A and $[1, 1, 1, 1]$ to Group B. This breaks equality or changes magnitude for any PGM that uses $g \in \text{GC}$. Concisely,

$$\left(\exists g \in \vec{g} : g \in \text{GC}\right) \implies M \not\models \text{Participation}.$$

□

The consequent of Lemmas 4.7 and 4.8 is that metrics in PGM only satisfy Participation if their intra-group metrics are restricted to Group Proportions.

Participation versus Consistency

Finally, we now prove the mutually exclusive result.

Theorem 4.1. *Across the PGM taxonomy, Consistency and Participation cannot be jointly guaranteed.*

Proof. The incompatibility arises from the fact that Participation and Consistency promote opposite structural directions.

Lemmas 4.7 and 4.8 prove that Participation may only be satisfied if every $g \in \vec{g}$ is an instance of GP. We now show that these same same branches cannot guarantee Consistency.

In Section 4.6.2, we prove that no SEP with $g \in \text{GP}$ satisfies Consistency. To extend these examples to MEP, we show that every MEP branch contains a one-coordinate degeneration. Let $z = \varphi\left(g_1(A), g_1(B)\right)$ be the result on one GP coordinate, and let all remaining coordinates be neutral. Then each branching class in MEP collapses accordingly:

$$
\begin{aligned}
\text{MEP-I}: &\quad \bigwedge(\mathbf{1}[z=0], 1, \ldots, 1) = \mathbf{1}[z=0], \\
\text{MEP-M-}\Sigma: &\quad \sum_{k=1}^{d} w_k z_k = w_1 z \quad \text{if } z_2 = \cdots = z_d = 0, \text{ and } w_1 > 0, \\
\text{MEP-T-aff}: &\quad \sum_{k=1}^{d} w_k z_k = w_1 z \quad \text{if } z_2 = \cdots = z_d = 0, \text{ and } w_1 < 0, \\
\text{MEP-M-max}: &\quad \max(z, 0, \ldots, 0) = z, \\
\text{MEP-T-norm}: &\quad \|(z, 0, \ldots, 0)\|_p = z \quad \text{if } p \text{ is odd.}
\end{aligned}
$$

□

Thus, we prove that no $M \in$ PGM can jointly satisfy Consistency and Participation.

4.7 Conclusions and Future Work

We reframe confusion-matrix fairness metrics as social-choice-style aggregation rules over groups and introduce a fairness taxonomy rooted in the Parity-Gap Metrics (PGM). We complement this taxonomy with a decision-tree summary. Furthermore, we adapt or extend social-choice theoretic principles into thirteen properties for group-based fairness metrics in binary-classification settings. For classes that contain the majority of fairness metrics, we prove their property failure or satisfaction over all thirteen properties. A central takeaway from our work is that metric choice has important effects. When one realizes their preferred effects, then one can prune the space of eligible metrics. Even so, some desiderata are mutually exclusive (e.g., Consistency and Participation). The resulting summaries and decision diagram allow practitioners an explicit, auditable checklist tied to deployment goals.

Future work includes proving our characterized social-choice axioms to the full MEP taxonomy, and extending our work to multiclass settings and ranking-based audits. We also look forward to incorporating statistical uncertainty (confidence intervals and small-sample corrections) into axiomatic behavior and validating how the axioms align with stakeholder judgments in real audits and governance workflows.

Chapter 5

ADDRESSING EVALUATION BIASES IN CLASSIFICATION: SAMPLE-SIZE-INDUCED BIAS

Evaluating machine learning models is crucial not only for determining their technical accuracy but also for assessing their potential societal implications. While the potential for sample size-induced bias in algorithms is well known, we demonstrate that many widely used classification metrics also exhibit this bias. This revelation challenges the efficacy of these metrics in assessing bias with high resolution, especially when comparing groups of disparate sizes, which frequently arise in social applications. We provide analyses of the bias that appears in several commonly applied metrics and propose a correction technique. Additionally, we explore the often-overlooked issue of undefined behaviors in metric calculations, which can lead to ambiguous or misleading evaluations. This work illuminates the previously unrecognized challenge of jaggedness in standard evaluation practices, hoping to advance the community's approach for performing equitable and trustworthy classification methods.

5.1 Introduction

Classification metrics derived from confusion matrices are fundamental tools in evaluating machine learning models, particularly in binary classification tasks. Metrics such as accuracy, precision, recall, and Matthews Correlation Coefficient (MCC) provide critical insights into model performance. However, an often overlooked issue is the impact of sample size on these metrics, especially when comparing performance across different subgroups or datasets of varying sizes.

Small sample sizes introduce significant variability and jaggedness in classification metrics due to the discrete and combinatorial nature of confusion matrices. This variability can lead to misleading interpretations of a model's performance and fairness. For instance, minor changes in the counts of true positives or false negatives can cause disproportionate shifts in metric values, complicating comparisons across groups and potentially obscuring biases or disparities.

Moreover, certain metrics become undefined under specific conditions, such as divisions by zero. These undefined cases, or "holes", in the metric space further challenge the reliability of performance assessments, especially in subgroups with limited data.

Despite the widespread use of confusion-matrix metrics, there has been limited attention to the systematic biases and inconsistencies induced by sample size variations. Existing literature often assumes large sample sizes or overlooks the combinatorial complexities that small samples introduce, leaving a gap in understanding how to accurately assess model performance.

We address these challenges by providing a comprehensive analysis of sample-size-induced biases in confusion-matrix metrics. This chapter's contributions include:

- **Demonstrating Metric Variability**: We present both theoretical and empirical evidence of the jaggedness and variability in classification metrics caused by small sample sizes. By illustrating how these effects distort performance evaluations, we underscore the need for careful interpretation of metrics in small-sample scenarios.
- **Quantifying Undefined Cases**: We systematically count the number of "holes" (situations where metrics are undefined) in several commonly used classification metrics. This quantification reveals the extent to which certain metrics may be unreliable or misleading under specific sample configurations.
- **Metric Alignment Trial for Checking Homogeneity (MATCH) Test**: We introduce a statistical test that assesses the significance of a subgroup's metric score by comparing it to the distribution of scores from a reference group. This approach allows us to determine whether observed differences are due to genuine performance disparities or merely artifacts of sample size variability.
- **Cross-Prior Smoothing (CPS)**: We propose a smoothing technique that incorporates prior information from other groups to enhance metric reliability. By adjusting confusion

matrix counts using cross-group priors, CPS uniformly reduces variability and improves the stability of metric estimates.

This work sheds light on a critical issue in the use of confusion-matrix metrics and offers practical solutions to improve their reliability. By addressing the biases introduced by sample size, we aim to enhance the fairness and accuracy of model evaluations, particularly in applications where subgroup comparisons are essential.

5.2 Related Work

As machine learning models are increasingly used in high-stakes domains, ensuring fairness across subgroups is critical. However, the impact of sample size on classification metrics such as accuracy, precision, and recall remains underexplored. This study examines how these metrics evolve with changing sample sizes, revealing potential biases that may exacerbate disparities between subgroups. Understanding these distributional shifts can help develop more equitable AI systems, particularly in settings where sample sizes differ significantly. Metric definitions are provided in Section 2.1.2.

Confusion matrices are fundamental tools for assessing classification performance, providing the basis for scalar metrics like accuracy and MCC. While these metrics are widely used, they can exhibit biases related to sample size—a phenomenon we refer to as *score distribution shift*. This bias affects common fairness metrics, including disparate impact (Feldman et al., 2015), equalized odds (Hardt et al., 2016), and predictive parity (Das et al., 2021). By highlighting this issue, we aim to enhance the robustness of metric comparisons across datasets with varying sample sizes.

In social data analysis, accurate measurement of fairness is crucial, particularly in legal contexts where metrics like disparate impact assess potential discrimination. For instance, in employment or housing discrimination cases (Dennis J. Aigner and Wiles, 2024; Gastwirth and Miao, 2009) and recidivism prediction (Dressel and Farid, 2018; Moore et al., 2023), datasets often feature small or imbalanced sample sizes. Recognizing how distributional shifts in metrics affect evaluations is vital for ensuring fair outcomes. The recently introduced Objective Fairness

Index (Briscoe and Gebremedhin, 2024) is also susceptible to such shifts.

Recent work by Feng et al. (2024) introduces a method to assess model calibration across subgroups, ensuring that models are reliable for different populations, particularly in high-dimensional settings with limited data.

Statistical methods designed for small sample sizes are relevant to this approach. Techniques like Laplacian smoothing, commonly used in natural language processing to handle unobserved events (Eisenstein, 2019), can be adapted to confusion matrices where certain classes may not be represented. Discussions around the use of non-informative or weakly informative priors in Bayesian estimation for small samples (He et al., 2021; Lemoine, 2019) inform this methodology, particularly in developing methods robust under data limitations.

Sokolova and Lapalme (2009) provides a comprehensive analysis of various performance measures used in classification tasks, focusing on their invariance properties with respect to changes in the confusion matrix. Their work introduces the concept of "measure invariance", which refers to a metric's ability to maintain consistent evaluations despite transformations in the underlying data distribution.

Additionally, Goutte and Gaussier (2005) introduces a probabilistic framework that interprets precision, recall, and the F_1-score by using a symmetric beta distribution and Monte Carlo sampling techniques. This approach moves beyond simple point estimates by estimating the likelihood that one algorithm would outperform another based on these scores. Part of this work extends this idea by generalizing precision and recall to a broader set of metrics, grouped as *Joint Ratio Metrics*.

The issue of metric unreliability in small sample sizes is often overlooked, with practitioners sometimes attributing it to the need for more data. Chicco and Jurman (2020) specifically addresses this issue along with imbalanced data by comparing MCC against the more widely used accuracy and F_1-score. They argue that MCC provides a more reliable assessment of classifier performance in binary tasks, particularly when there is class imbalance, because MCC incorporates all four confusion matrix categories. In contrast, accuracy and F_1 score can give misleadingly high values in imbalanced datasets, failing to capture poor performance in one class.

In another work, Rudner et al. (2024) adjusts model parameters with Group-Aware Priors to improve model robustness under subpopulation shifts. Aghbalou et al. (2024) examines imbalanced classification and derive sharp error bounds under class imbalance. Their focus is on improving classification performance when one class is underrepresented, offering new theoretical insights on error rates when class probabilities approach zero.

This work builds on these foundations by examining the effect of small sample sizes on a broader range of classification metrics. We demonstrate that metrics like accuracy, precision, recall, and even MCC not only exhibit increased variability, but also have fluctuating distributions (jaggedness) as the sample size scales, potentially leading to misleading evaluations. For instance, as Chicco and Jurman present, while MCC is robust in imbalanced scenarios, it can still fluctuate in extreme cases of imbalance or small samples, an issue the Cross-Prior Smoothing (CPS) technique aims to mitigate by using prior information from reference groups. Additionally, the introduction of the MATCH Test allows for a rigorous assessment of whether observed metric scores in small subgroups are meaningful or merely artifacts of sampling variability.

By addressing these gaps, we contribute to the growing literature on improving the robustness and fairness of classification metrics. This approach offers both theoretical insights and practical tools to ensure more reliable evaluations across diverse subgroups, particularly when sample sizes are limited or imbalanced.

5.3 Distribution Shifts

The variability and jaggedness in the predicted positive rates, as illustrated in Figure 5.1, is a direct consequence of both the probabilistic nature of small sample sizes and the combinatorial explosion of the confusion matrix space. This section explains the underlying phenomena contributing to these effects through a probabilistic lens.

Let us begin by reviewing the binary confusion matrix, CM, from Definition 2.1. CM is a 2×2 matrix of the non-negative counts of true positives (TP), false negatives (FN), false positives (FP), and true negatives (TN). Each of these counts is referred to as a *cell* c. The total number

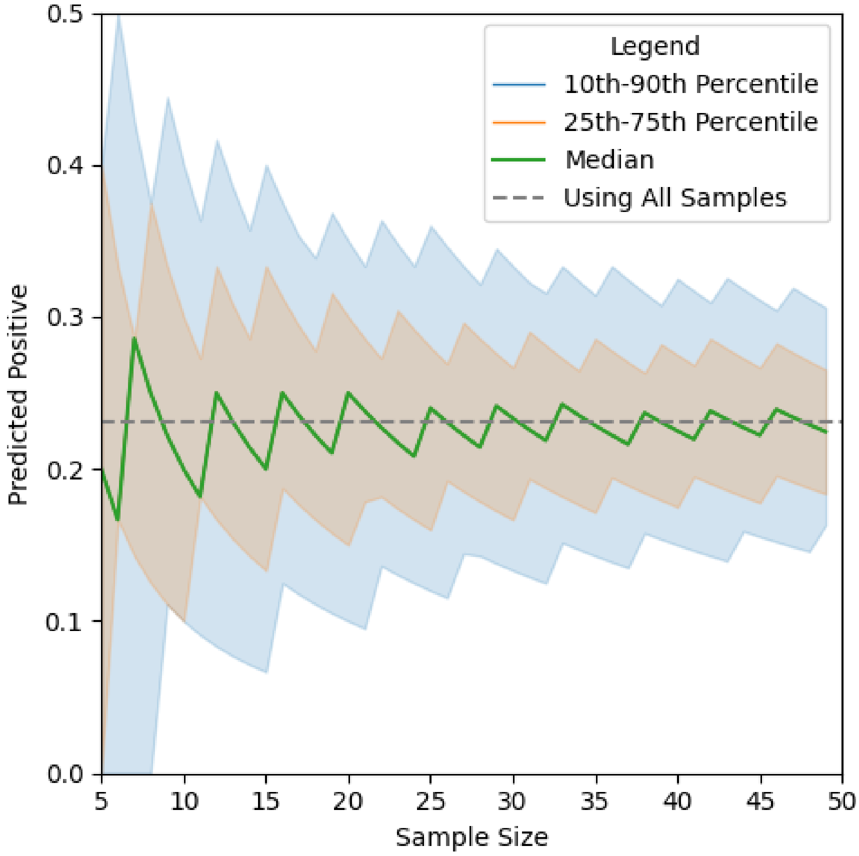


Figure 5.1: Variability of Positive Predictive Rate for Wealth Classification Among Multiracial Individuals: A Monte Carlo Simulation Study Based on Sample Size.

of samples n is partitioned among these four categories, such that $n = \mathrm{TP} + \mathrm{FN} + \mathrm{FP} + \mathrm{TN}$. For a given sample size n, we denote the set of all possible confusion matrices as $\mathcal{M}(n)$.

The narrowing of the distribution observed in Figure 5.1 can be attributed to the combinatorial expansion of $\mathcal{M}(n)$ with increasing n. For small sample sizes, the number of possible confusion matrices is limited, causing each configuration (i.e., unique confusion matrix) to represent a significant fraction of the total space—including those yielding extreme metric values. As n grows, the number of possible confusion matrices increases rapidly, and $\mathcal{M}(n)$ becomes densely populated. Specifically, the cardinality of $\mathcal{M}(n)$ is given by $|\mathcal{M}(n)| = \binom{n+3}{3}$ using the stars and bars method (Theorem 5.1). Hence, the number of possible confusion matrices grows cubically with n (Corollary 5.1).

Theorem 5.1. *The cardinality of $\mathcal{M}(n)$ (the space of all possible confusion matrices given size n) is $\mathcal{N}(n) = \binom{n+3}{3} = (n+1)(n+2)(n+3)/6$.*

Proof. We prove this using the stars and bars method, which is defined as:

$$\binom{n^* + k - 1}{k - 1}, \tag{5.1}$$

where n^* represents the number of samples, and k denotes the number of buckets.

For our case, we have $n^* = n$ samples and $k = 4$ buckets, which corresponds to $k - 1 = 3$ bars. Applying the formula, we obtain:

$$|\mathcal{M}(n)| = \mathcal{N}(n) = \binom{n + 4 - 1}{4 - 1} = \frac{(n+1)(n+2)(n+3)}{6}. \tag{5.2}$$

□

Corollary 5.1. *From Equation 5.2, the complexity of $\mathcal{N}(n)$ is $O(n^3)$, and more precisely, $\Theta(n^3)$.*

Empirically, these configurations are influenced by the underlying probability distribution of classification outcomes, denoted as $(p_{\text{TP}}, p_{\text{FN}}, p_{\text{FP}}, p_{\text{TN}})$. Using k_i as the integer count of i, we use the multinomial distribution definition to define the probability of any CM configuration:

$$P(CM) = \frac{n!}{k_{\text{TP}}!\, k_{\text{FP}}!\, k_{\text{FN}}!\, k_{\text{TN}}!}\, p_{\text{TP}}^{k_{\text{TP}}}\, p_{\text{FP}}^{k_{\text{FP}}}\, p_{\text{FN}}^{k_{\text{FN}}}\, p_{\text{TN}}^{k_{\text{TN}}}. \tag{5.3}$$

Assumption 5.1. *Using the multinomial distribution assumes that 1) the data is sampled i.i.d., and 2) that the model's classifications are independent.*

Due to the factorial terms in the multinomial coefficient from Equation 5.3, even a single-unit increase in n can significantly disrupt the probability mass distribution for small sample sizes, as depicted in Figure 5.1.

The distribution shifts in Figure 5.2 are not a probabilistic artifact as *all* possible configurations are considered. As n increases, the discrete space of possible confusion matrices becomes more finely granulated, and the number of configurations corresponding to certain metric scores increases at different rates.

As the sample size increases, the probability masses of metric scores tend to converge to their expected values. In Figure 5.1, the expected value of the predicted positive rate converges to $1365/5956 \approx 0.23$. For any metric M, the expected value is $\mathbb{E}[M(p_{\text{TP}}, p_{\text{FN}}, p_{\text{FP}}, p_{\text{TN}})]$.

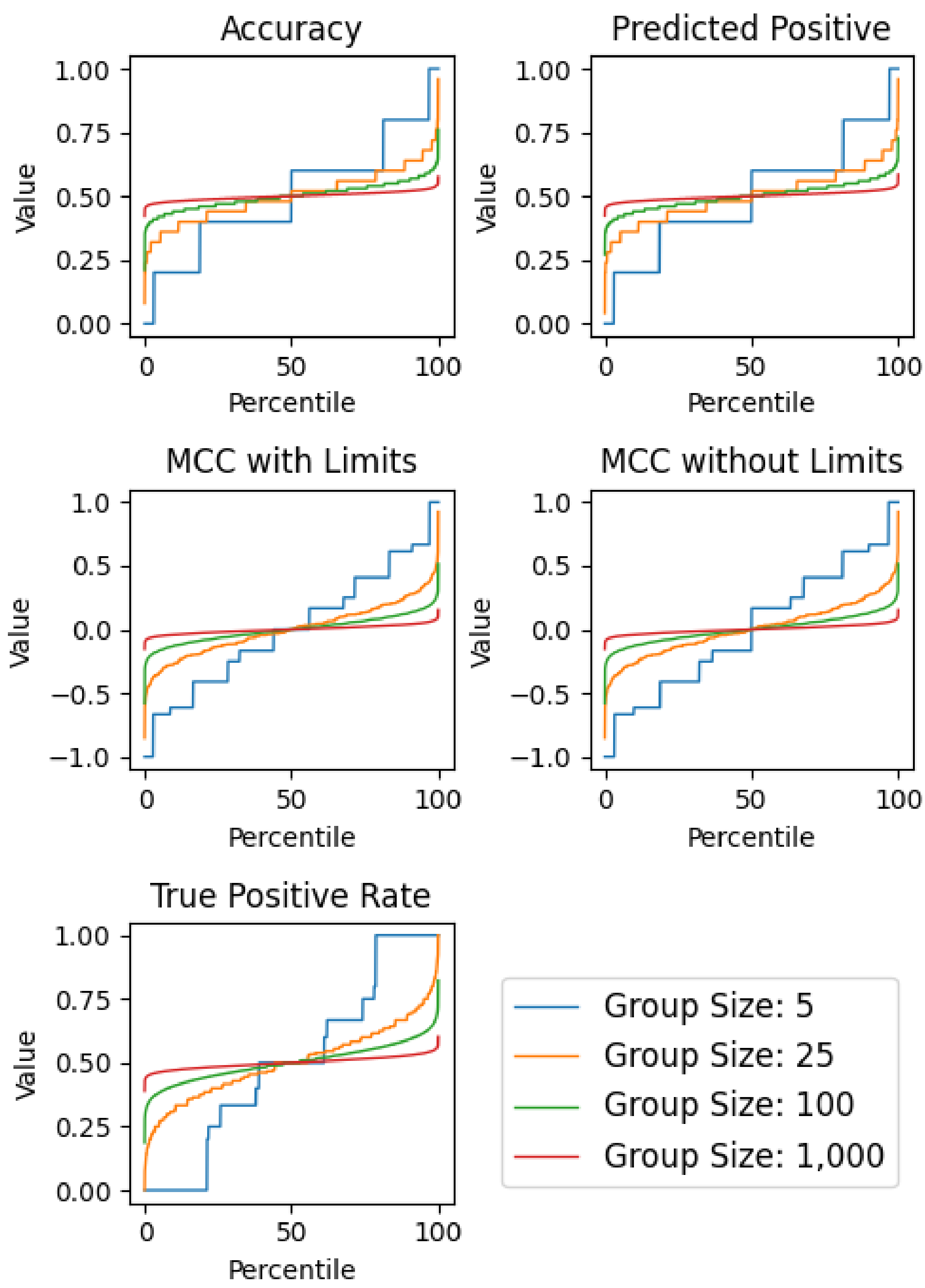


Figure 5.2: Cumulative distribution functions of common classification metrics as sample size increases.

For example, Figure 5.2 uses all configurations, so all probabilities are 0.25. We see that accuracy and predicted positive converge to (0.25+0.25)/1=0.5; MCC (with or without limits) converges to $\frac{0.25\cdot 0.25-0.25\cdot 0.25}{\sqrt{(0.25+0.25)\cdot(0.25+0.25)\cdot(0.25+0.25)\cdot(0.25+0.25)}} = 0$; and TPR goes to 0.25/(0.25+0.25)=0.5.

By understanding discrete distribution shifts, we highlight the importance of considering sample size effects when interpreting classification metrics, particularly where small groups are compared. Ignoring these effects can lead to misleading conclusions about performance and fairness across different populations.

5.4 Undefined Cases

Classification metrics derived from confusion matrices can exhibit undefined behaviors, or "holes", under certain conditions—particularly when denominators in their formulas become zero. In Table 5.1, we find that for all considered metrics, the asymptotic growth of undefined cases is less than the cubic growth of total possible configurations.

We generalize metrics of the form $M = \frac{c_i+c_j}{n}$ as *Binomial Metrics*, which include common measures like accuracy, prevalence, and predicted positive rate. Furthermore, we categorize metrics with form $M = \frac{c_i}{c_i+c_j}$, as *Joint Ratio Metrics* (JRMs), which include common measures like true positive rate (TPR), false positive rate (FPR), precision (PPV), and others. We now prove each undefined case count from Table 5.1.

Theorem 5.2. ***Binomial Metrics*** *have zero holes.*

Proof. With form $(c_i + c_j)/n$, these metrics are defined for all $n \geq 1$. □

Theorem 5.3. ***Marginal Benefit*** *has zero holes.*

Proof. Defined by $\mathcal{B} = \frac{\mathrm{FP}-\mathrm{FN}}{n}$, this metric is defined for all $n \geq 1$. □

Theorem 5.4. ***Objective Fairness Index*** *has zero holes.*

Proof. Defined as $\mathcal{B}_i - \mathcal{B}_j$ for groups i and j, OFI is defined for all $n \geq 1$. □

Theorem 5.5. ***Joint Ratio Metrics*** *have $n + 1$ holes for each metric.*

Table 5.1: Undefined Cases Across Metrics Compared to Total Configuration Counts for all $n \geq 3$.

Metric	**Asymptotic Growth**	**Undefined Case Count**
Binomial Metrics (ACC, PREV, ...)	$\Theta(1)$	0
Marginal Benefit	$\Theta(1)$	0
Objective Fairness Index	$\Theta(1)$	0
Simplified F_1 Score	$\Theta(1)$	1
Joint Ratio Metrics (TPR, FPR, ...)	$\Theta(n)$	$n+1$
Matthews Correlation Coefficient	$\Theta(n)$	$4n$
Prevalence Threshold	$\Omega(n)$ **&** $O(n \log\log n)$	$\geq 2n+2$ and $< e^{\gamma} n \log\log n + \frac{0.6483n}{\log\log n}$
Treatment Equality	$\Theta(n^2)$	$\binom{n_1+2}{2} + \binom{n_2+2}{2} - 1$
Original F_1 Score	$\Theta(n^2)$	$\binom{n+2}{2}$
Unique Confusion-Matrices Count	$\Theta(n^3)$	Count of All Possible CMs: $\binom{n+3}{3}$

Proof. With form $c_i/(c_i + c_j)$, these metrics are only undefined when $c_i + c_j = 0$ (i.e. one row or column sums to zero). For example, consider matrices where the first row sums to zero (i.e. $TP + FN = 0$). For these specific matrices, the metrics $TPR = TP/(TP + FN)$ and $FNR = FN/(FN + TP)$ are undefined. This implies that the other entries TN & FP must sum to n. So, picking TN to be any non-negative integer forces $FP = n - TN$. Since we are restricted to choosing $TN = 0, 1, 2, \ldots, n$, there are $n + 1$ options for the pair TN and FP. As such, there are $n + 1$ holes for the metrics TPR and FNR.

More examples include the cases where the second row sums to zero (holes at FPR & TNR), the first column sums to zero (holes at PPV & FDR), and the second column sums to zero (holes at NPV & FOR). If we assume c_k and c_ℓ are the entries in the zero row or column while c_i and c_j are the others, then we can count using the same restriction from above, $c_i = n - c_j$. Since there are $n + 1$ options for the pair c_i and c_j, there are $n + 1$ holes for the metric undefined when $c_k + c_\ell = 0$. Consequently, each JRM has $n + 1$ matrices at which they will be undefined. □

Theorem 5.6. *The* ***original*** $\mathbf{F_1}$ ***Score*** *has* $\binom{n+2}{2}$ *holes.*

Proof. This score relies on the inverse of precision and recall being defined. As such, it is undefined when TP = 0. By Theorem **??**, some CM cell (e.g., TP) has $\binom{n+2}{2}$ counts of zero over all possible CM configurations given n. □

Theorem 5.7. *The **simplified** $\mathbf{F_1}$ **Score** has one hole.*

Proof. Introduced as the harmonic mean of precision and recall, modern ML benchmarks often use the simplified version: 2TP/(2TP+FP+FN). Here, F_1 is only undefined when 2TP+FP+FN=0, giving 1 hole. □

Theorem 5.8. *The **Matthews Correlation Coefficient** has* $4n$ *holes.*

Proof. Defined as $\frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$, MCC is only undefined when any term in the denominator's product is zero (e.g., TP+FN=0). From the JRM proof, there are n+1 configurations for c_i+c_j=0. However, there are four overlapping cases, all when some $c_i = n$. Hence, there are $4(n+1) - 4 = 4n$ holes. □

Theorem 5.9. *The **Prevalence Threshold** has at least* $2n+2$ *holes and less than* $e^{\gamma} n \log \log n + \frac{0.6483n}{\log \log n}$ *holes for all* $n \geq 3$.

Note: *Since e is Euler's number and* γ *is the Euler-Mascheroni constant,* $e^{\gamma} \approx 1.78$.

Proof. Defined as $\frac{\sqrt{TPR \cdot FPR} - FPR}{TPR - FPR}$, PT is undefined if TP+FN=0 (TPR), or FP+TN=0 (FPR), or TPR−FPR=0. TPR and FPR are JRM metrics, hence have n+1 holes each. We now move to TPR−FPR=TP/(TP+FN)−FP/(FP+TN)=0.

$$TP/(TP+FN) - FP/(FP+TN) = 0 \tag{5.4}$$

$$\iff \quad TP(FP+TN) - FP(TP+FN) = 0 \tag{5.5}$$

$$\iff \quad TP \cdot TN - FP \cdot FN = 0 \tag{5.6}$$

Equation 5.6 only occurs if one row or column is a multiple of the other. For a matrix with a zero row or column, this is the set of MCC holes described above. Otherwise, there is at most one multiple of that column such that the sum of the entries is n. For example, if the predicted-negative column is an integer (k) multiple of the predicted-positive column, then we have k·(TP+FN)=n and hence (TP+FN)$|n$. Thus, we can bound the number of holes using the sum of divisor function. Then, by Robin's Theorem (Robin, 1984), there are strictly less than $e^{\gamma} n \log \log n + \frac{0.6483n}{\log \log n}$ holes for $n \geq 3$.

Furthermore, since the total holes from FPR and TPR equals $2n + 2$, the lower bound is $\Omega(n)$. □

Theorem 5.10. *The **Treatment Equality** metric has* $\binom{n_1+2}{2} + \binom{n_2+2}{2} - 1$ *holes.*

Proof. Recall that TE is defined as the difference of the ratios $\frac{FN_1}{FP_1} - \frac{FN_2}{FP_2}$ for two groups within the dataset: 1, 2. The metric is undefined when either $FP_1 = 0$ or $FP_2 = 0$. For each group, the number of ways for $FP_i = 0$ is $\binom{n_i+2}{2}$, where n_i is the sample size of group i. Summing over both groups and subtracting the case where both $FP_1 = FP_2 = 0$ (to avoid double-counting), the total number of holes is $\binom{n_1+2}{2} + \binom{n_2+2}{2} - 1$. □

Corollary 5.2. *Comparing these holes with the cardinality of possible configurations,* $\Theta(n^3)$ *from Corollary 5.1, we see that the frequencies of holes diminish toward zero as n increases.*

By analyzing the limiting behavior of metrics when they are undefined, we begin to understand their implications. Next, we briefly discuss the limits of JRMs and MCC.

JRMs are indeterminate when $c_i + c_j = 0$. This occurs because as $c_i + c_j \to 0$, the ratio $M = \frac{c_i}{c_i+c_j}$ approaches an indeterminate form $0/0$. Depending on how c_i and c_j approach zero, M can take any value in $(-\infty, \infty)$. We formally prove this in Theorem 5.11.

Theorem 5.11. *Joint Ratio Metrics (JRMs) are indeterminate when undefined.*

Proof. Defined by $c_i/(c_i + c_j)$, a JRM is undefined when $c_i + c_j = 0$. We manipulate this equation with the ratio: $c_i = rc_j$. Now, $JRM \triangleq rc_j/(rc_j + c_j) = r/(r + 1)$.

We consider $r \to -1$. As $r = -1^-$, $c_i \to -1^+c_j$, JRM $\to -\infty$. Conversely, if $r = -1^+$, $c_i \to -1^-c_j$, JRM $\to \infty$. As such, a JRM is indeterminate for $c_i + c_j = 0$.

However, one may argue that it is impossible to have a negative ratio of counts. We consider the problem of defining JRM $= 0/0$ using FPR $=$ FP$/($FP $+$ TN$)$. On one hand, FP equals the count of negatives $N =$ FP $+$ TN, suggesting $N/N = 0/0 = 1$. However, the algorithm also made no incorrect predictions in the negative set, suggesting $0/N = 0/0 = 0$. Finally, we consider the naïve prior: $\text{FPR}_\epsilon = \frac{\text{FP}+1}{N+2}$. Here, $c_i + c_j = 0 \implies 0.5$. With similar interpretations yielding different results, we can conclude that a JRM is indeterminate for $c_i + c_j = 0$. □

For $c_i + c_j = 0$, Chicco and Jurman (2020) finds that MCC tends to zero. However, they neglect the four cases of $c = n$. We find that all four cases are indeterminate since perturbations (relocating one instance to another cell) either remains a hole (tending to zero) if a single entry in the same row or column is increased, or a valid value of ± 1 if the increased cell does not share a row or column with c.

5.5 MATCH Test

The inherent variability and jaggedness in classification metrics complicates the assessment of model performance across groups of different sizes. Small variations in the sample size n can lead to disproportionately large changes in metric values, making direct comparisons potentially misleading. This issue is particularly acute in fairness evaluations, where comparing metrics across groups (e.g., different demographic groups) is essential.

To address this challenge, we propose a *Metric Alignment Trial for Checking Homogeneity (MATCH) Test* that quantifies the likelihood of observing a given metric score under a reference distribution. Specifically, we assess whether an observed metric score, $M(\mathrm{CM}_i)$, computed from a confusion matrix CM_i for group i, is consistent with what would be expected if this group followed the same performance distribution as a reference group. By comparing the observed score from group i against the cumulative distribution function (CDF) derived from the reference group, we determine the percentile rank of the observed score within this distribution. This method allows us to evaluate the statistical significance of observed metrics, facilitating more informed interpretations when sample size variations could skew direct comparisons.

In this section, we focus on several classification metrics: the six Binomial Metrics, Marginal Benefit, and the eight Joint-Ratio Metrics. For each metric, we derive methods to compute the cumulative probability of an observed score S_{obs} under the assumption that it is drawn from the reference distribution.

To prepare, we now define several terms. Let $k_i \geq 0$ denote the integer count of c_i, p_i is the probability of c_i, $q_i = 1 - p_i$ is the probability of not c_i, S_{obs} is the observed score of a metric M, and $P(S \leq S_{\mathrm{obs}}) \in [0, 1]$ is the probability that $S \leq S_{\mathrm{obs}}$. The subscript $i + j$ means i or j.

5.5.1 Binomial Metrics

We find distributions for all Binomial Metrics that have the form $(c_i + c_j)/n$. This includes accuracy, prevalence, and predicted positive rate, inaccuracy, negative prevalence, and predicted negative rate. We define and describe these metrics in Table 2.2.

Theorem 5.12. *For Binomial Metrics, the cumulative probability is given by the binomial cumulative distribution function (CDF):*

$$P(S \leq S_{\text{obs}}) = \sum_{f=0}^{k_{i+j}} \binom{n}{f} p^f q^{n-f} \tag{5.7}$$

Proof. The total count k_{i+j} corresponds to the number of successes in n trials, each with success probability p. Thus, k_{i+j} follows a binomial distribution with parameters n and p. □

When $np \geq 5$ and $nq \geq 5$, we can approximate $P(S \leq S_{\text{obs}})$ using the normal distribution due to the Central Limit Theorem (Ye et al., 2024; Freund et al., 2014). This is done in constant time. By definition, the mean $\mu = np_{i+j}$ and standard deviation $\sigma = \sqrt{np_{i+j}q_{i+j}}$.

$$\text{Standardize: } z = \frac{k_{i+j} + 0.5 - \mu}{\sigma} \tag{5.8}$$

The +0.5 is the continuity correction as we approximate this discrete distribution with a continuous one. Then, we use the standard normal CDF with z: $\Phi(z)$.

For smaller n, exact computation using the binomial CDF is feasible in $O(n^2)$ time. Otherwise, if the approximation criteria does not hold, one can use efficient approximations such as the Lanczos approximation for $O(n \log n)$ time (Lanczos, 1964).

Example. Consider $n = 100$, $p = 0.75$, and observed score $S_{\text{obs}} = 0.80$. Then $k_{i+j} = nS_{\text{obs}} = 80$, $\mu = 75$, $\sigma \approx 4.33$, and $z \approx 1.27$. Thus, $P(S \leq 0.80) \approx \Phi(1.27) \approx 0.90$, so the observed score is higher than approximately 90% of the expected outcomes under the reference distribution.

5.5.2 Marginal Benefit

The marginal benefit is defined as $\mathcal{B} = (\mathrm{FP} - \mathrm{FN})/n$, and is designed to measure the net benefit or cost to a group (Briscoe and Gebremedhin, 2024). Unlike Binomial Metrics, $\mathcal{B}$ involves the difference of counts, making its distribution symmetric around zero when $p_{\mathrm{FP}} = p_{\mathrm{FN}}$. In this section, we provide a method to approximate the cumulative probability for an estimate of $\mathcal{B}$ in Theorem 5.13. Next, we are able to compute the exact probability by deriving the probability mass function (PMF) in Theorem 5.14 and CDF in Theorem 5.15.

Theorem 5.13. *The cumulative probability for $\mathcal{B}$ can be approximated using the normal distribution:*

$$F(S \leq S_{\mathrm{obs}}) \approx \Phi\left(\frac{k_{FP} - k_{FN} - (p_+ - p_-)}{\sqrt{(p_+ + p_-) - (p_+ - p_-)^2}}\right) \tag{5.9}$$

Proof. $\mathcal{B}$ can be expressed as the average of n i.i.d. random variables X_i: $\frac{1}{n}\sum_{i=1}^{n} X_i$. For

$$X_i = \begin{cases} +1 & \text{with probability } p_+ = p_{FP} \\ -1 & \text{with probability } p_- = p_{FN} \\ 0 & \text{with probability } p_0 = p_{TN+TP}. \end{cases} \tag{5.10}$$

Now we calculate the moments of $\mathcal{B}$.

$$\mu = \mathbb{E}[X_i] = p_+ - p_- + 0 \cdot p_0 = p_+ - p_- \tag{5.11}$$

For σ^2, we use the expanded definition: σ^2=$\mathbb{E}[X_i^2]$-μ^2.

$$\mathbb{E}[X_i^2] = (+1)^2 p_+ + (-1)^2 p_- + (0)^2 p_0 \tag{5.12}$$

$$\text{Thus, } \sigma^2 = (p_+ + p_-) - (p_+ - p_-)^2. \tag{5.13}$$

Then we calculate the z-score and use the normal CDF Φ. Note that the continuity correction δ cancels out: $z = ((k_{FP} + \delta) - (k_{FN} + \delta) - \mu)/\sigma$. □

The Berry-Esseen Theorem also guarantees that this error is within $O(n^{-1/2})$ (Berry, 1941; Esseen, 1942).

Now, we provide exact solutions.

Theorem 5.14 (PMF of $\mathcal{B}$)**.** *We find the PMF of $\mathcal{B} = \frac{k}{n}$ by summing over the probabilities of all counts FP and FN such that $FP - FN = k$.*

$$P\left(\mathcal{B} = \frac{k}{n}\right) = \sum_{n_{-1}=n_{-1}^{\min}}^{n_{-1}^{\max}} \left(\underbrace{\frac{n!}{(k+n_{-1})!\, n_{-1}!\, (n-k-2n_{-1})!}}_{\textit{Multinomial Coefficient}} \times \underbrace{p_{\mathrm{FP}}^{k+n_{-1}}\, p_{\mathrm{FN}}^{n_{-1}}\, p_0^{n-k-2n_{-1}}}_{\textit{Joint Probability Term}} \right) \tag{5.14}$$

where: $\quad n_{-1}^{\min} = \max(0, -k), \quad n_{-1}^{\max} = \left\lfloor \frac{n-k}{2} \right\rfloor,$ *and k ranges over the integers $-n$ and n since $k = FP - FN \in [-n, n]$.*

Theorem 5.15 (CDF of $\mathcal{B}$)**.** *The cumulative distribution function of $\mathcal{B}$ is:*

$$P(\mathcal{B} \le b) = P(S \le nb) = \sum_{k=-n}^{\lfloor nb \rfloor} \underbrace{\sum_{n_{-1}=n_{-1}^{\min}(k)}^{n_{-1}^{\max}(k)} \left(\frac{n!}{(k+n_{-1})!\, n_{-1}!\, (n-k-2n_{-1})!} \times p_{\mathrm{FP}}^{k+n_{-1}}\, p_{\mathrm{FN}}^{n_{-1}}\, p_0^{n-k-2n_{-1}} \right)}_{\textit{Theorem 5.14}} \tag{5.15}$$

where b is a real number ranging from -1 to 1 and k is an integer ranging from $-n$ to $\lfloor nb \rfloor$.

Proof of Theorem 5.14. Recall that $\mathcal{B}$ can be expressed as the average of n i.i.d. random variables X_i: $\frac{1}{n}\sum_{i=1}^{n} X_i$. For

$$X_i = \begin{cases} +1 & \text{with probability } p_+ = p_{FP} \\ -1 & \text{with probability } p_- = p_{FN} \\ 0 & \text{with probability } p_0 = p_{TN+TP}. \end{cases} \tag{5.16}$$

We proceed with the proof in two parts. First, we determine the bounds, and then we derive the corresponding multinomial probability.

The Bounds of the PMF:

Let n_{+1} be the number of times $X_i = +1$ (number of false positives), n_{-1} be the number of times $X_i = -1$ (number of false negatives), and n_0 be the number of times $X_i = 0$ (number of true positives and true negatives). Then:

$$
\begin{aligned}
n &= n_{+1} + n_{-1} + n_0 \\
S = k &= n_{+1} \cdot (+1) + n_{-1} \cdot (-1) + n_0 \cdot 0 = n_{+1} - n_{-1}
\end{aligned}
$$

$$\text{We highlight } k = n_{+1} - n_{-1} \text{ from above.} \tag{5.17}$$

It follows that $n_{+1} = k + n_{-1}$. We use this to redefine n_0 in terms of n, k, and n_{-1}:

$$n_0 = n - (k + n_{-1}) - n_{-1} = n - k - 2n_{-1}. \tag{5.18}$$

Our bounds are constrained by all cells being non-negative. As such, the counts n_{+1}, n_{-1}, and n_0 must satisfy:

$$
\begin{aligned}
n_{+1} &\geq 0 \implies k + n_{-1} \geq 0 \implies n_{-1} \geq -k \\
n_{-1} &\geq 0 \\
n_0 &\geq 0 \implies n - k - 2n_{-1} \geq 0 \implies n_{-1} \leq \frac{n-k}{2}
\end{aligned}
$$

Hence, the valid range for n_{-1} is:

$$n_{-1}^{\min} = \max(0, -k), \quad n_{-1}^{\max} = \left\lfloor \frac{n-k}{2} \right\rfloor \tag{5.19}$$

The Multinomial Probability:

By definition, the multinomial probability of observing counts (n_{+1}, n_{-1}, n_0) is:

$$P(n_{+1}, n_{-1}, n_0) = \underbrace{\frac{n!}{n_{+1}!\, n_{-1}!\, n_0!}}_{\text{Multinomial Coefficient}} \times \underbrace{p_{\text{FP}}^{n_{+1}}\, p_{\text{FN}}^{n_{-1}}\, p_0^{n_0}}_{\text{Joint Probability Term.}} \tag{5.20}$$

Then by substituting $n_{+1} = k + n_{-1}$ (Equation 5.17) and n_0 with Equation 5.18 into $P(S = k)$, we have:

$$P(S = k) = P\left(\mathcal{B} = \frac{k}{n}\right).$$

This concludes the proof of Theorem 5.14's formula for $\mathcal{B}$'s PMF. □

Proof of Theorem 5.15. We start by noting that $\mathcal{B} = \frac{S}{n}$, where $S = \text{FP} - \text{FN} = \sum_{i=1}^{n} X_i$ is the sum of independent random variables X_i defined in Theorem 5.14 (Equation 5.16). Furthermore, we reframe $P(\mathcal{B} < b)$ as $P(\mathcal{B} \leq b) = P\left(\frac{S}{n} \leq b\right) = P\left(S \leq nb\right)$.

Since $S = FP - FN$ is an integer ranging from $-n$ to n, the possible values of $\mathcal{B}$ are $\frac{k}{n}$ for integer $k \in [-n, n]$. Thus, $\mathcal{B}$'s CDF is:

$$P\left(\mathcal{B} \leq b\right) = \sum_{k=-n}^{\lfloor nb \rfloor} P(S = k).$$

□

5.5.3 Joint Ratio Metrics

Joint Ratio Metrics (JRMs) are expressed as $c_i/(c_i + c_j)$. Computing the cumulative probability for JRMs presents additional challenges due to the ratio of random variables involved. Theorem 5.16 extends the work of Goutte and Gaussier (2005) by introducing a generalized framework for these metrics, along with an analysis of computational complexity. For simplicity in the following discussion, we define $k_{i+j} = k$ and $p_{i+j} = p$.

Theorem 5.16. *The cumulative probability for a JRM score S_{obs} is given by:*

$$\begin{aligned} P(S \leq S_{\text{obs}}) &= \sum_{k=1}^{n} P(c_{i+j} = k) P\left(\frac{k_i}{k} \leq S_{\text{obs}} \middle| c_{i+j} = k\right) \\ &= \sum_{k=1}^{n} \left(\binom{n}{k} p^k (1-p)^{n-k} \cdot \sum_{k_i=0}^{k_i^{\max}} \binom{k}{k_i} \theta^{k_i} (1-\theta)^{k-k_i}\right) \end{aligned} \tag{5.21}$$

Proof. From our analyses in binomial metrics, we have c_{i+j} and c_i following binomial distributions:

$$c_{i+j} \sim Binomial(n, p_{i+j}), \quad c_i \sim Binomial(n, p_i).$$

Since S is only defined for $k_i + k_j = k > 0$, we focus on $k \geq 1$. We begin with the simplest solution in Equation 5.22 which sums all probabilities for scores lower than S_{obs}.

$$P(S \leq S_{\text{obs}}) = \sum_{k=1}^{n} \sum_{k_i=0}^{n} \begin{cases} P(k_i \big| c_{i+j} = k) & \text{if } \frac{k_i}{k} \leq S_{\text{obs}} \\ 0 & \text{otherwise.} \end{cases} \tag{5.22}$$

By definition, if c_i is sampled, then c_{i+j} is also sampled, implying that $P(c_{i+j} \mid c_i) = 1$. Using Bayes' rule, we express $P(c_i \mid c_{i+j})$ as:

$$\theta = P(c_i \mid c_{i+j}) = \frac{P(c_{i+j} \mid c_i) \cdot P(c_i)}{P(c_{i+j})} = \frac{p_i}{p_{i+j}}. \tag{5.23}$$

Consequently, we model the conditional sampling from a JRM M as:

$$c_i \mid c_{i+j} \sim \text{Binomial}(c_{i+j}, \theta).$$

We now proceed to derive the PMFs to construct the CDFs. Since the events are binomially distributed, the PMFs are given by:

$$P(c_{i+j}) = \binom{n}{c_{i+j}} p_{i+j}^{c_{i+j}} (1 - p_{i+j})^{n - c_{i+j}} \tag{5.24}$$

$$\text{and} \quad P(c_i | c_{i+j}) = \binom{c_{i+j}}{c_i} \theta^{c_i} (1 - \theta)^{c_{i+j} - c_i}. \tag{5.25}$$

Then, the CDF of Binomial(c_{i+j}, θ) can be expressed as:

$$\begin{aligned} P\left(\frac{k_i}{k} \leq S_{obs} \middle| c_{i+j} = k\right) &= P\left(k_i \leq S_{obs} \cdot k \middle| c_{i+j} = k\right) \\ &= \sum_{k_i=0}^{k_i^{\max}} \binom{k}{k_i} \theta^{k_i} (1 - \theta)^{k - k_i} \end{aligned} \tag{5.26}$$

where $k_i^{\max} = \lfloor S_{obs} \cdot c_{i+j} \rfloor$.

By expanding the PMF of c_{i+j} and combining it with the CDF of $c_i \mid c_{i+j}$, we obtain a calculable CDF of any JRM:

$$
\begin{aligned}
P(S \leq S_{\text{obs}}) &= \sum_{k=1}^{n} P(c_{i+j} = k) P\left(\frac{k_i}{k} \leq S_{\text{obs}} \middle| c_{i+j} = k\right) \\
&= \sum_{k=1}^{n} \left(\binom{n}{k} p^k (1-p)^{n-k} \cdot \sum_{k_i=0}^{k_i^{\max}} \binom{k}{k_i} \theta^{k_i} (1-\theta)^{k-k_i} \right)
\end{aligned}
\tag{5.27}
$$

□

Computing this sum directly has computational complexity $O(n^2)$. However, we can improve efficiency by approximating the distribution of the JRM using the beta distribution. When applying a Bayesian approach with a uniform prior (equivalent to adding pseudo-counts $\lambda = 1$), the posterior distribution of the JRM is a beta distribution with parameters $k_i + \lambda$ and $k_j + \lambda$.

Using the beta distribution, the cumulative probability is approximated as:

$$
P(S \leq S_{\text{obs}}) \approx I_{S_{\text{obs}}}(k_i + \lambda, k - k_i + \lambda) \tag{5.28}
$$

where I is the regularized incomplete beta function. This improves the time complexity to $O(n)$ since I integrates once over an independent variable. Beta tables allow constant time if they are available. Otherwise, efficient continued fraction and recursive approximations, or quadrature approximations of integrals may allow further improvements as the integrand is well-behaved.

5.5.4 More on the Normal Approximation

The normal approximation is a powerful tool for approximating the distribution of Binomial Metrics and $\mathcal{B}$. However, the approximation's precision is limited by the sample size n. In this section, we showcase the normal approximation's validity and precision as n scales, and provide ways to improve the approximation's precision.

Justifying the Normal Approximation

To justify the use of the normal approximation in our MATCH Test for binomial metrics and the marginal benefit metric $\mathcal{B}$, we perform empirical validations. Figure 5.3 illustrates the probability density functions (PDFs) of these metrics as the sample size n increases, demonstrating their convergence towards the normal distribution.

In our simulations, we consider non-uniform probabilities to reflect realistic scenarios where class distributions are imbalanced. Specifically, we set $p_{\text{TP}} = 0.3$, $p_{\text{TP}} = 0.2$, $p_{\text{FP}} = 0.1$, and $p_{\text{TN}} = 0.4$. We generate one million samples for each sample size n to observe how the distribution of the metrics evolves.

As shown in Figure 5.3, both the accuracy and marginal benefit metrics exhibit decreasing skewness and increasing kurtosis as n grows. This behavior indicates that the distributions are becoming more symmetric and peaked, aligning with the characteristics of the normal distribution. The convergence towards normality supports the validity of using the normal approximation for these metrics in our MATCH Test, particularly for larger sample sizes.

Assessing and Improving the Normal Approximation's Precision

While the normal approximation is generally acceptable for large sample sizes, it may not provide sufficient accuracy for small n. To address this, we examine theoretical bounds on the approximation error and explore more precise approximations applicable to smaller sample sizes.

The Berry-Esseen theorem provides an upper bound of $O\left(n^{-1/2}\right)$ on the error of the normal approximation to the distribution of a sum of independent random variables (Berry, 1941; Esseen, 1942). Specifically, it states that the maximum difference between the cumulative distribution function $F_n(x)$ of the standardized sum and the standard normal cumulative distribution function $\Phi(x)$ is bounded by:

$$|F_n(x) - \Phi(x)| \leq \frac{C_0 \cdot \gamma}{\sigma^3 \sqrt{n}} \tag{5.29}$$

where C_0 is a constant between 0.4097 and 0.5600 (gustav Esseen, 1956; Shevtsova, 2010), γ is the third absolute central moment of the summands, σ^2 is the variance of the summands, and n

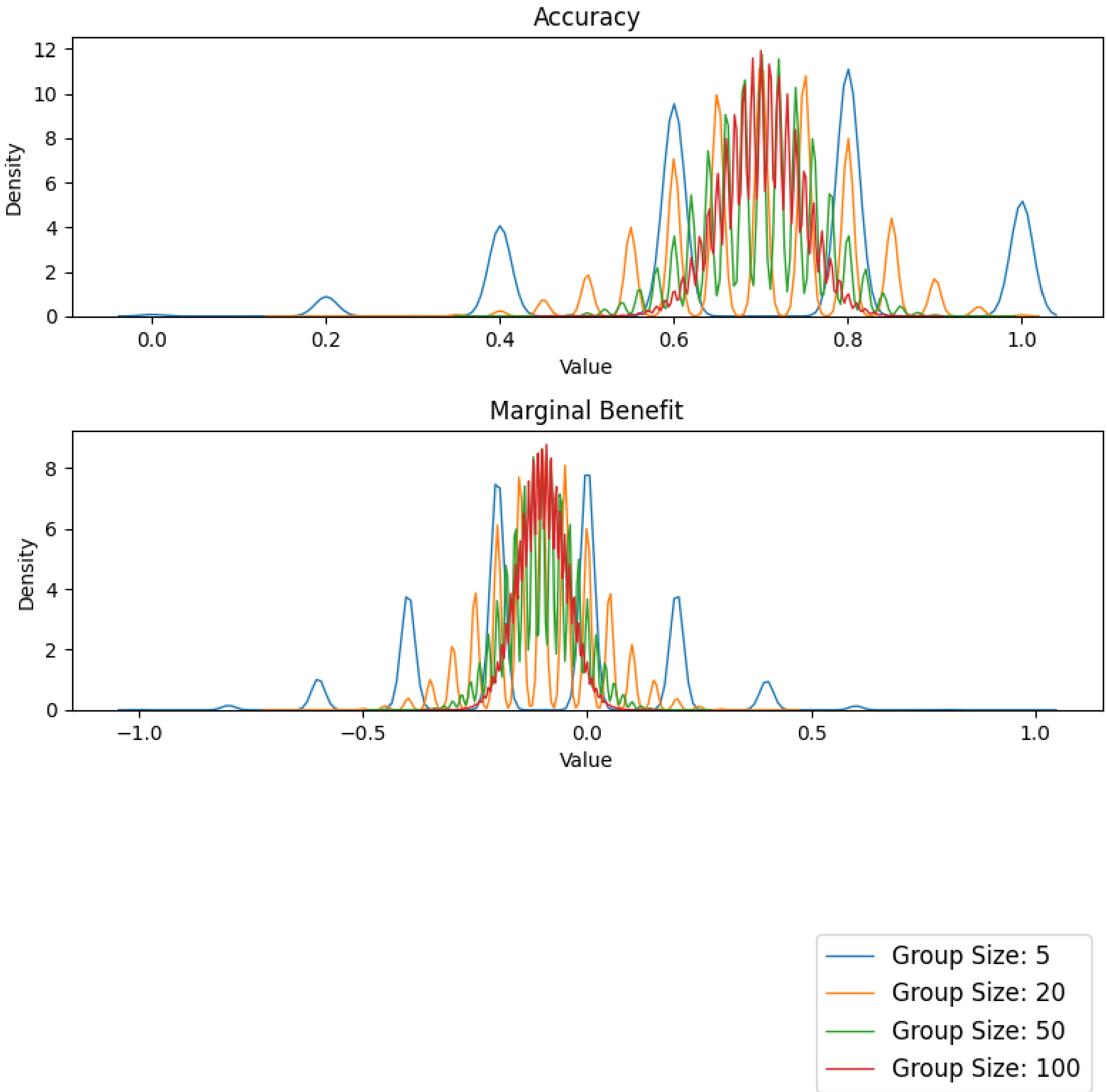


Figure 5.3: Probability density functions of accuracy and marginal benefit metrics for varying sample sizes n. As n increases, both metrics converge towards the normal distribution.

is the number of summands.

To reduce approximation error, we can use Peizer and Pratt's (1968) normal approximation for the binomial distribution. This is far more complicated, but the error becomes less than 0.1% for $\min(x + 1, n - x) \geq 2$ (Johnson et al., 2005).

By having binomial metrics and $\mathcal{B}$ analogous to binomial distributions, we improve the bounds of what can be approximated within 0.1% by a better z-score approximation. Peizer and Pratt recommend approximating the z-score as:

$$\frac{x + \frac{2}{3} - \left(n + \frac{1}{3}\right) p}{\sqrt{\left(n + \frac{1}{6}\right) pq}} \times \frac{\sqrt{npq}}{\left(x + \frac{1}{2} - np\right)} \times \left\{2 \left[\left(x + \frac{1}{2}\right) \ln \left(\frac{x + \frac{1}{2}}{np}\right) + \left(n - x - \frac{1}{2}\right) \ln \left(\frac{n - x - \frac{1}{2}}{nq}\right)\right]\right\}^{1/2}. \tag{5.30}$$

This gives good results, but for marginal improvement, add

$$\frac{1}{50} \left[(x + 1)^{-1} q - (n - x)^{-1} p + (n + 1)^{-1} \left(q - \frac{1}{2}\right)\right] \tag{5.31}$$

to the expression $x + \frac{2}{3} - (n + \frac{1}{3})p$. Altogether, we obtain the precision of strictly less than 0.1% for all x, n such that $\min(x + 1, n - x) \geq 2$.

The proper parameters for z_p for binomial metrics and $\mathcal{B}$ are given in Table 5.2.

Table 5.2: Proper Parameters for Peizer and Pratt's z-score Approximation.

Metrics	x	p	q	n
Binomial metrics	k_{i+j}	p	q	n
$\mathcal{B}$	$k_{\text{FP}} - k_{\text{FN}}$	$p_+ - p_-$	p_0	n

5.6 Cross-Prior Smoothing

In this section, we introduce *Cross-Prior Smoothing* (CPS), a novel correction method designed to enhance the reliability of CM metrics, particularly when comparing groups of varying sizes or distributions. CPS addresses the inherent variability and instability in classification metrics that arise due to sample size disparities, while preserving the consistency of bias assessments across different subgroups. Additionally, CPS drastically reduces the likelihood of encountering zero counts in corresponding cells of CM_i, which can lead to undefined scores. Furthermore, we empirically demonstrate the effectiveness of CPS in improving the reliability of classification metrics.

5.6.1 The Method

CPS leverages information from a reference group's confusion matrix, $\mathrm{CM}_{\mathrm{ref}}$, to inform and smooth the metrics derived from a target group's confusion matrix, CM_i. This approach is especially pertinent in fairness analysis, where both CM_i and $\mathrm{CM}_{\mathrm{ref}}$ are generated by the same underlying algorithm but may represent different demographic groups.

Let $\mathrm{CM}_{\mathrm{total}}$ denote the confusion matrix derived from the entire dataset, and $\mathrm{CM}_i \subset \mathrm{CM}_{\mathrm{total}}$ be the confusion matrix for subgroup i. In our experiments, we define the reference confusion matrix, $\mathrm{CM}_{\mathrm{ref}}$, as the portion of $\mathrm{CM}_{\mathrm{total}}$ excluding the data from group i: $\mathrm{CM}_{\mathrm{ref}} = \mathrm{CM}_{\mathrm{total}} \setminus \mathrm{CM}_i$.

Assumption 5.2. *The reference confusion matrix* $\mathrm{CM}_{\mathrm{ref}}$ *provides a sufficiently informative prior for correcting the metrics derived from* CM_i*.*

As we demonstrate in the following experiments, this assumption is natural in practice, particularly given that once the parameters are fixed for a given model, the multinomial parameters are simple to estimate.

We define CPS in Algorithm 1. We normalize $\mathrm{CM}_{\mathrm{ref}}$ to prevent it from dominating the target group that has small sample sizes. Furthermore, the choice of λ is crucial for balancing the contribution of the reference group's data. We set $\lambda = 5$ for all experiments.

This algorithm is inspired by the Dirichlet distribution, which is a conjugate prior to the multinomial distribution. However, this approach extends beyond a basic Bayesian estimate with a non-informative prior. By incorporating a reference group's data, the CPS technique provides a more robust estimate, particularly when the sample size n is small. This smoothing method significantly improves the stability and reliability of metric estimates by reducing the variability and jaggedness that arises in small-sample scenarios.

Algorithm 1: Cross-Prior Smoothing (CPS)

Input: Confusion matrix for group i, CM_i; Normalized reference confusion matrix, $\hat{\text{CM}}_{\text{ref}}$; Smoothing constant λ

Output: Smoothed confusion matrix CM_{smooth}

Create the concentrations (α) using the prior;

foreach $c \in \text{CM}_i$, $c' \in \hat{\text{CM}}_{\text{ref}}$ **do**

  $\alpha_c = c + \lambda \cdot c'$;

foreach $c_{\text{smooth}} \in \text{CM}_{\text{smooth}}$ **do**

  Normalize the posterior distribution;

  $c_{\text{smooth}} = \alpha_c / (\sum_{\forall\alpha} \alpha)$;

  Scale back for size-dependent metrics;

  $c_{\text{smooth}} \leftarrow c_{\text{smooth}} \cdot \mid CM_i \mid$;

return Smoothed confusion matrix CM_{smooth};

5.6.2 Experiments and Results

We conduct extensive experiments across fifteen commonly used classification metrics, including the eight JRMs, accuracy, prevalence, predicted positive rate, marginal benefit, Matthews Correlation Coefficient, F_1 score, and prevalence threshold. The experiments are performed on two datasets: the COMPAS dataset and the Folktables income dataset.

COMPAS Dataset: The COMPAS dataset comprises risk assessments used in the U.S. judicial system, which have been scrutinized for potential bias (Angwin et al., 2016b; Larson et al., 2016; ProPublica, 2016). We utilize the confusion matrices reported in ProPublica to evaluate the effectiveness of CPS in a real-world fairness context.

Folktables' Income Dataset: We also employ the Folktables income dataset, training a random forest classifier to predict whether individuals earn more than \$50k per year based on features such as marital status, race, and education. For bias assessment, we partitioned the data into eight racial groups and computed the corresponding confusion matrices.

For each metric, we consider various subgroup sample sizes, ranging from $n = 5$ to $n = 150$. To ensure statistical robustness, we conduct one million random samples for each sample size and metric combination. We compare the performance of the original metrics, metrics with additive smoothing ("bashful" smoothing with $\varepsilon = 1 \times 10^{-10}$ and the non-informative prior $\varepsilon = 1$), and metrics smoothed using CPS with the priori scales $\lambda = 5$, $\lambda = 10$, and $\lambda = 20$.

To obtain real-valued results, our experimental analyses excludes any undefined values.

However, this could make direct comparisons between CPS and the original metrics (which often include such undefined cases) potentially unfair. As such, we include "bashful" smoothing ($\varepsilon = 1e-10$) to maintain scores that closely resemble the original metrics. We improve over all original metrics. In Figures 5.4 and 5.5, we zoom into four different metrics with $\lambda \in [5, 20]$ and $\varepsilon \in [0, 1e-10]$. The improvement in MCC is particularly encouraging, as this metric is well-recognized for its strong informational value (Chicco and Jurman, 2023).

We also explore the effect of a the more common non-informative prior, $\varepsilon = 1$, with prevalence and FNR in Figure 5.6. While additive smoothing with $\varepsilon = 1$ reduces errors for some metrics, it leads to inconsistent and less predictable results. In some cases, smoothing with $\varepsilon = 1$ performs worse than no smoothing (e.g., False Omission Rate (FOR), Negative Predictive Value (NPV), and prevalence). In other cases, it performs better than CPS with $\lambda = 10$ for certain metrics (e.g., False Negative Rate (FNR) and TPR). However, CPS with higher λ values still outperforms additive smoothing with $\varepsilon = 1$ overall.

With a 95% confidence interval, the error bands in our experiments confirm statistical significance of CPS' improvement. The error bands are narrow and often indistinguishable, indicating high consistency across datasets and subgroups. This consistency is crucial for ensuring that the bias assessments remain reliable and robust across different groups.

5.6.3 CPS Gains with Minimal Data: Comparing Prior Assumptions

To illustrate the flexibility of CPS in a challenging scenario, we consider a confusion matrix from the COMPAS dataset that reflects only the Black group. This group's confusion matrix is historically cited for its divergence from other racial groups and has invoked great discussion (Washington, 2019; ProPublica, 2016).

Assume we have only a small subset of the Black group data and need a prior to estimate the full Black group's confusion matrix for more reliable metric evaluations. Table 5.3 shows that while the Black group's confusion matrix does indeed differ from other racial groups, the variance is still small enough that a data-driven prior (CPS) offers a more faithful correction than assuming no prior or a uniform prior (such as the "add-one" prior). By showing that CPS differs the least among all priors, we demonstrate that CPS is applicable even in this controversial case.

Table 5.3: Evaluating Prior Assumptions for COMPAS' Black Group When Data is Sparse or Unavailable

Category	TP	FN	FP	TN
Black (Hidden)	0.09	0.33	0.05	0.53
All Other Races	0.06	0.25	0.05	0.64
Possible Priors for the Black Group				
No Prior Assumption	0.00	0.00	0.00	0.00
CPS Estimate	0.06	0.25	0.05	0.64
Uniform Prior	0.25	0.25	0.25	0.25
Absolute Differences from the Black Group				
No Prior Assumption	0.09	0.33	0.05	0.53
CPS Estimate	**0.03**	**0.08**	**0.00**	**0.11**
Uniform Prior	0.16	**0.08**	0.20	0.28

5.6.4 Bias-Variance Trade-Off

We investigate the bias-variance trade-off of CPS by analyzing the underlying multinomial distribution of CMs.

Theorem 5.17. *Given the Cross-Prior Smoothing (CPS) estimate $\hat{p}_c^{CPS}$, the bias and variance have the following asymptotic behavior with respect to n and λ:*

$$Bias(\hat{p}_c^{CPS}) \in O\left(\frac{\lambda}{\lambda + n}\right) \quad and \quad Var(\hat{p}_c^{CPS}) \in O\left(\frac{n}{(n+\lambda)^2}\right). \tag{5.32}$$

Proof. Consider a cell with true probability p_c and reference probability p_c'.

- Without smoothing, $\hat{p}_c = C/n$.
- With CPS, $\hat{p}_c^{CPS} = (C + \lambda p_c')/(n + \lambda)$.

Substituting $\mathbb{E}[C] = np_c$ yields $\mathbb{E}[\hat{p}_c^{CPS}] = (np_c + \lambda p_c')/(n + \lambda)$, so the bias is:

$$\text{Bias}(\hat{p}_c^{CPS}) = \mathbb{E}[\hat{p}_c^{CPS}] - p_c = \frac{\lambda}{n+\lambda}(p_c' - p_c).$$

As a binomial distribution, $\text{Var}(C) = np_c(1 - p_c)$. Since $\hat{p}_c^{CPS}$ is a sample proportion normalized by $n + \lambda$:

$$\mathrm{Var}(\hat{p}_c^{CPS}) = \frac{\mathrm{Var}(C)}{(n+\lambda)^2} = \frac{np_c(1-p_c)}{(n+\lambda)^2}.$$

In summary,

$$\mathrm{Bias}(\hat{p}_c^{CPS}) = \frac{\lambda}{n+\lambda}(p_c' - p_c), \quad \mathrm{Var}(\hat{p}_c^{CPS}) = \frac{np_c(1-p_c)}{(n+\lambda)^2}. \tag{5.33}$$

□

Corollary 5.3. *Fixing n, the bias and variance of $\hat{p}_c^{CPS}$ with respect to λ satisfies:*

$$Bias(\hat{p}_c^{CPS}) \in O(1) \quad and \quad Var(\hat{p}_c^{CPS}) \in O\left(\frac{1}{\lambda^2}\right).$$

For a given n, CPS effectively addresses the *sample-size-induced bias* arising from the discrete, multinomial nature of confusion-matrix metrics (see Figure 5.1) by adjusting the smoothing parameter λ. Specifically, and with a constant n, as λ increases, the variance of each cell decreases on the order of $O(1/\lambda^2)$, while the cell estimates are biased toward p_c' on the order of $O(1)$. This behavior parallels well-known shrinkage approaches such as the James–Stein estimator, wherein one trades off smaller variance for additional bias. It is important to note, however, that this small-sample "bias"—an artifact of limited data and discrete counting—does *not* correspond to the classical bias term in the bias-variance trade-off for confusion-matrix cells. Instead, it reflects how some values of n can systematically shift metric scores away from their population-level score.

5.6.5 Discussion

CPS offers a practical and effective solution for correcting classification metrics in the presence of sample-size-induced bias. By using the reference group's confusion matrix as a prior, we can achieve more stable and reliable metric estimates without compromising the integrity of bias assessments. However, it is important to note that the success of CPS hinges on the validity of Assumption 5.2. By experimenting with popular bias datasets, we provide supporting evidence of CPS's broad applicability. Nevertheless, we recommend following this practical guide for

CPS:

1. **Reasonable Prior Assumption**: CPS is most effective when the reference confusion matrix (the prior) is reasonably expected to be more informative than a uniform prior (e.g., adding one to each cell). Domain knowledge can often justify this; for instance, in the COMPAS dataset, we know that most criminals will not reoffend.
2. **Reference Size**: The reference group should have $n \geq 100$. This is due to the reference having sample-sized-induced bias itself. Our experiments show there is significant bias for $n < 50$, and the bias becomes negligible once $n \approx 100$.
3. **Choosing λ**: Values in $[5, 10, 20]$ worked well empirically. Users can tune λ by iterative refinement.

Alternatively, methods such as synthetic stress tests (down-sampling and cross-validation), marginal distribution comparisons, and divergence measures (e.g., KL divergence) can support CPS' applicability to the given problem.

To evaluate CPS under divergent conditions, we analyze one of the most problematic CMs in the COMPAS dataset, whose systematic bias was publicized by ProPublica and has become a staple in the bias/fairness problems of machine learning.

5.7 Conclusions

In this chapter, we present a comprehensive analysis of sample-size-induced bias in confusion-matrix metrics, highlighting significant implications for evaluating classification performance and fairness across groups of varying sizes. Our theoretical exploration revealed that small sample sizes lead to increased variability and jaggedness in metric scores due to the discrete and combinatorial nature of confusion matrices.

To address these challenges, we propose two novel approaches. First, we introduce the *MATCH Test* that assesses the statistical significance of an observed metric score relative to a reference distribution. This test enables more informed comparisons across groups by accounting for the variability introduced by different sample sizes. Second, we develop *Cross-Prior Smoothing* (CPS), a method that leverages prior information from a reference group's confusion matrix to correct and stabilize metrics derived from smaller groups. Empirically, we show that CPS improves metric reliability by reducing the error rates for all 15 metrics tested.

Our findings underscore the importance of accounting for sample-size-induced bias when interpreting classification metrics, especially in fairness assessments where group comparisons are critical.

Future work could explore optimizing sample sizes to minimize metric variability and potential exploitation. Investigating the most advantageous configurations that could be misused and refining the MATCH Test by incorporating uncertainty in reference probabilities are also promising directions. Another avenue may explore behavior for different metrics; we have noticed that binomial metrics exhibit the most jaggedness. Additionally, extending the theoretical foundations of CPS and examining its applicability to a broader range of metrics and settings (such as multiclass classification) would further enhance its utility.

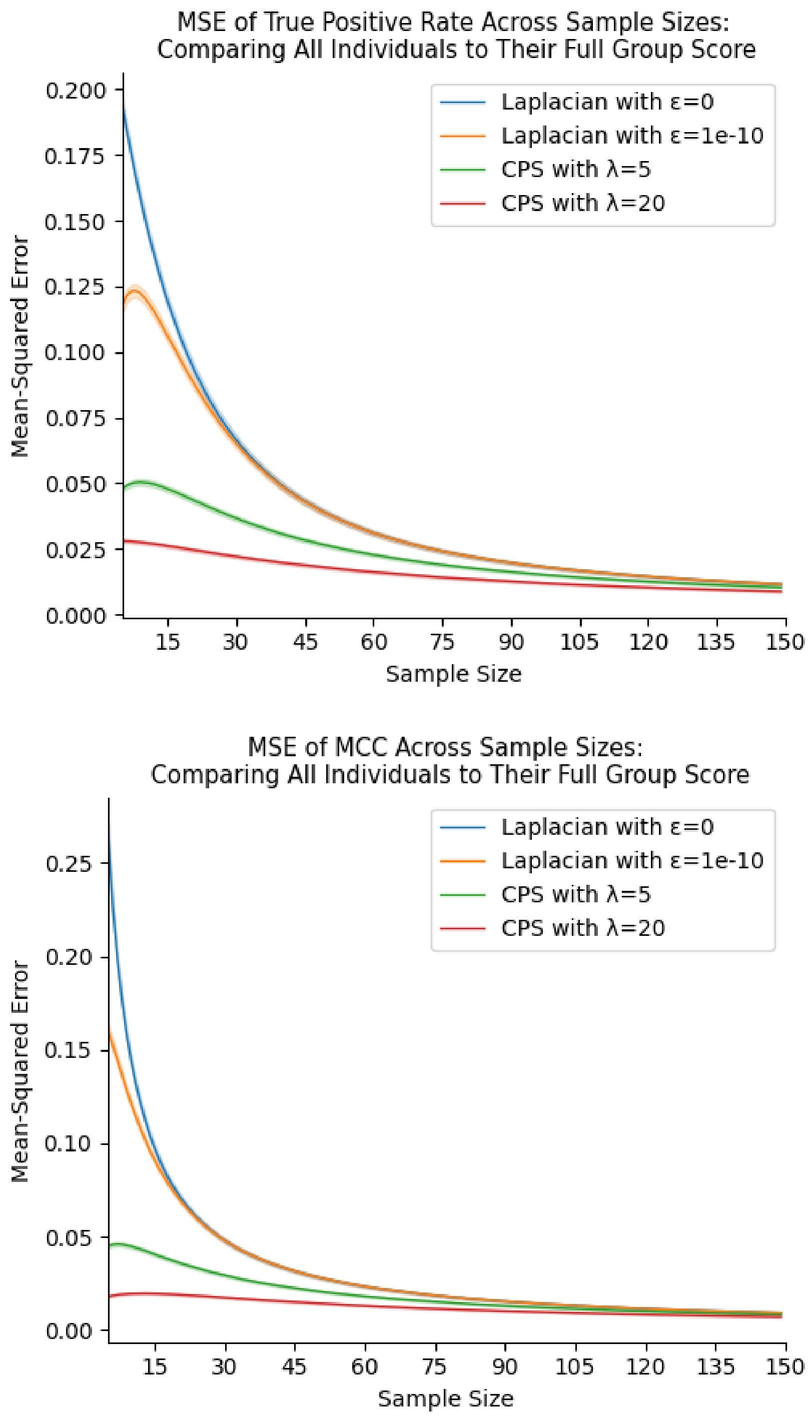


Figure 5.4: Effect of smoothing techniques on metrics with undefined values. Bashful smoothing ($\varepsilon = 1e^{-10}$) reduces errors compared to no smoothing ($\varepsilon = 0$). Cross-Prior Smoothing (CPS) offers further improvements, with the stronger prior ($\lambda = 20$) outperforming the weaker prior ($\lambda = 5$), indicating that CPS uses sufficiently informative priors.

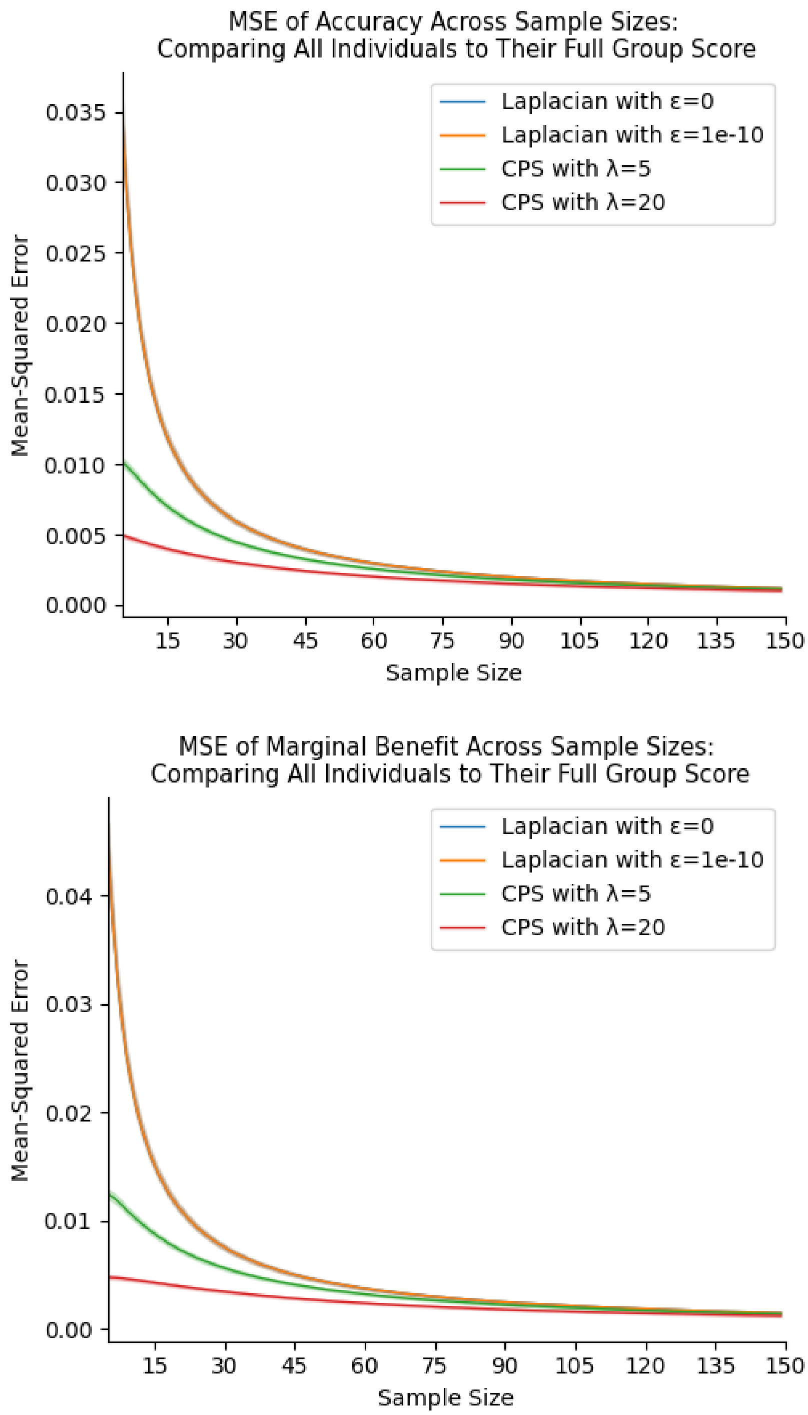


Figure 5.5: Comparison of smoothing effects on metrics without holes. Results show that applying a small smoothing factor ($\varepsilon = 1e^{-10}$) has minimal impact compared to no smoothing. CPS continues to reduce errors, following the same trend observed in Figure 5.4.

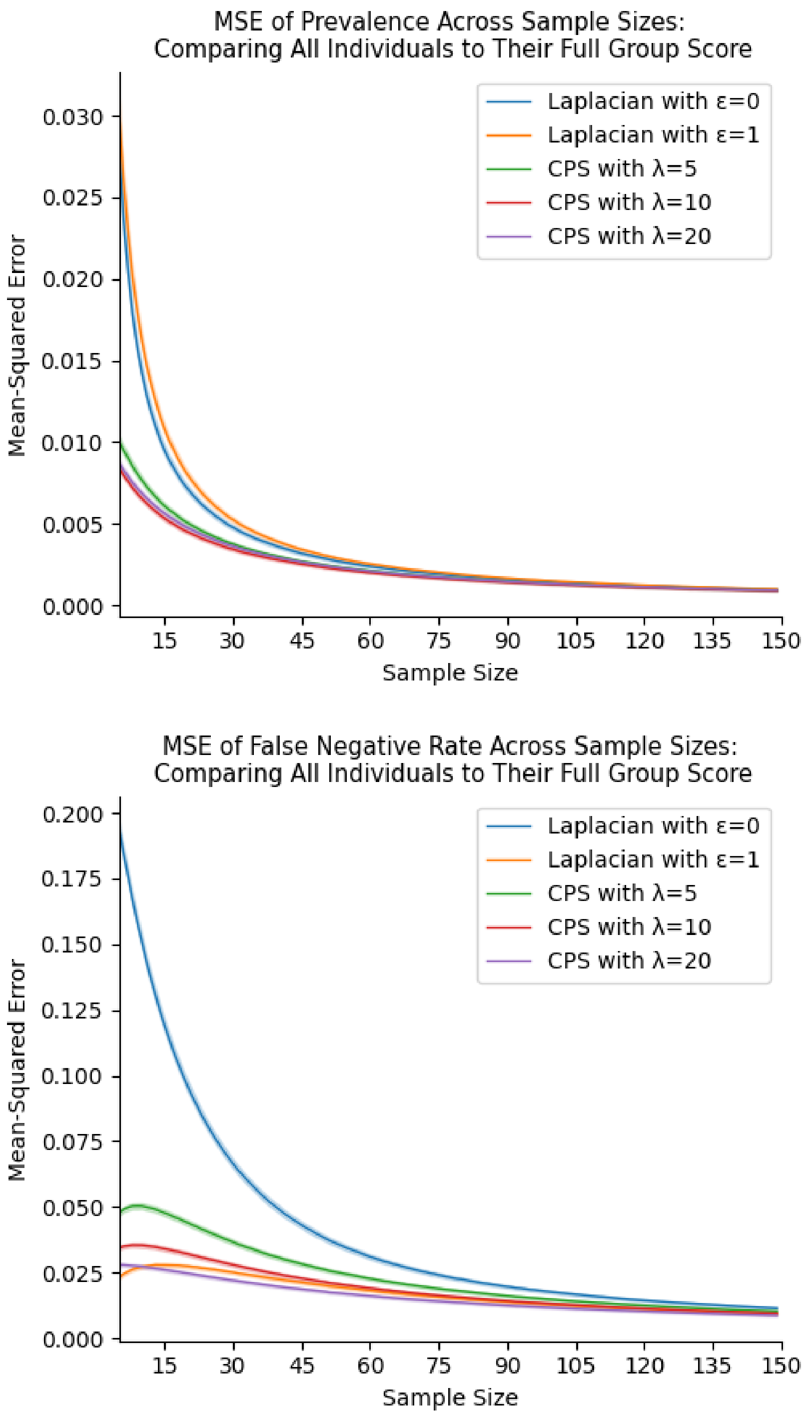


Figure 5.6: Smoothing with $\varepsilon = 1$ yields inconsistent results. In some metrics (left), it performs worse than no smoothing, while in others (right), it outperforms CPS with $\lambda = 10$.

Chapter 6

BEYOND 1-NN ACCURACY: OBJECTIVE-INFORMED BENCHMARKING OF TIME SERIES SIMILARITY MEASURES

Choosing a time series similarity (or distance) measure can strongly shape the performance and reliability of *learning algorithms*. However, most benchmarking approaches rank measures with a single diagnostic—typically leave-one-out 1-Nearest Neighbor (1-NN)—implicitly generalizing the rankings to other downstream tasks. We ask an external-validity question: do measure rankings obtained under 1-NN persist when the same measures drive other algorithms? To test this, we benchmark a set of measures over three algorithmic families: supervised classification, unsupervised partitioning, and structural separability. We utilize the Friedman-Nemenyi pipeline over 81 binary-label datasets to demonstrate that 1-NN accuracy rankings provide poor generalizations. Additionally, we find that Matthews Correlation Coefficient consistently yields larger between-measure score dispersion across datasets, as measured by statistical power in the Friedman-Iman-Davenport test. Across diagnostics, measure rankings are strongly objective-dependent, with winner groups shifting as the diagnostic changes. Together, these findings caution against treating 1-NN accuracy as a surrogate for "best" distance.

6.1 Introduction

Time series data is significant for practical and computational problems in fields such as biology, healthcare, economics, industry, and weather planning (Caldeira et al., 2025; Hess and Feuerriegel, 2024; Wald et al., 2025; Wu et al., 2024; Rohbeck et al., 2025; Briscoe et al., 2022; Steinberg et al., 2023). Similarity (or distance) measures are critical in analyses of algorithms, loss functions, or benchmarking objectives, yet their selection is too often asserted rather than justified. The assertions are implicitly traditional adherence or some rule-of-thumb instead of being based on research in ranking measures (measure benchmarks). While insightful, we find the measure benchmark literature is often based on a single classifier–this is dominantly "leave-one-out" 1-nearest-neighbor classification (1-NN).

Measure benchmarking parallels classifier benchmarking. Demšar (2006) ranks classifiers of tabular datasets, and Bagnall et al. (2017) extends Demšar's work to ranking time series classifiers. Both works are driven by accuracy and use the Friedman-Nemenyi pipeline—a pipeline that checks if a set of rankings indicates statistical significance over order (the Friedman test), then conducts pairwise comparisons to derive statistically indistinguishable groups (Nemenyi post-hoc analysis). Later, Liu and Paparrizos (2024) ranks time-series algorithms by their anomaly-detection abilities.

In 2025, Shao et al. seek to unify time-series forecasting benchmarking under a standard pipeline, suggesting that dataset heterogeneity causes conflicting results in the literature. Similarly, Ragab et al. (2023) seek to standardize evaluation through ADATIME, a benchmarking suite for time-series domain adaptation.

Paparrizos et al. (2023) offer a recent time series measure benchmark. They conduct accuracy and runtime analyses of time series distance measures partitioned by measure type (e.g., lock-step, elastic, kernel). Here, they use 1-NN to find KDTW (a kernel variant of dynamic time warping) to have the highest accuracy and cost.

Wang et al. (2013) perform a similar benchmark comparing accuracies with the 1-NN classifier, but without the Friedman-Nemenyi pipeline. They discover that elastic distances best lock-step distances on small datasets and that a window-constrained DTW is a safe default. Wang et al. found no universally superior measure–but found it to be dataset dependent. In their future work, Paparrizos et al. (2023) echo the belief that the best time series measure is data dependent. Our results reinforce this conclusion.

Recent work in measures for multivariate time series settings often preserves the same evaluation proxy of 1-NN with accuracy. For example, Shifaz et al. (2023) contribute multivariate variants of several elastic measures. This work suggests that the measures support many applications such as clustering, classification, anomaly detection, indexing, subsequence search, and segmentation, yet the empirical comparisons are still anchored to classification–especially the one-nearest neighbor classifier.

Dau et al. (2018) visualize cross-dataset results with *critical difference diagrams* (CDDs) derived from 1-NN classifier accuracy. CDDs visualize average rankings with a black bar that groups rankings as statistically equivalent. A central caution in this work is the Texas-sharpshooter fallacy: retroactively framing a method as "for this subset of datasets" after observing outcomes, thereby overstating generality or drawing post-hoc domain boundaries. However, using just one diagnostic also risks overstating generality, as we show in our analyses. We design our framework to guard against such statements.

Across this literature, the common way to ground *"which measure is best?"* is treating the measure quality as 1-NN classification performance, evaluated by accuracy and using the leave-one-out method. Later works (Paparrizos et al., 2023; Górecki et al., 2024) combine the 1-NN method with statistical rigors such as the Friedman test and Nemenyi's post-hoc analysis. The Friedman test is *global* and asks if the ranking differences are significant, while Nemenyi's post-hoc analysis is *pairwise*—asking if this measure's rank significantly differs from another's. The appeal for 1-NN is pragmatic: it is parameter free, contains no training biases, and leverages the entire dataset (besides the test data point). However, the intuition for using the 1-NN diagnostic deserves scrutiny: does label-matching the closest datum really indicate a "good" measure? This diagnostic can convey if a measure has label-informative value; however, only considering the closest neighbor biases to micro-scale geometry. Perhaps a more robust assessment requires evaluating measures across multiple algorithmic families that probe different scales of structure—including local neighborhoods, global partitions, and density-based separations. Our work is centered around the question: "does 1-NN with accuracy yield similar results as other diagnostics, such as spectral clustering or a support-vector classifier?" It makes the following contributions:

- We propose objective-informed benchmarking across 3 diagnostic families.
- We find that 1-NN with accuracy has poor generalizability to other diagnostics.
- We show that Matthews Correlation Coefficient (MCC) consistently yields higher rank-

discrimination power than accuracy, using empirical tests on 81 datasets from a wide spectrum of applications.

- We present the top-performing measures and the critical difference diagrams for each diagnostic. We find that distance-measure rankings are strongly objective-dependent.

6.2 Background

We discuss the selection of time series measures and datasets included in our empirical analysis. Here, we also introduce the relevant rank matrix that is passed into the Friedman-Nemenyi pipeline.

6.2.1 Measures

We evaluate seventeen measures selected for their diversity in geometric assumptions. These measures are organized into five families:

- *L_p Minkowski*: captures standard norm-based geometries.
- *L_1 distance*: various normalizations for absolute differences.
- *Squared-chord*: assumes data resembles an empirical probability distribution.
- *Squared L_2*: Euclidean variants with scaling, normalizing, or censoring adjustments.
- *"Other"*: four miscellaneous measures.

We provide full details (names, abbreviations, and formal definitions) in Tables 6.1 and 6.2.

Since our experiments are computationally expensive, we restrict our chosen measures to those with a runtime exactly Θ(number of time steps). Our goal is to find the generalizability of the 1-NN classifier based on an accuracy diagnostic, so this selection is acceptable and currently necessary. Including measures such as dynamic time warping is an interesting direction for future work. For measures that are undefined for identical segments, we insert a distance of zero when identical segments are seen.

Table 6.1: Measures used in this work, grouped by family.

Family	Abbrev.	Metric	Function $M(a,b)$
Lp Minkowski	MD	Manhattan distance	$\sum_{i=1}^{d} \lvert a_i - b_i \rvert$
	CD	Chebyshev distance	$\max_{1\leq i\leq d} \lvert a_i - b_i \rvert$
	ED	Euclidean distance	$\sqrt{\sum_{i=1}^{d} (a_i - b_i)^2}$
L_1 distance	LD	Lorentzian distance	$\sum_{i=1}^{d} \ln\big(1 + \lvert a_i - b_i \rvert\big)$
	CanD	Canberra distance	$\sum_{i=1}^{d} \frac{\lvert a_i - b_i \rvert}{\lvert a_i \rvert + \lvert b_i \rvert}$
	SoD	Soergel distance	$\frac{\sum_{i=1}^{d} \lvert a_i - b_i \rvert}{\sum_{i=1}^{d} \max(a_i, b_i)}$
	MCD	Mean Character distance	$\frac{1}{d} \sum_{i=1}^{d} \lvert a_i - b_i \rvert$
	NID	Non Intersection distance	$\frac{1}{2} \sum_{i=1}^{d} \lvert a_i - b_i \rvert$
Squared-chord	MatD	Matusita distance	$\sqrt{\sum_{i=1}^{d} \left(\sqrt{a_i} - \sqrt{b_i}\right)^2}$
	HeD	Hellinger distance	$\sqrt{2 \sum_{i=1}^{d} \left(\sqrt{a_i} - \sqrt{b_i}\right)^2}$
Squared L_2	ClaD	Clark distance	$\sqrt{\sum_{i=1}^{d} \left(\frac{a_i - b_i}{\lvert a_i \rvert + \lvert b_i \rvert}\right)^2}$
	AD	Average (Euclidean) distance	$\sqrt{\frac{1}{d} \sum_{i=1}^{d} (a_i - b_i)^2}$
	MCED	Mean Censored Euclidean distance	$\sqrt{\frac{\sum_{i=1}^{d} (a_i - b_i)^2}{\sum_{i=1}^{d} \mathbf{1}\left[a_i^2 + b_i^2 \neq 0\right]}}$

6.2.2 Datasets

We evaluate distance measures on 81 univariate time series datasets. This collection spans diverse application domains including healthcare monitoring (ECG, EMG, EOG), human activity recognition, spectroscopy, environmental sensing, and synthetic benchmarks. The datasets are described in Tables 6.4 and 6.5, and their sizes are summarized in Table 6.3.

Dataset selection prioritized a sufficient sample size (1K–30K observations) to promote representativeness and reliability for our analyses. All datasets are cast as binary classification problems to ensure methodological consistency across our algorithmic families and their corresponding diagnostic objectives. Regression targets are binarized at the median, while multiclass labels are cast as the majority class or any other class.

Preprocessing emphasizes temporal continuity over sample retention. We apply standard missing value imputation with linear interpolation for short gaps, but critically, longer gaps

Table 6.2: Other measures used in this work.

Abbrev.	Metric	Function $M(a,b)$ for $a,b \in \mathbb{R}^d$
JacD	Jaccard distance	$\frac{\sum_{i=1}^d (a_i-b_i)^2}{\sum_{i=1}^d a_i^2+\sum_{i=1}^d b_i^2-\sum_{i=1}^d a_i b_i}$
HamD	Hamming distance	$\sum_{i=1}^d \mathbf{1}[a_i \neq b_i]$
WIAD	Whittaker's index of association	$\frac{1}{2}\sum_{i=1}^d \left\lvert \frac{a_i}{\sum_{k=1}^d a_k} - \frac{b_i}{\sum_{k=1}^d b_k} \right\rvert$
HasD	Hassanat distance	$\sum_{i=1}^d \begin{cases} 1-\frac{1+\min(x,y)}{1+\max(x,y)}, & \min(x,y) \geq 0, \\ 1-\frac{1+\min(x,y)+\lvert\min(x,y)\rvert}{1+\max(x,y)+\lvert\min(x,y)\rvert}, & \min(x,y) < 0. \end{cases}$

Table 6.3: Dataset characteristics

Sample Count	**Datasets**
1K – 2K	16
2K – 5K	30
5K – 10K	16
10K – 20K	13
20K – 30K	6
Total	**81**

define segment boundaries rather than being imputed—arbitrary interpolation across extended periods destroys the sequential structure required for meaningful similarity computation. After splitting at null boundaries, we retain only sufficiently long segments and apply per-segment min-max normalization (between zero and one; chosen to preclude negative square root errors). This segment-wise approach ensures scale-invariance across diverse measurement units while preserving within-segment temporal dynamics. For distance computation, preprocessed segments are windowed into fixed-length observations of 32 time steps.

6.2.3 Rank Matrix

Let $\mathcal{M} = \{m_1, \ldots, m_j, \ldots m_C\}$ be the list of distance measures and $\mathcal{D} = \{D_1, \ldots, D_i, \ldots, D_B\}$ be the list of datasets. Then for a given algorithm A, we construct a rank matrix $\mathbf{R} \in \mathbb{R}^{B\times C}$ where the ranking of measure m_j given dataset D_i is $1 \leq \mathbf{R}_{i,j} \leq C$.

Rankings are populated by evaluating each measure m_j with a fixed algorithm A on dataset

D_i with appropriate performance metric P (e.g., accuracy). A lower rank indicates better performance. Following Garcia and Herrera (2008), ties are handled by average rank; for example, scores of (1, 2, 2, 3) produce rankings of (1, 2.5, 2.5, 4).

When an algorithm such as spectral clustering expects a similarity measure, we map the distance measures with the Gaussian kernel as done in Bodó et al. (2011).

6.2.4 Friedman-Nemenyi Pipeline

The Friedman test is a non-parametric test that operates on the average ranks of the measures. Under this null hypothesis, all measures have the same expected rank across datasets. We adopt the Iman-Davenport refinement of the Friedman test since the original test is undesirably conservative Iman and Davenport (1980); Demšar (2006). Following Demšar's recommendation, we reject the null hypothesis for $p_{\text{Friedman}} \leq 0.10$.

Table 6.4: Dataset collection organized by domain and modality (81 univariate time series). Most datasets are obtained from Dau et al. (2019). For datasets obtained from other sources, the references are provided next to the dataset. Continued in Table 6.5.

Domain	Modality	Datasets	Primary Task
Acoustic	Speech	Phoneme	Phoneme classification from short audio segments
Biomedical	Cardiac	ECG 5000, ECG Five Days, CinC ECG Torso, Non-Invasive Fetal ECG (2)	Cardiac rhythm classification and arrhythmia detection
	EMG/EOG	Semg Hand (3), EOG Signal (2)	Muscle- and eye-movement–based gesture and gaze classification
	Clinical	Diabetes Pedregosa et al. (2011)	Binary disease-state classification from glucose measurements
Energy	Appliances	Computers, Large Kitchen Appliances, Refrigeration Devices, Screen Type, Power Cons, House Twenty, Freezer (2)	Device-state and appliance identification from power consumption
	Building	Energy Consumption Hebrail and Berard (2012)	Consumption-pattern classification in building energy usage
Environmental	Geophysical	Earthquakes, Lightning (2), Rock, Sunspot Seabold and Perktold (2010)	Event detection and regime classification in geophysical time series
	Biological	Insect Wingbeat Sound, Insect EPG (2), Fungi, Pig Vitals (3)	Species identification and physiological state classification
Infrastructure	Transport	Ford (2), Car, Dodger Loop (3)	Vehicle identification and traffic-pattern classification
	Sensors & HVAC	Mote Strain, Room Temperature Candanedo (2016), Azure Predictive Maintenance Microsoft Azure (2016)	Structural state classification and equipment health monitoring

Table 6.5: Dataset collection organized by domain and modality (81 univariate time series). Most datasets are obtained from Dau et al. (2019). For datasets obtained from other sources, the references are provided next to the dataset. Continued from Table 6.4.

Domain	**Modality**	**Datasets**	**Primary Task**
Motion & Gesture	Wearable	All Gesture Wiimote (3), Gesture Pebble (2), Pickup Gesture Wiimote Z, Gesture Mid-Air (3)	Gesture classification from wearable and depth-based motion signals
	Action	Gun Point (4), Cricket (3), Inline Skate, Haptics	Human action classification from pose and trajectory sequences
Shape-Based	Trajectory, Visual	Arrow Head, Fish, Plane, Fifty Words, Adiac	Shape and trajectory classification from temporal contours
Socioeconomic	Public, Economic	UK Driver Deaths Seabold and Perktold (2010), US Acc Deaths Seabold and Perktold (2010), California Housing Pedregosa et al. (2011)	Temporal trend classification in public and economic indicators
Spectral	Spectroscopy	Beef, Coffee, Ham, Meat, Olive Oil, Chlorine Concentration, Ethanol Level	Material identification and concentration discrimination from spectra
Synthetic	Benchmarks	ACSF1, BME, CBF, Mallat, Mixed Shapes (2)	Controlled pattern classification under known generative structure

If the Friedman test finds the rankings to be statistically significant, we proceed with Nemenyi's post-hoc analysis. Nemenyi's post-hoc analysis does pairwise comparisons (pairwise over measures) to define a "critical difference" scalar. Measures whose respective ranks are within the range of this critical difference are statistically indistinguishable (Nemenyi, 1963).

6.3 Methods

We present the best distance measures for several algorithms using the rigorous Friedman-Nemenyi pipeline. This pipeline requires a collection of ranked lists (a ranked matrix) before deciding if the measure rankings are statistically significant, then identifying statistically indistinguishable groupings. We first explain the rank matrix construction and our Friedman-Nemenyi adoption, then we reason performance metric and null-hypothesis significance test choices based on the algorithmic family.

Algorithm 2 articulates our adoption of the Friedman-Nemenyi pipeline. We highlight the creation of the rank matrix with the appropriate performance metric P (corresponding to algorithm A's family), demonstrate the gate-keeping affect of the Friedman test–failure indicates all measures are statistically indistinguishable, then we quantify the critical difference we use in Nemenyi's post-hoc analysis.

All algorithms (diagnostics) we test learn a concept that computes a pairwise distance matrix (structure recording distances between every point pair) and assigns each data point a class, cluster, or scalar value. The nuance of assignment affects the diagnostic's objective, as we summarize in Table 6.6. We partition these diagnostics into three families: supervised (1-NN, support-vector classifier), unsupervised (spectral clustering, k-medoids, aggregate clustering), and label-separation (RNLDD, weighted $(n-1)$-NN).

6.3.1 Supervised Pipeline

The supervised pipeline applies to all interested in supervised classification and estimates a common applicable concern of practitioners: *can this measure provide reliable predictions?* Statistically, this question is better worded as *how strong is the label-prediction agreement?*

Algorithm 2: Objective-Informed Friedman–Nemenyi Pipeline

Input: Algorithm A, performance metric P, datasets $\mathcal{D} = \{D_1, \ldots, D_B\}$, measures $\mathcal{M} = \{m_1, \ldots, m_C\}$

Output: Rank-ordered list of statistically indistinguishable best measures $\mathcal{M}^*$

`// Populate rank matrix`

Define the $\mathcal{D} \times \mathcal{M}$ rank matrix as $\mathbf{R} \in [1, C]_{\mathbb{R}}^{B \times C}$;

foreach *dataset* $D_i \in \mathcal{D}$ **do**

  Define the performance cache as $s \in \mathbb{R}^C$;

  foreach *measure* $m_j \in \mathcal{M}$ **do**

    $s_j \leftarrow \mathrm{P}(\mathrm{A}(D_i; m_j))$;

  $\mathbf{R}_i \leftarrow \mathrm{rank}(s)$; `// Lower rank is better`

`// Begin Friedman--Nemenyi pipeline`

$p_{\text{Fried}} \leftarrow$ Iman-Davenport-Friedman-Test($\mathbf{R}$);

if $p_{Fried} > 0.10$ **then**

  return $\mathcal{M}$; `// All measures are statistically indistinguishable`

Let $\bar{\mathbf{r}} \in [1, C]_{\mathbb{R}}^{C}$ denote the average rank of each measure over $\mathcal{D}$;

$\mathrm{CD} \leftarrow q_{\alpha,C}\sqrt{\frac{C(C+1)}{6B}}$; `// Uses the Studentized-range statistic` $q_{\alpha,C}$

$\mathcal{M}^* \leftarrow \{\, m \in \mathcal{M} : \bar{\mathbf{r}}_j \leq \min \bar{\mathbf{r}} + \mathrm{CD} \,\}$; `// Measures tied with the best`

return sort($\mathcal{M}^*$ by $\bar{\mathbf{r}}_j \uparrow$); `// Winner group, ordered by average rank`

than *what proportion of predictions are correct?*.

We choose Matthew's Correlation Coefficient (MCC) over accuracy (ACC) since MCC answers the former while ACC answers the latter. The range of MCC values $\in [-1, 1]$, where MCC $= -1$ indicates perfect disagreement, MCC $= 0$ signifies the results are no better than random guessing, and MCC $= 1$ indicates perfect agreement. As supporting evidence of MCC's strengths over accuracy–including disparate majority/minority class sizes, we recommend Chicco et al. (2021). For a short intuition, MCC is a rigorously defined coefficient that is uniquely sensitive to perturbations of any confusion-matrix cell, while accuracy is a ratio that is reducible to a binary (correct/incorrect) function (Briscoe et al., 2025).

Since paired disagreement is the null, our complementary null hypothesis test is the McNemar test (McNemar, 1947). For this significance test, our baseline is the majority label.

6.3.2 Unsupervised Pipeline

The unsupervised pipeline applies whenever the practitioner cares about how well some distance-induced partition of the data aligns with the ground-truth labels, without giving the learning

Table 6.6: Components of the three diagnostic families used in the Friedman-Nemenyi ranking pipeline.

Family	Diagnostic Objective	Performance Metric	Null Hypothesis Test
Supervised	Prediction–Label Agreement	MCC	McNemar Test
Unsupervised	Partition–Label Agreement	ARI	Fisher Test
Structural	Label–Distribution Separability	AUC	Mann–Whitney U Test

algorithm access to those labels. In other words, instead of asking *“can this measure drive a good classifier?”* (supervised), we ask *“if we cluster in this measure’s geometry, do the resulting groups coincide with the labels?”* This may be useful in applications where data comes before the label or when concept drift occurs.

We score partition–label agreement using the chance-corrected clustering metric Adjusted Rand Index (ARI). ARI analyzes pairwise agreements between labels and cluster assignments then rescales to ARI $\in [-1, 1]$ where random partitions have expectation 0, perfect agreement yields ARI = 1, and worse-than-random partitions have ARI < 0. Unlike accuracy on clusters, ARI does not require cluster-label assignment and inherently compensates for label imbalance.

For statistical significance, we use the two-sided Fisher’s test to check for the null hypothesis—which is that there is no association between the two clusters and two labels being compared (Hubert and Arabie, 1985).

6.3.3 Structural Pipeline

The structural (label–separation) pipeline applies whenever the practitioner has a scalar score derived from the distance structure (e.g., a density score or a weighted nearest-neighbor score) and cares about how cleanly this score separates the two label classes. Here the question is not *“can we assign a hard class label?”* or *“do clusters match labels?”* but instead *“can the measure induce a distribution on which positive and negative examples occupy distinguishable regions?”*

We quantify label–distribution separability with AUC (Area Under the Curve). Given a scalar score for each instance and a binary label, AUC assesses how well the groups are separated by pair-wise comparisons. It is inherently threshold-free and invariant to any strictly monotone transformation of the scores (such as scaling). In other words, AUC has no specific pivot and cares about order instead of magnitudes. For insight, we review AUC in Equation 6.1 for classes C_0 and C_1, and structural score function S.

$$\mathrm{AUC}(S) = \frac{\sum_{t_0 \in C_0} \sum_{t_1 \in C_1} \mathbf{1}[S(t_0) < S(t_1)]}{|C_0| \cdot |C_1|} \tag{6.1}$$

However, our label-separability objective is independent of which class typically yields lower scores—$S(t_0) < S(t_1)$ or $S(t_0) > S(t_1)$ is immaterial. So for label invariance, we symmetrize AUC by the transformation AUC $\leftarrow \max(\mathrm{AUC}, 1-\mathrm{AUC})$. A value near 0.5 indicates no useful separation, whereas AUC close to 1 indicates strong separation of the two label distributions.

We use the p-value from the Mann-Whitney U test since this null is that the two label distributions are identical and any apparent separation is due to sampling noise (AUC ≈ 0.5). The Mann-Whitney U test is natural since AUC is equivalent to the normalized Mann-Whitney U statistic (Cortes and Mohri, 2003).

Weighted $(n-1)$-NN.

Our large-scale nearest-neighbor diagnostic considers every data point in its neighborhood $\mathcal{N}(k)$ except k which is the point being predicted:

$$\mathcal{N}(k) = \{1, \ldots, n\} \setminus \{k\}.$$

This neighborhood is then partitioned by label $c \in 0, 1$: $\mathcal{N}(k) = \mathcal{N}_0(k) \cup \mathcal{N}_1(k)$. For each class c, we compute the average distance from k to that class,

$$\mathrm{avg}\left(k, \mathcal{N}_c(k)\right) = \frac{1}{|\mathcal{N}_c(k)|} \sum_{g \in \mathcal{N}_c(k)} \mathbf{D}_{k,g}, \tag{6.2}$$

and define the structural score

$$S_k = \mathrm{avg}\left(k, \mathcal{N}_1(k)\right) - \mathrm{avg}\left(k, \mathcal{N}_0(k)\right). \tag{6.3}$$

Thus, S_k is the signed margin of k's average distance between the two classes. $S_k < 0$ means k averages closer to class 0.

Weighted (n-1)-NN is useful for scenarios demanding a holistic view of class separability. In our studies, it is useful for insight into the tested measures. For applications, it gives insight for anomaly detection studies such as in cyber security.

RNLDD

Our local density-based diagnostic is Rank-Normalized Local Density Divergence (RNLDD). RNLDD is parameterized by $q \in (0, 1)$, the proportion of data defined as local (the neighborhood). With the pair-wise distance matrix $\mathbf{D}$, nearest-neighbors set $\mathcal{N}(k)$, and population size n, we calculate the local reachability density of point k as ρ_k. Next, we obtain the relative density divergence (S_k) by ranking, where the densest point receives rank one:

$$\rho_k = \frac{|\mathcal{N}_q(k)|}{\sum_{\ell \in \mathcal{N}_q(k)} \mathbf{D}_{k,\ell}}, \quad S_k = \frac{\operatorname{rank}(\rho_1, \ldots, \rho_n)_k - 1}{n - 1}.$$

In our studies, RNLDD gives insight on class separability by local densities. In applications, RNLDD could give information in anomaly detection applications by harnessing the local density manifold to identify deviations from normal patterns. For example, a high-value, fraudulent credit card transaction among a cluster of local, low-value purchases.

6.3.4 Diagnostic-Dataset Compatibility Check

When the Friedman test fails to reject the null hypothesis ($p_{\text{Friedman}} > 0.10$) for algorithm A, we check if this failure is influenced by algorithm/dataset incompatibility. Such incompatibility occurs when the algorithm cannot draw a boundary better than random guess or than majority-label guessing for any tested measure.

Algorithm A has a p-value for every dataset/measure pair which can check that the given dataset/measure pair is label informative for A. To check if the dataset is label informative for A, we fuse all p-values over the measure dimension using the harmonic mean to control for family-wise error under dependence (Wilson, 2019). Each dataset j then has a corresponding harmonic-mean p-value called HMP_j.

6.4 Results

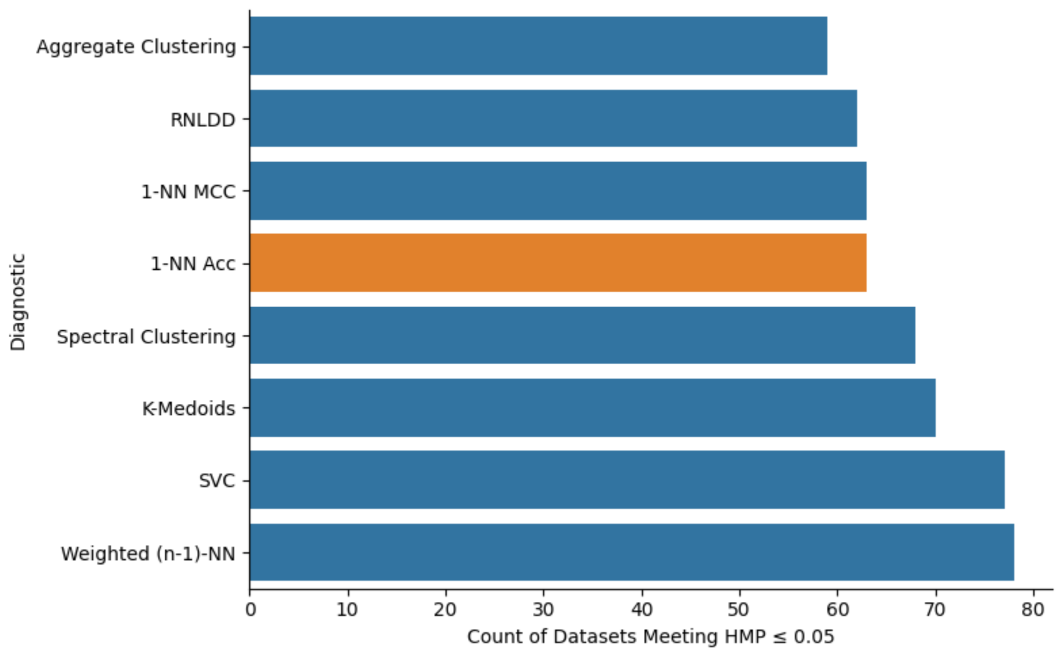


Figure 6.1: Comparing dataset compatibility for each diagnostic. We assess by comparing each dataset's harmonic mean p-value (HMP) to 0.05.

We begin by reviewing diagnostic-dataset compatibilities for all algorithms and dataset compatibilities in Figure 6.1. Intuitively, this graph counts how many datasets were statistically significant in improvement over a trivial baseline such as random guessing or a majority label.

In Figure 6.2, we see that MCC consistently yields more Friedman statistical power than accuracy. This indicates that MCC provides sharper discrimination between distance measures. Next, we show that 1-NN with MCC yields more statistical power than all other objectives in Figure 6.3.

Figure 6.4 illustrates each diagnostic's similarity to the baseline "1-NN with accuracy" by comparing how quick the baseline's top five measures (the "target-bag") are reached over each diagnostic. The x-axis represents the "X" highest ranking measures of the respective diagnostic, and the y-axis is the percent of measures from the target-bag also seen in the diagnostic's "Top-X". For example, "1-NN with MCC" has the coordinate (3,60); this means that 60% of the

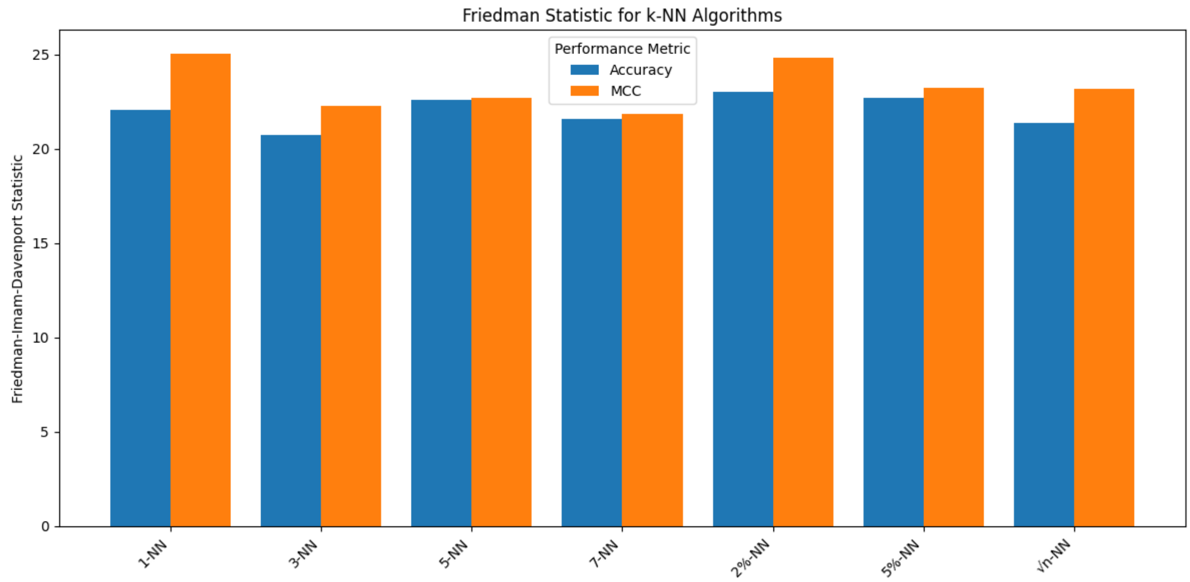


Figure 6.2: Comparison of Friedman test outcomes for k-NN diagnostics for $k \in \{1, 3, 5, 7, 0.02n, 0.05n, \sqrt{n}\}$. MCC consistently has more statistical power than accuracy.

target-bag's measures are found in this diagnostic's top 3 measures.

We present the rankings by the traditional 1-NN with accuracy in Figure 6.5. This figure is a "critical difference diagram" (CDD). In a CDD, the average ranking of each measure over all datasets is presented along a ranking dimension, and "critical differences" (obtained via Nemenyi's post-hoc analysis) are shown by black bars.

For the supervised family, we highlight the critical difference diagrams for 1-NN and Support Vector Classifier in Figure 6.6. Both algorithms are assessed with MCC over "leave-one-out" cross validation. For the unsupervised family, we present the CDD for K-Medoids in Figure 6.7, the CDD for aggregate clustering in Figure 6.8, and the CDD for spectral clustering in Figure 6.9. For the structural family, we present RNLDD and weighted (n-1)-NN in Figure 6.10.

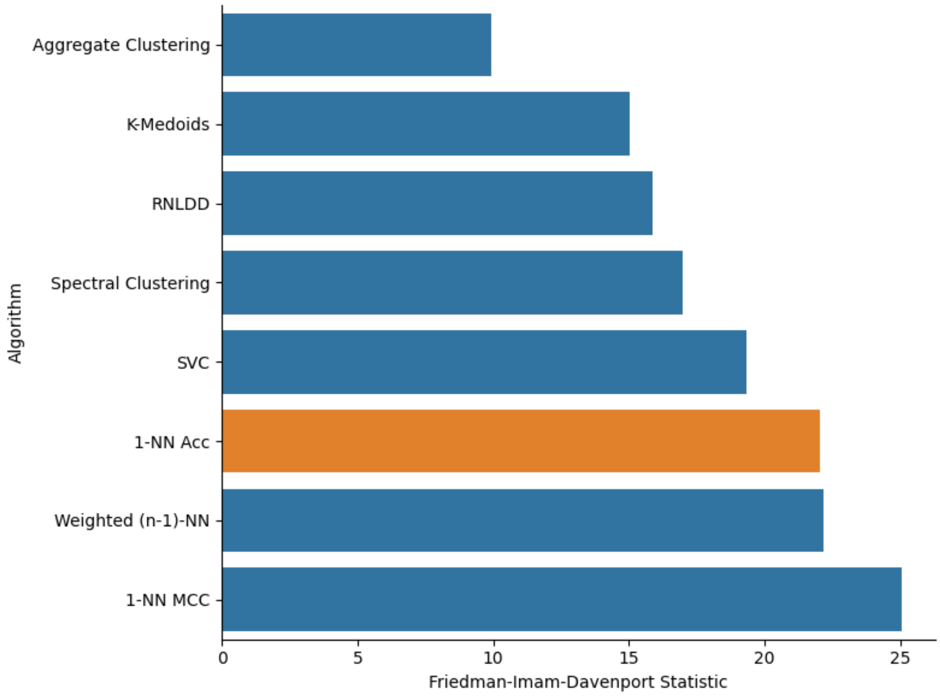


Figure 6.3: Comparison of Friedman-Imam-Davenport statistics for various diagnostics.

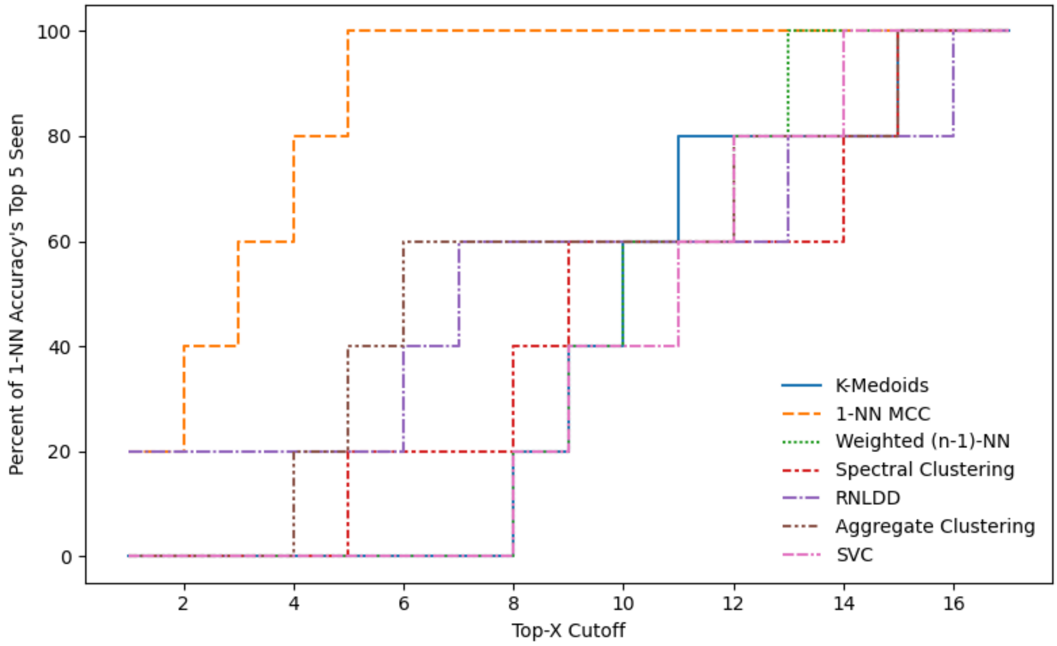


Figure 6.4: Top-X overlap with the top five measures from the 1-NN with Accuracy diagnostic baseline. The curve shows the fraction of measures in the baseline's top 5 that also appears within the given diagnostic's top-X measures. Lines that rise earlier (high overlap at small X) indicate stronger agreement with the baseline, while late-rising curves indicate poorer transferability from the baseline's top rankings.

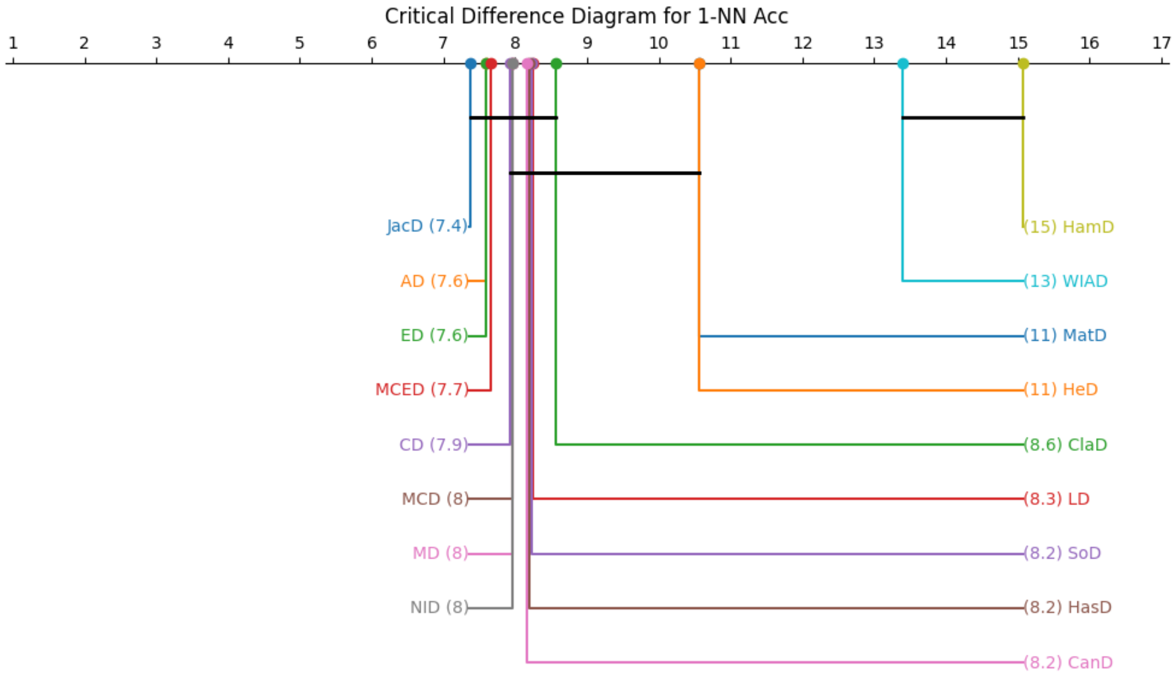
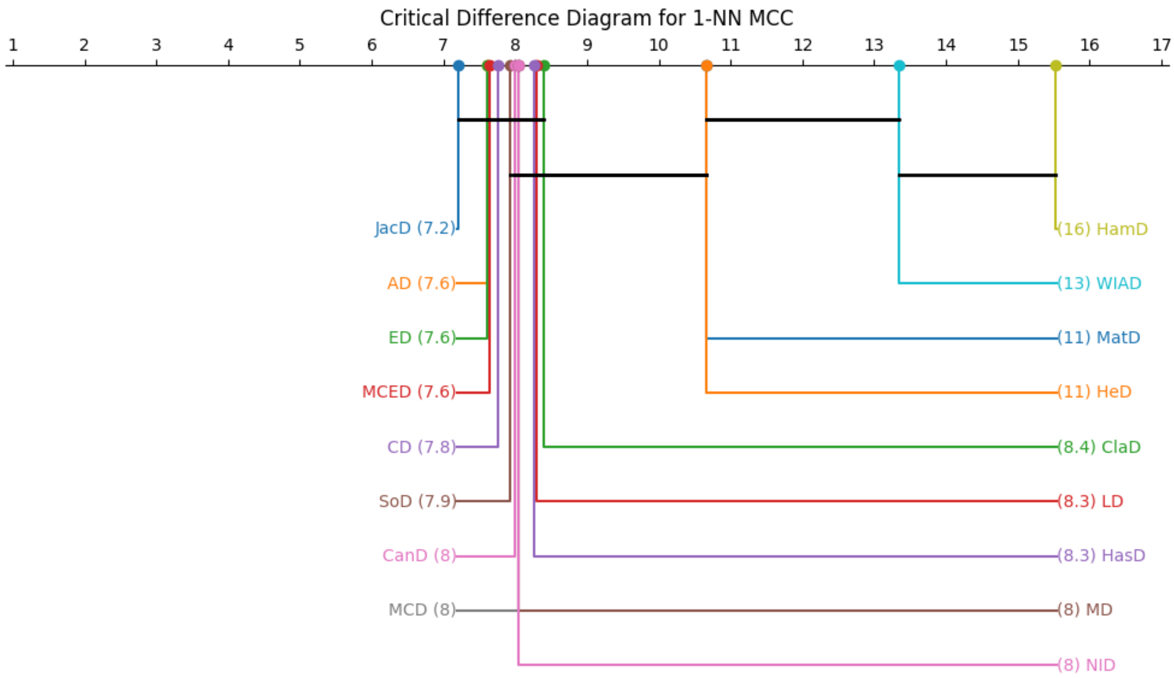
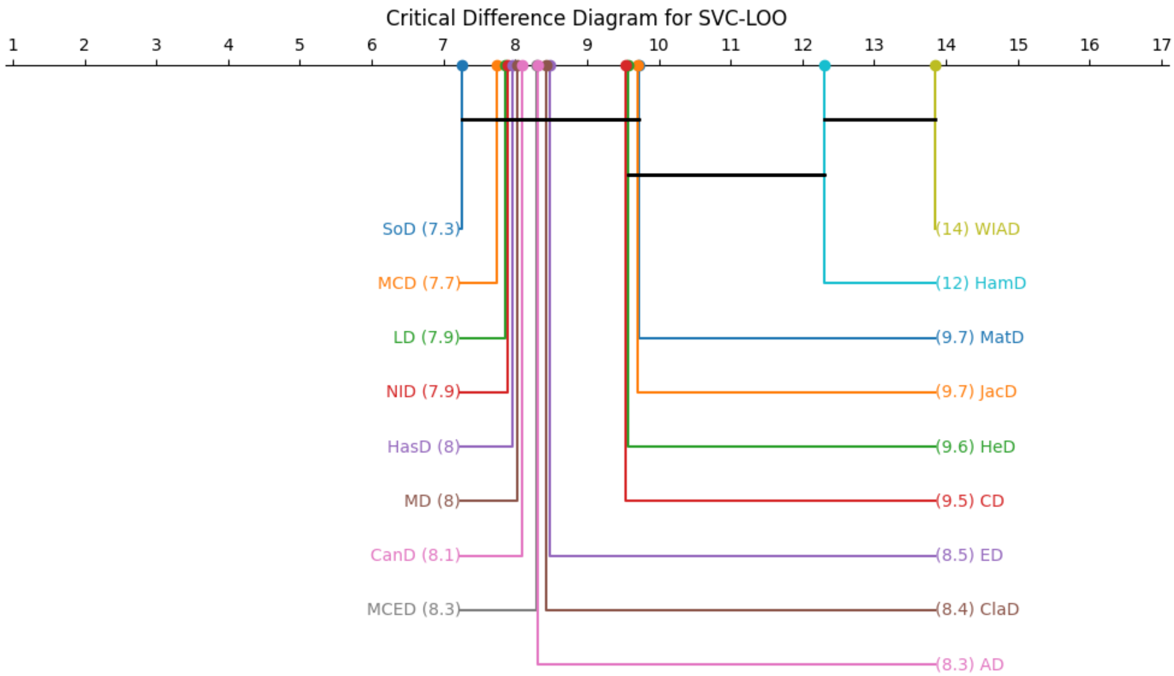


Figure 6.5: The Critical Difference Diagram for the standard 1-NN with Accuracy Diagnostic. The top 13 measures are statistically indistinguishable per Nemenyi's critical difference.



(a) Leave-One-Out 1-Nearest Neighbor with MCC Diagnostic



(b) Leave-One-Out Support Vector Classifier with MCC Diagnostic

Figure 6.6: The critical difference diagrams for two supervised diagnostics using the MCC performance metric. Each subfigure shows the average measure rank over 81 datasets. The black bars are statistically indistinguishable groups.

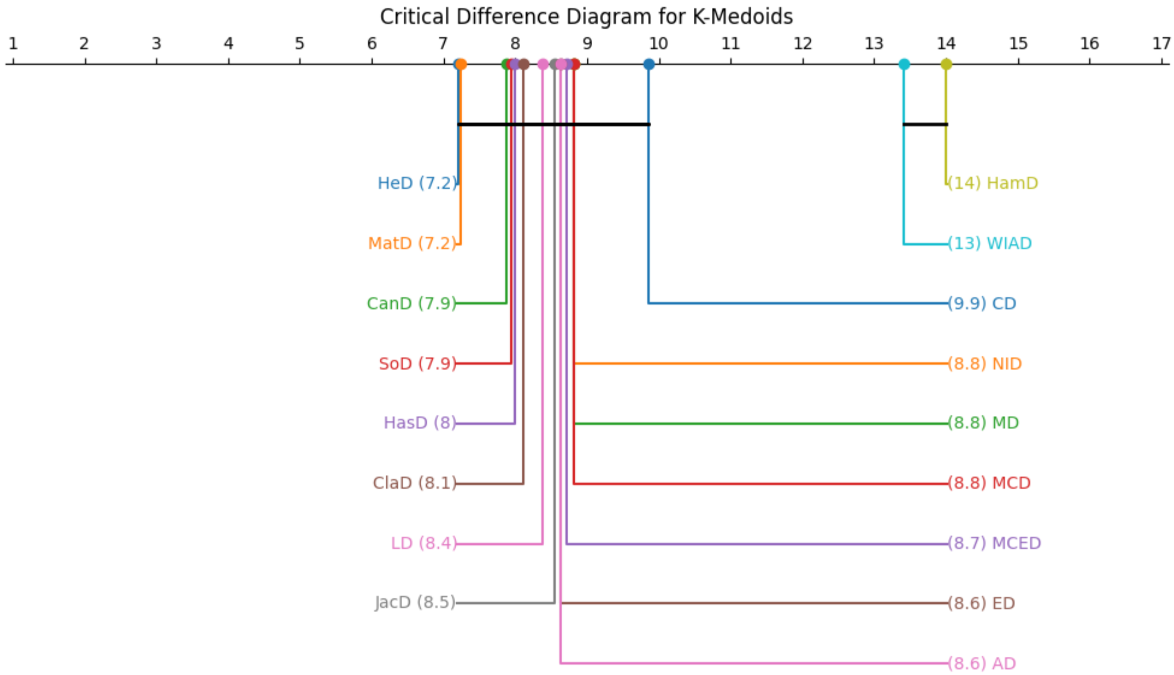


Figure 6.7: The critical difference diagram for the "k-medoids with ARI" diagnostic. Each measure's average rank over 81 datasets is given. The black bars are statistically indistinguishable groups.

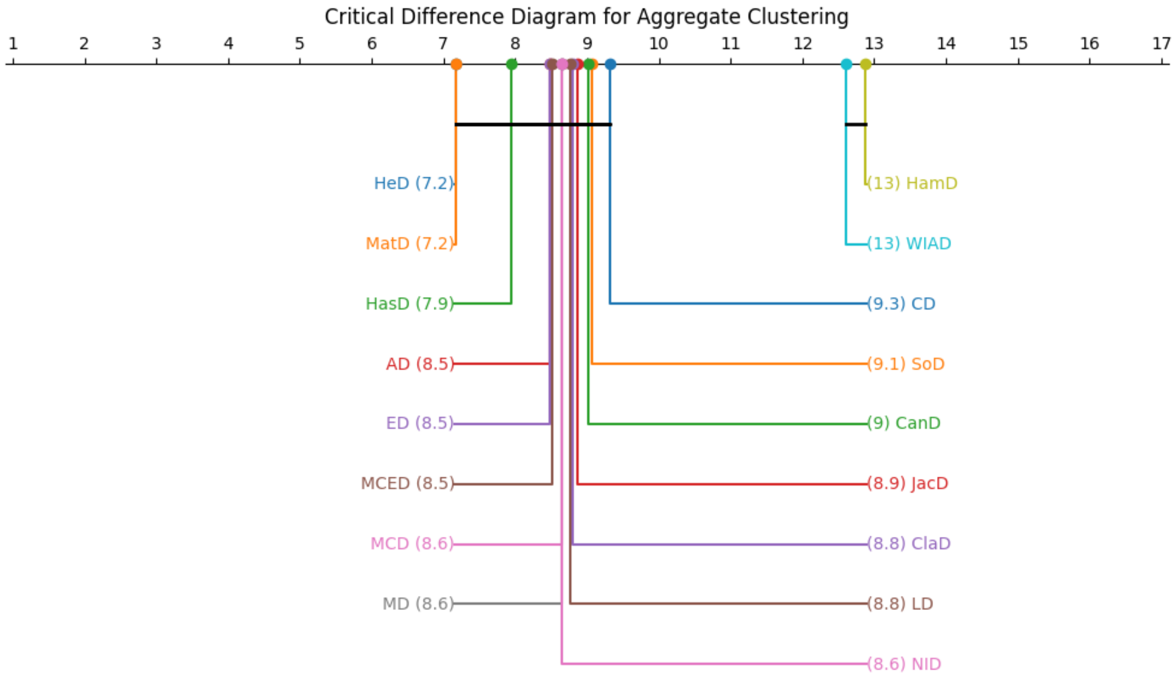


Figure 6.8: The critical difference diagram for "aggregate clustering with ARI" diagnostic. Each measure's average rank over 81 datasets is given. The black bars are statistically indistinguishable groups.

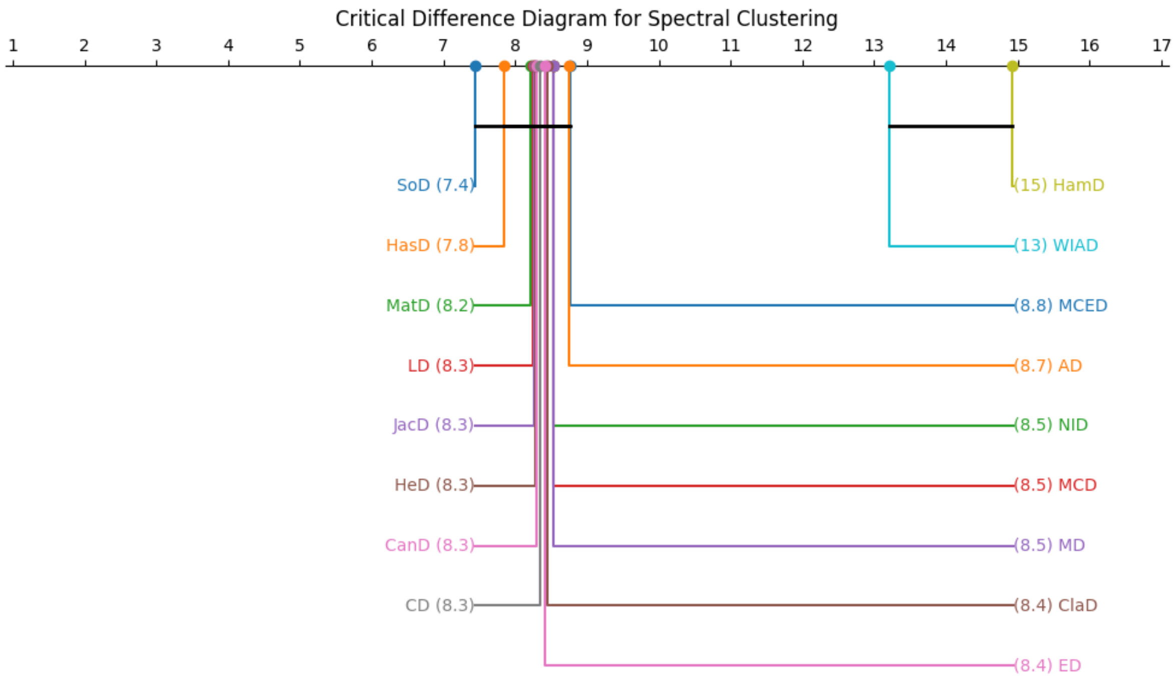
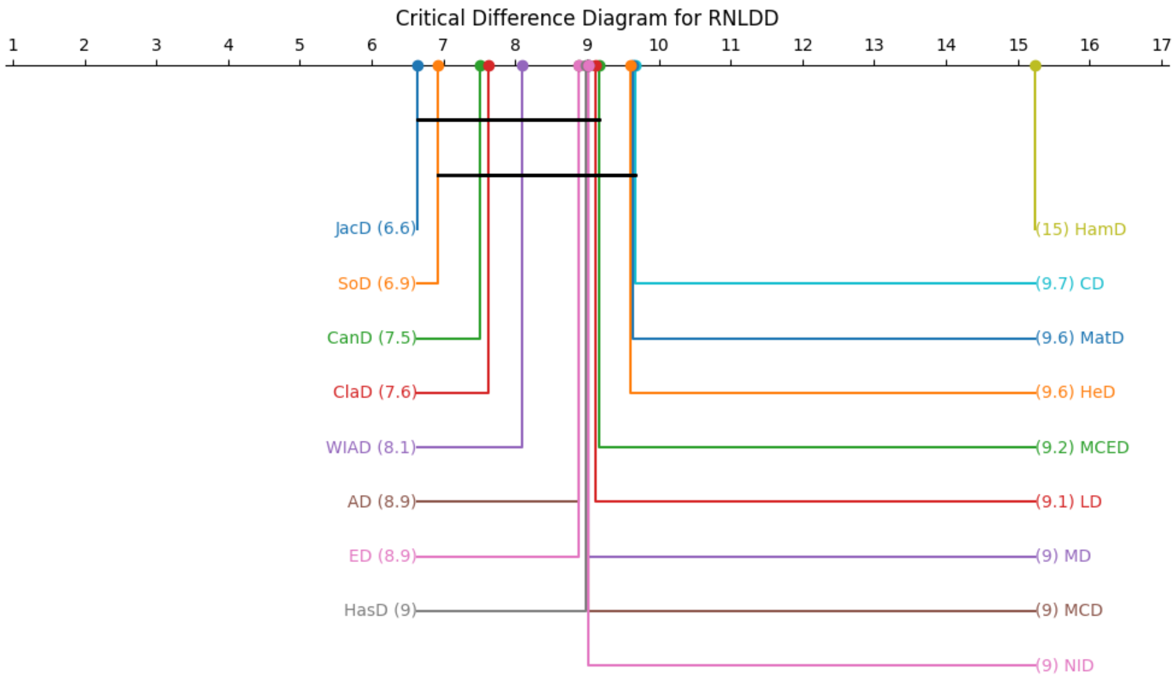
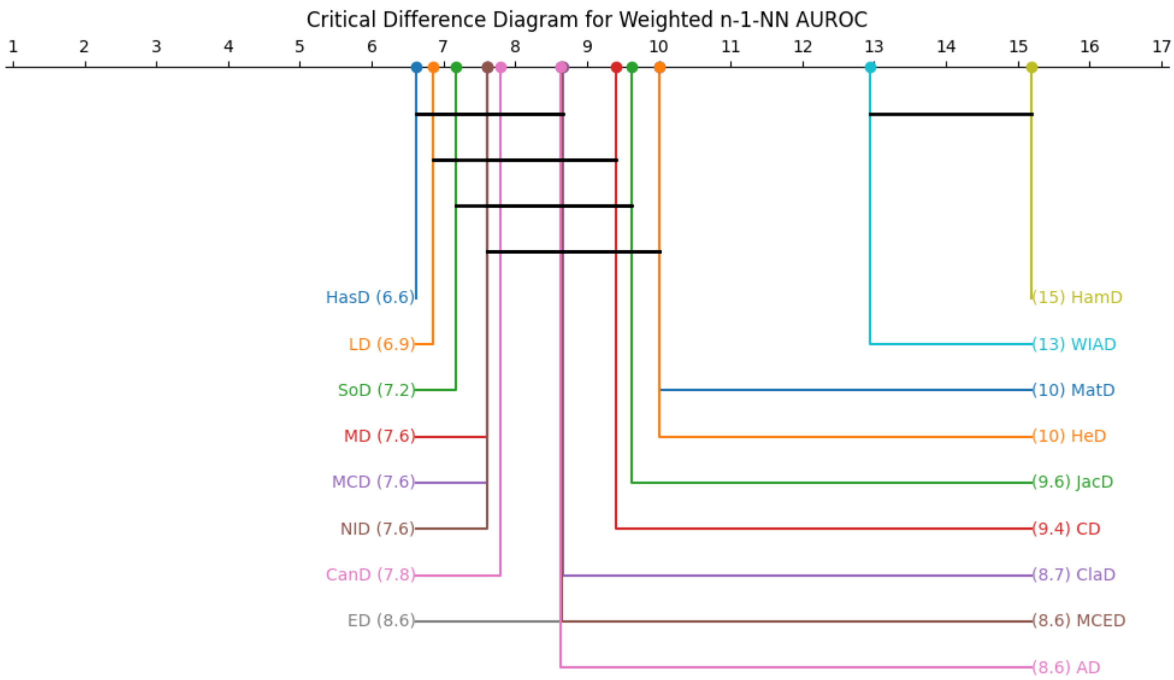


Figure 6.9: The critical difference diagram for "spectral clustering with ARI" diagnostic. Each measure's average rank over 81 datasets is given. The black bars are statistically indistinguishable groups.



(a) RNLDD with AUC Diagnostic. RNLDD is a local density-based algorithm.



(b) Leave-One-Out Weighted $(n-1)$-NN with AUC Diagnostic

Figure 6.10: The critical difference diagrams for two structural diagnostics using the AUC performance metric. Each subfigure shows the average measure rank over 81 datasets. The black bars are statistically indistinguishable groups.

6.5 Discussion

Recall that a diagnostic is compatible with a dataset if we can reject the joint-null hypothesis (HMP$\leq$ 0.05). In other words, after considering all measures, the diagnostic likely performs better than the relevant baseline (random chance or majority-label guessing). In Figure 6.1, we see that all diagnostics are compatible with at least 59/81 (>71%) datasets. Aggregate clustering (59) and RNLDD (62) are compatible with the fewest datasets, while SVC (77) and Weighted (n-1)-NN (78) are compatible with the most. As for the other null-hypothesis testing, the global Friedman test that checks if rankings are reasonably far away from the expected average ranking, all diagnostics passed.

As demonstrated in Figure 6.2, MCC consistently yields more statistical power than accuracy for all k-NN tested (for $k \in \{1, 3, 5, 7, 0.02n, 0.05n, \sqrt{n}\}$). Figure 6.3 reflects on statistical power for each selected algorithm in our pipeline; here, only 1-NN with MCC and Weighted (n-1)-NN have more statistical power (measure-rank separability) than 1-NN with accuracy.

In Figure 6.4, we see that the top five MCC values for 1-NN are identical to those based on 1-NN accuracy. However, the other six diagnostics are more distant: only K-Medoids has *any* of the target-bag (1-NN with accuracy's top five measures) in its top three, and the first algorithm to see the entire target-bag is Weighted (n-1)-NN, achieving only after most measures (13/17) are seen.

Now let us consider a null hypothesis: that diagnostics have no similarity to 1-NN with accuracy. This dissimilarity would be apparently random. If we expect a uniform-random distribution in Figure 6.4 with 17 measures, we expect about 1/17 increase in the available y-axis for every step on the x-axis. That is, we expect $1/17 \approx 5.9\%$ of the target-bag to fill for every increase of the top X measures of the respective diagnostic (seen by line color). For example, we would expect to see about 50% of the target-bag once half the measures (8.5) are seen. Figure 6.4 paints a similar picture–the other diagnostics cluster around (8.5, 50). Additionally, other key points suggesting a random distribution such as ($0.4 \times 17 = 6.8$, 40) and ($0.6 \times 17 = 10.2$, 60) are near the center mass of these non-1-NN diagnostics. Finally, there doesn't appear to be consistency in one diagnostic being left or right of the center mass–RNLDD begins as the leftmost (most similar in early Top-X trials) and ends as the rightmost (least similar in late Top-X trials). The other diagnostic with high similarity in the early Top-X trials (seeing 60% of the target-bag by X=6), aggregate clustering, similarly becomes more dissimilar when compared to other diagnostics (100% at X=15).

Reflecting on the time series measures over the CDDs (Figures 6.5–6.10), Jaccard distance (JacD) performs best for 1-NN under both accuracy (average rank $\bar{r}$ = 7.4) and MCC ($\bar{r}$ = 7.2), while Soergel distance (SoD) is best for the support vector classifier ($\bar{r}$ = 7.3). For the structural diagnostics (evaluated by AUC), the winner for the global-scale diagnostic, weighted $(n - 1)$-NN, is Hassanat distance (HasD; $\bar{r}$ = 6.6); while the local density diagnostic, RNLDD, is JacD ($\bar{r}$ = 6.6), with SoD closely following as second ($\bar{r}$ = 6.9). For unsupervised diagnostics, Hellinger (HeD) and Matusita (MatD) distances are K-medoids' *and* aggregate clustering's best performers (all with $\bar{r}$ = 7.2); while SoD leads spectral clustering ($\bar{r}$ = 7.4). At the bottom end, Hamming distance (HamD) is consistently in last or second-to-last place.

6.6 Conclusion

In conclusion, the "best" measure is clearly diagnostic-dependent. Rankings induced by leave-one-out (LOO) 1-NN with accuracy do not reliably transfer to other diagnostic objectives in our benchmarking. We recommend careful consideration of your learning downstream task before accepting benchmark results. This includes diagnostic-dataset compatibility: by considering fused p-values for each dataset, we find that LOO 1-NN is compatible with fewer datasets than LOO support-vector classifier or LOO weighted $(n - 1)$-NN diagnostics.

As for future benchmarking driven by 1-NN, we recommend tracking MCC estimates and reporting these induced rankings alongside accuracy. We find that MCC consistently yields more Friedman statistical power than accuracy. In other words, MCC causes a more decisive rank separation among measures–potentially reducing statistically indistinguishable ambiguity. More broadly, we suggest running objective-informed rankings across multiple diagnostic families to provide a safer basis for measure selection.

Our results signal several directions for further research. First, analyses can be done in multivariate or multiclass settings. However, computational expense is a hurdle as using just one diagnostic, LOO 1-NN, benchmarking can take months of server compute. Another next step is incorporating more computationally expensive measures, such as kernel dynamic time warping. Furthermore, one could investigate partitioning more diagnostics in different ways (besides supervised/unsupervised/structural) to theorize the generalizability of the respective partitions.

Overall, our central takeaway is that measure benchmarking should be objective informed with a proper performance metric. One cannot assume 1-NN generalizes—despite the intuitive idea of “the closest data should typically have the same label”.

Chapter 7

CONCLUSIONS

Throughout this dissertation, the underlying theme argues that accountability in artificial intelligence depends on more than a model's predictive success. It must also depend on whether fairness is defined in a principled way, whether evaluation metrics remain reliable under realistic data conditions, whether metric choices are proper and justified, and whether uncertainty is material or known. Furthermore, this dissertation holds the common theme that models should not be the only objects of scrutiny. The measurements, comparisons, and conclusions built around them must also be critically examined.

The dissertation addresses this broader problem through four connected contributions. First, Chapter 3 introduces the *Objective Fairness Index* (OFI), a fairness metric grounded in the legal logic of objective testing. OFI distinguishes between disparity caused by the algorithm and disparity that may instead reflect external (possibly societal) influences. Accordingly, we provide a clearer rationale for fairness assessment in binary classification settings where the central question is whether the system itself is acting discriminatorily.

Second, Chapter 4 reframes fairness metrics through the lens of social choice theory. By treating fairness metrics as aggregation rules over groups, we develop a taxonomy of fairness metrics, adapt social-choice-style axioms to binary classification, and identify important structural tensions. The resulting analysis shows that fairness metrics inherently carry assumptions or pathologies. They encode different commitments, satisfy different properties, and can exhibit pathologies such as aggregation reversals or impossibility trade-offs. This work provides a more principled basis for selecting fairness metrics.

Third, Chapter 5 shows that evaluation metrics themselves can become unreliable in small or imbalanced groups. Many confusion-matrix-based metrics exhibit instability, jaggedness, or undefined behavior when sample sizes are limited. Because these are exactly the conditions that often arise in subgroup fairness analysis, this is more than a technical inconvenience; it is a hazard to trustworthy evaluation. To address this problem, the chapter introduces the MATCH test and Cross-Prior Smoothing. The MATCH test gives the probability of achieving a given score with a reference distribution. Cross-Prior Smoothing uses a reference distribution to increase metric reliability in small sample sizes. These contributions shift attention from raw metric values alone to whether those values are sufficiently reliable to support a conclusion.

Fourth, Chapter 6 examines accountability in benchmarking itself. In that literature, leave-one-out 1-nearest-neighbor evaluated by accuracy is often treated as a default proxy for identifying strong time-series similarity measures. Here, we show that such rankings generalize poorly across other objectives, and that conclusions about the "best" measure depend strongly on the utilized diagnostic. we also show that Matthews Correlation Coefficient often yields sharper statistical discrimination than accuracy when ranking time-series similarity measures. The broader implication is that benchmarking pipelines, like fairness assessments, must be justified relative to the problem at hand.

In summary, these contributions support a single claim: accountable AI requires accountability in evaluation. A model cannot be considered accountable merely because it scores well on a familiar benchmark or satisfies some fairness metric. One must also ask whether the fairness construct is appropriate, whether the metric behaves reliably, whether its formal properties are acceptable, and whether uncertainty has been honestly represented. In this sense, accountability is not only a property of systems, but also of the evaluative frameworks used to judge them.

Several practical lessons follow from this work. Fairness assessment should begin with the conscious consideration of what kind of unfairness is material. Group comparisons should be treated cautiously when their sizes are small, since apparent differences may reflect metric instability of discrete values rather than genuine disparities. Benchmarking studies should carefully consider the diagnostic in use, especially when alternative objectives lead to materially different rankings. More broadly, uncertainty should be consistently considered in analyses.

This dissertation also suggests several directions for future work. OFI can be extended beyond binary classification to multiclass, regression, and ranking settings. The social-choice-theoretic analyses can be pushed further into multiclass fairness, recommender systems, and ensemble decision settings, where aggregation also applies. The theory and practice of MATCH and CPS can be extended with stronger approximations, additional uncertainty modeling, and broader empirical validation. Objective-informed benchmarking can likewise be expanded to multivariate time series, more computationally expensive similarity measures, and other diagnostic families beyond the ones studied here. A broader long-term direction is to unify these lines of work into a more general meta-science theory of evaluation for machine learning. This is a theory that studies not only models, but also the assumptions and failure modes of the comparative metrics used.

In closing, the central message is simple. Artificial intelligence becomes more accountable when its evaluations become more accountable. Fairness definitions should be principled, metric choices should be justified, empirical comparisons should remain reliable under realistic conditions, and uncertainty should be central in analyses. These directions contribute to shaping what researchers, practitioners, and institutions are entitled to conclude from the systems they build.

APPENDIX

Chapter A

DEFERRED PROOFS FOR CHAPTER 4

This chapter presents the necessary proofs for the results in Chapter 4.

A.1 Intra-Groups Metrics and Properties

Unless otherwise specified, we use $\vec{w} := [1, -1, 0, 0]$ for NMC and $\vec{w} := [1, 1, 0, 0]$ for MC.

We write $A = (\mathrm{TP}, \mathrm{FN}, \mathrm{FP}, \mathrm{TN})$ and, where useful, $A^+ = A + e_c$ for the matrix obtained by adding a single instance to cell $c \in \{\mathrm{TP}, \mathrm{FN}, \mathrm{FP}, \mathrm{TN}\}$. Recall the intra-group classes from Group Proportions:

$$\begin{aligned}
\mathrm{JRM} &= \left\{\frac{c_i}{c_i + c_j} : i \neq j\right\},\\
\mathrm{CPD} &= \left\{\frac{c_i + c_j}{n} : i \neq j\right\},\\
\mathrm{CPT} &= \left\{\frac{c_i - c_j}{n} : i \neq j\right\},
\end{aligned}$$

and the intra-group classes from Group Counts:

$$\begin{aligned}
\mathrm{MC} &= \left\{\vec{c} \cdot \vec{w} : \vec{w} \in \mathbb{R}^4_{\geq 0}\right\},\\
\mathrm{NMC} &= \{\vec{c} \cdot \vec{w} : \vec{w} \in \mathbb{R}^4 \setminus \mathbb{R}^4_{\geq 0}\}.
\end{aligned}$$

A.1.1 π-Involution Symmetries.

Recall the three self-inverse permutations

$$\begin{aligned}
\pi_{\text{corr}} &: (\text{TP}, \text{FN}, \text{FP}, \text{TN}) \mapsto (\text{TN}, \text{FP}, \text{FN}, \text{TP}),\\
\pi_{\text{pred}} &: (\text{TP}, \text{FN}, \text{FP}, \text{TN}) \mapsto (\text{FN}, \text{TP}, \text{TN}, \text{FP}),\\
\pi_{\text{label}} &: (\text{TP}, \text{FN}, \text{FP}, \text{TN}) \mapsto (\text{FP}, \text{TN}, \text{TP}, \text{FN}),
\end{aligned}$$

and the polarity map polrev (sign flip for signed metrics, $1 - x$ for $[0, 1]$ metrics).

Lemma A.1 (JRM: when complements appear)**.** *Fix* $g(A) = \dfrac{c_i}{c_i + c_j} \in \text{JRM}$ *with midpoint* $m = \frac{1}{2}$*. For any involution* $\pi \in \{\pi_{corr}, \pi_{pred}, \pi_{label}\}$*,*

$$g(\pi(A)) = \text{polrev}\big(g(A)\big) \iff \{c_i, c_j\} \text{ is mapped by } \pi \text{ to } \{c_j, c_i\} \text{ (pairwise swap).} \tag{A.1}$$

Proof. If π swaps the pair (c_i, c_j), then $g(\pi(A)) = \dfrac{c_j}{c_j + c_i} = 1 - \dfrac{c_i}{c_i + c_j} = \text{polrev}(g(A))$. Conversely, if π leaves the ordered pair unchanged or maps it to a different pair, then $g(\pi(A)) \in \{g(A),\ c_{i'}/(c_{i'} + c_{j'})\}$ with $\{i', j'\} \neq \{i, j\}$; in neither case is $g(\pi(A))$ identically $1 - g(A)$ over the domain. □

Corollary A.1. *We identify all unordered cell pairs necessary as arguments for a JRM to satisfy the given symmetry in Table A.1. Each symmetry contains exactly four unique possibilities of ordered cell arguments.*

Corollary A.2. *Similarly, any metric in CPT or CPD satisfies the given symmetry provided by Table A.1.*

Group Counts Metrics in MC or NMC are undefined for symmetric properties since their polarity reversal is undefined. Polarity reversal is defined for GP since all are normalized by definition. However, with scale being material, it is not just to define a default polarity reversal at all—how could one argue that $f = \text{TP} + \text{FN}$ has a polarity reversal without domain context?

Table A.1: A metric in GP $\triangleq$ JRM $\cup$ CPT $\cup$ CPD satisfies the given symmetry iff it is a function of one of the following unordered cell pairs.

Symmetry Held	Unordered Cell Pairs
π_{corr}	(TP, TN) or (FP, FN)
π_{pred}	(TP, FN) or (TN, FP)
π_{label}	(TN, FN) or (TP, FP)

A.1.2 Data-Augmentation Stability / Participation

Lemma A.2. *Participation only holds if one can replace cell counts with normalized rates.*

Corollary A.3. *All g in Group Proportions satisfy participation.*

Corollary A.4. *No g in Group Counts satisfies participation.*

A.1.3 Universal Domain

Recall that we restrict our analyses to $n \geq 1$—at least one cell in the confusion matrix must be positive.

Lemma A.3. *Any rate where the denominator must be greater than zero satisfies UD.*

Corollary A.5. *CPD and CPT satisfy UD; JRM does not.*

Lemma A.4. *Real numbers are closed under addition and multiplication.*

Corollary A.6. *MC and NMC satisfy UD.*

A.1.4 Non-Dictatorship

Lemma A.5. *CPT satisfies non-dictatorship.*

Proof. Defined with the form $(c_i - c_j)/n \in [-1, 1]$, and $m = 0$, it stands that

$$g = m \iff c_i = c_j, \tag{A.2}$$

$$g < m \iff c_i < c_j, \tag{A.3}$$

$$g > m \iff c_i > c_j. \tag{A.4}$$

Since an individual must be assigned to only one cell, assigning the individual to cells c_l or c_k is independent of the side of m that g resides.

Furthermore, by discreteness,

$$\max(c_i \mid c_i < c_j) = c_j - 1 \quad \max(c_j \mid c_i > c_j) = c_i - 1.$$

Therefore, an increment of c_i for $c_i < c_j$ can at most cause $c_i = c_j$. Similarly for c_j.

Finally, CPT satisfies Universal Domain so $g = \perp$ is an impossible antecedent. □

Lemma A.6. *CPD satisfies non-dictatorship.*

Proof. Defined with the form $(c_i + c_j)/n \in [0, 1]$, and $m = 0.5$, it stands that

$$g = m \iff c_i + c_j = c_k + c_l, \tag{A.5}$$

$$g < m \iff c_i + c_j < c_k + c_l, \tag{A.6}$$

$$g > m \iff c_i + c_j > c_k + c_l. \tag{A.7}$$

Let $x = c_i + c_j$ and $y = c_k + c_l$. Then by discreteness,

$$\max(x \mid x < y) = y - 1, \quad \max(y \mid x > y) = x - 1$$

Therefore, an increment in x when $x < y$ can at most cause $x = y$. Similarly for an increment in y when $x > y$. □

Corollary A.7. *JRM does satisfies Non-Dictatorship.*

Lemma A.7. *Group Counts do not satisfy Non-Dictatorship.*

Proof. MC and NMC are both defined as linear combinations with weights having no upper bound. So, if $g(A) < m$ and a cell with a weight greater than $g(A) - m$ is incremented, then $g(A^+) > m$. □

A.2 Inter-Group Metrics and Properties

In this section, we review all SEP inter-group metrics for CPD, CPT, JRM, and MC intra-group classes. MEP inter-group metrics and the NMC intra-group class is left for future work.

A.2.1 Direction

First, we assume that all intra-group metrics are indicative of bias since metrics in PGM are intended to identify or quantify bias.

Lemma A.8. *SEP-E do not satisfy Direction.*

Proof. By definition, a metric M cannot satisfy Direction if $M(A, B) > p$ or $M(A, B) < p$ is impossible. □

Lemma A.9. *SEP-D and SEP-R satisfy Direction.*

Proof. SEP-D has constant $p = 0$ and SEP-R has constant $p = 1$. If $g(A) > g(B)$, SEP-D and SEP-R resolve as $M > p$. If $g(A) < g(B)$, SEP-D and SEP-R resolve as $M < p$. □

A.2.2 Skew-Symmetry

Lemma A.10. *SEP-E and SEP-R do not satisfy Skew-Symmetry.*

Proof. These classes can produce positive values but never can produce a negative value. □

Lemma A.11. *SEP-D satisfies Skew-Symmetry.*

Proof. By definition, $\varphi(A, B; g) = g(A) - g(B)$. Hence, $M(A, B) = g(A) - g(B)$ and $M(B, A) = g(B) - g(A)$. Therefore, $M(A, B) = -M(B, A)$. □

A.2.3 Non-Imposition

Lemma A.12. *SEP-D and SEP-R satisfy Non-Imposition.*

Proof. All intra-group metrics have infinitely many distinct values as dataset size n goes to infinity. Intuition: for Group Proportions, set the numerator then increment a cell that is uniquely in the denominator. For Monotonic Counts, increment any cell with a non-zero weight.

Now, let us fix some reference group i (a fixed confusion matrix). Define π as a strict ranking of the other groups $\{G_1, \ldots, G_m\}$. We know that g attains infinitely many defined values from above, so we can select m confusion matrices such that

$$g(G_{\pi[1]} > g(G_{\pi[2]}) > \cdots > G_{\pi[m]},$$

all with $g \neq \perp$. Since $g(i)$ is constant, either subracting (SEP-D) or dividing (SEP-R) by $g(i)$ preserves strict order. We give an example for SEP-D:

$$M(G_{\pi[1]}, i) = g(G_{\pi[1]}) - g(i) > \cdots\cdots > g(G_{\pi[m]}) - g(i) = M(G_{\pi[m]}, i). \tag{A.8}$$

□

Lemma A.13. *SEP-E fails Non-Imposition.*

Proof. By definition, $M \in$ SEP-E has range $\{0, 1, \perp\}$. Hence, with reference group i, a strict ordering is not possible if there is more than two non-reference groups. □

A.2.4 Non-Dictatorship

Lemma A.14. *SEP-D and SEP-R fail non-dictatorship for GP metrics.*

Proof. Non-Dictatorship disallows a single increment to flip the direction of bias. In SEP-D and SEP-R, this is violated when $g(A^+) < g(B) < g(A)$. Here is an example for SEP-D (recall that SEP-D has $p = 0$).

$$M(A, B) = g(A) - g(B) \implies \begin{cases} M(A, B) > p \iff g(A) > g(B) M(A, B) < p \iff g(B) > g(A). \end{cases} \tag{A.9}$$

So, the bias direction flips if $g(A) > g(B)$ and $g(B) > g(A^+)$ (or vice versa). This case occurs when each group's estimate g is sufficiently close. For example, let us consider $M = \frac{\mathrm{TP}+\mathrm{TN}}{n} \in \mathrm{CPD}$ and fix any integer $X \geq 1$.

$$A = (\mathrm{TP} = X, \mathrm{FN} = 0, \mathrm{FP} = 0, \mathrm{TN} = 0),$$
$$A^+ = (\mathrm{TP} = X, \mathrm{FN} = 1, \mathrm{FP} = 0, \mathrm{TN} = 0),$$
$$B = (\mathrm{TP} = 2X + 1, \mathrm{FN} = 1, \mathrm{FP} = 0, \mathrm{TN} = 0).$$

$$\begin{aligned} \text{Now, } g(A^+) &= \frac{X}{X+1} = \frac{X+0.0}{X+1} < 1, \\ g(B) &= \frac{2X+1}{2X+2} = \frac{X+0.5}{X+1} < 1, \\ \text{and } g(A) &= \frac{X}{X} = 1. \end{aligned}$$

Thus, $g(A^+) < g(B) < g(A)$.

This logic is applicable to all GP intra-group metrics for SEP-D and SEP-R. □

Lemma A.15. *SEP-D and SEP-R fail non-dictatorship for MC metrics.*

Proof. We prove by counterexample. For integer $X \geq 0$, let $M(A, B) = 3\mathrm{TP} + \mathrm{FN}$.

$$A = (\mathrm{TP} = 0, \mathrm{FN} = 0, \mathrm{FP} = 0, \mathrm{TN} = X),$$
$$A^+ = (\mathrm{TP} = 1, \mathrm{FN} = 0, \mathrm{FP} = 0, \mathrm{TN} = X),$$
$$B = (\mathrm{TP} = 0, \mathrm{FN} = 1, \mathrm{FP} = 0, \mathrm{TN} = X).$$

Thus, $g(A) = 0 < g(B) = 1 < g(A^+) = 3$. □

Lemma A.16. *Non-Dictatorship is vacuously satisfied for SEP-E.*

Proof. SEP-E has $M \in \{0, 1, \perp\}$ and $p = 1$. So, the antecedent $M(A, B) > p$ is never held and $M(A, B) < p \implies M(A^+, B) \leq p$ is held trivially. □

A.2.5 Condorcet Consistency

Recall Definition 4.11: for mutually distinct groups $A, B, C \in \mathcal{G}$,

$$M(A,B) > p \ \wedge \ M(B,C) > p \implies M(A,C) > p,$$

whenever the three comparisons are defined.

Lemma A.17. *SEP-D is Condorcet consistent (whenever defined).*

Proof. Assume $M(A,B) > 0$ and $M(B,C) > 0$ (antecedents). Then, Condorcet consistency is the consequent of $M(A,C) > 0$. Using the definition of SEP-D, we expand the antecedents: $g(A) - g(B) > 0$ and $g(B) - g(C) > 0$. Then, adding the inequalities yields

$$\big(g(A) - g(B)\big) + \big(g(B) - g(C)\big) \ = \ g(A) - g(C) \ > \ 0.$$

Since $M(A,C) = g(A) - g(C)$, we prove Condorcet consistency. □

Lemma A.18. *SEP-R is Condorcet consistent (whenever defined).*

Proof. Recall that SEP-R has $p = 1$. Assume $M(A,B) > 1$ and $M(B,C) > 1$. Expanding terms: $\frac{g(A)}{g(B)} > 1$ and $\frac{g(B)}{g(C)} > 1$ (both are strictly positive). To obtain $\frac{g(A)}{g(C)}$, let us multiply the inequalities:

$$\frac{g(A)}{g(B)} \cdot \frac{g(B)}{g(C)} > 1 \cdot 1 \implies \frac{g(A)}{g(C)} > 1.$$

□

Lemma A.19. *Condorcet Consistency does not apply to SEP-E.*

Proof. SEP-E has $M \in \{0, 1, \perp\}$ and $p = 1$. Hence, the Condorcet consistent antecedent of $M(A,B) > p \wedge M(B,C) > p$ cannot be met. □

A.2.6 Independence of Irrelevant Attributes

Lemma A.20. *All $M \in PGM$ satisfy IIA.*

Proof. PGM is defined by metrics dependent solely on groups A, B. Hence, a change in C cannot affect any $M \in$ PGM. $\square$

A.2.7 Anonymity

Lemma A.21. *All $M \in PGM$ satisfy Anonymity.*

Proof. All M are a function of confusion-matrix summaries–ordering of individuals is irrelevant. Furthermore, the confusion-matrix summaries do not depend on any attribute. Metrics in PGM should not be confused with the algorithms they evaluate. $\square$

Chapter B

PUBLICATIONS

Parts of this dissertation are based on these following works.

1. **Algorithmic Accountability in Small Data: Sample-Size-Induced Bias within Confusion-Matrix Metrics**

   Jarren Briscoe, Garrett Kepler, Daryl DeFord, Assefaw Gebremedhin

   Proceedings of The 28th International Conference on Artificial Intelligence and Statistics, 2025.

2. **Facets of Disparate Impact: Evaluating Legally Consistent Bias in Machine Learning**

   Jarren Briscoe, Assefaw Gebremedhin

   Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM 2024)

   DOI: 10.1145/3627673.3679925.

3. **Adversarial Creation of a Smart Home Testbed for Novelty Detection**

   Jarren Briscoe, Assefaw Gebremedhin, Lawrence B. Holder, Diane J. Cook

   AAAI Spring Symposium on Designing AI for Open Worlds, 2022.

4. **Reducing Sample Selection Bias in Clinical Data through Generation of Multi-Objective Synthetic Data**

   Jarren Briscoe, Chance DeSmet, Katherine Wuestney, Assefaw Gebremedhin, Roschelle Fritz, Diane J. Cook

   Proceedings of the 11th International Conference on Biomedical Engineering and Systems (ICBES 2024).

5. **Exploring Geriatric Clinical Data and Mitigating Bias with Multi-Objective Synthetic Data Generation for Equitable Health Predictions**

   Jarren Briscoe, Chance DeSmet, Katherine Wuestney, Assefaw Gebremedhin, Roschelle

Fritz, Diane J. Cook

Journal of Biomedical Engineering and Biosciences (JBEB), 2024.

6. **Fairness in Classification: Bridging Social Choice Theory and Machine Learning Fairness**

   Jarren Briscoe, Daryl DeFord, Diane J. Cook, Assefaw Gebremedhin

   Submitted to the Journal of Mechanism and Institution Design, 2026.

REFERENCES

Aghbalou, A., Sabourin, A., and Portier, F. (2024). Sharp error bounds for imbalanced classification: how many examples in the minority class? In *International Conference on Artificial Intelligence and Statistics*, pages 838–846. PMLR.

Amazon (2021). Amazon ai fairness and explainability whitepaper.

Aminikhanghahi, S. and Cook, D. J. (2019). Enhancing activity recognition using CPD-based activity segmentation. *Pervasive and Mobile Computing*, 53(75-89).

Angwin, J., Larson, J., Mattu, S., and Kirchner, L. (2016a). Machine bias. ProPublica. Available at: https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm.

Angwin, J., Larson, J., Mattu, S., and Kirchner, L. (2016b). Machine bias: There's software used across the country to predict future criminals. and it's biased against blacks. *ProPublica*. Accessed: 2024-10-08.

Arjovsky, M., Chintala, S., and Bottou, L. (2017). Wasserstein gan.

Arrow, K. J. (1950). A difficulty in the concept of social welfare. *Journal of Political Economy*, 58(4):328–346.

Arrow, K. J. (1951). *Social Choice and Individual Values*. Wiley, 2nd edition.

Bagnall, A., Lines, J., Bostrom, A., Large, J., and Keogh, E. (2017). The great time series classification bake off: a review and experimental evaluation of recent algorithmic advances. *Data mining and knowledge discovery*, 31(3):606–660.

Berry, A. C. (1941). The accuracy of the gaussian approximation to the sum of independent variates. *Transactions of the American Mathematical Society*, 49(1):122–136.

Biddle, D. (2017). *Adverse impact and test validation: A practitioner's guide to valid and defensible employment testing*. Routledge.

Bodó, Z., Minier, Z., and Csató, L. (2011). Active learning with clustering. In Guyon, I., Cawley, G., Dror, G., Lemaire, V., and Statnikov, A., editors, *Active Learning and Experimental Design workshop In conjunction with AISTATS 2010*, volume 16 of *Proceedings of Machine Learning Research*, pages 127–139, Sardinia, Italy. PMLR.

Brandt, F., Conitzer, V., Endriss, U., Lang, J., and Procaccia, A. D. (2016). *Handbook of computational social choice*. Cambridge University Press.

Brennan, T., Dieterich, W., and Ehret, B. (2009). Evaluating the predictive validity of the compas risk and needs assessment system. *Criminal Justice and behavior*, 36(1):21–40.

Briscoe, J., DeSmet, C., Wuestney, K., Gebremedhin, A., Fritz, R., and Cook, D. J. (2024a). Exploring geriatric clinical data and mitigating bias with multi-objective synthetic data generation for equitable health predictions. *Journal of Biomedical Engineering and Biosciences (JBEB)*, 11(1):33–43.

Briscoe, J., DeSmet, C., Wuestney, K., Gebremedhin, A., Fritz, R., and Cook, D. J. (2024b). Reducing sample selection bias in clinical data through generation of multi-objective synthetic data. In *Proceedings of the 10th World Congress on Electrical Engineering and Computer Systems and Sciences (EECSS'24)*.

Briscoe, J. and Gebremedhin, A. (2024). Facets of disparate impact: Evaluating legally consistent bias in machine learning. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, CIKM '24, page 3637–3641, New York, NY, USA. Association for Computing Machinery.

Briscoe, J., Gebremedhin, A., Holder, L. B., and Cook, D. J. (2022). Adversarial creation of a smart home testbed for novelty detection. In *AAAI Spring Symposium on Designing AI for Open Worlds*.

Briscoe, J., Kepler, G., DeFord, D. R., and Gebremedhin, A. (2025). Algorithmic accountability in small data: Sample-size-induced bias within classification metrics. In Li, Y., Mandt, S., Agrawal, S., and Khan, E., editors, *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*, volume 258 of *Proceedings of Machine Learning Research*, pages 4915–4923. PMLR.

Briscoe, J., Rague, B., Feuz, K., and Ball, R. (2021). Specialized neural network pruning for boolean abstractions. volume 2: KEOD of *IC3K 2021*, pages 178–185. Proceedings of the 13th International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management.

Caldeira, J. F., Cordeiro, W. C., Ruiz, E., and Santos, A. A. (2025). Forecasting the yield curve: the role of additional and time-varying decay parameters, conditional heteroscedasticity, and macro-economic factors. *Journal of Time Series Analysis*, 46(2):258–285.

Candanedo, L. (2016). Occupancy detection. UCI Machine Learning Repository.

Castelnovo, A., Crupi, R., Greco, G., Regoli, D., Penco, I. G., and Cosentini, A. C. (2022). A clarification of the nuances in the fairness metrics landscape. *Scientific Reports*, 12(1).

Caton, S. and Haas, C. (2024). Fairness in machine learning: A survey. *ACM Comput. Surv.*, 56(7).

Chicco, D. and Jurman, G. (2020). The advantages of the matthews correlation coefficient (mcc) over f1 score and accuracy in binary classification evaluation. *BMC genomics*, 21(1):1–13.

Chicco, D. and Jurman, G. (2023). The matthews correlation coefficient (mcc) should replace the roc auc as the standard metric for assessing binary classification. *BioData Mining*, 16(1):4.

Chicco, D., Tötsch, N., and Jurman, G. (2021). The matthews correlation coefficient (mcc) is more reliable than balanced accuracy, bookmaker informedness, and markedness in two-class confusion matrix evaluation. *BioData mining*, 14(1):13.

Chopin, I. and Germaine, C. (2017). A comparative analysis of non-discrimination law in europe: Section 3.1 genuine and determining occupational requirements. Research Report DS-

05-17-172-EN-N, European Commission, Directorate-General for Justice and Consumers, Brussels.

Conitzer, V., Freeman, R., Shah, N., and Vaughan, J. W. (2019). Group fairness for the allocation of indivisible goods. In *Proceedings of the Thirty-Third AAAI Conference on Artificial Intelligence and Thirty-First Innovative Applications of Artificial Intelligence Conference and Ninth AAAI Symposium on Educational Advances in Artificial Intelligence*, AAAI'19/IAAI'19/EAAI'19. AAAI Press.

Corbett-Davies, S. and Goel, S. (2018). The measure and mismeasure of fairness: A critical review of fair machine learning. *arXiv preprint arXiv:1808.00023*.

Cornacchia, G., Anelli, V. W., Narducci, F., Ragone, A., and Di Sciascio, E. (2023). Counterfactual reasoning for bias evaluation and detection in a fairness under unawareness setting. 372.

Cortes, C. and Mohri, M. (2003). Auc optimization vs. error rate minimization. 16.

Dai, Q., Li, H., Wu, P., Dong, Z., Zhou, X.-H., Zhang, R., Zhang, R., and Sun, J. (2022). A generalized doubly robust learning framework for debiasing post-click conversion rate prediction. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, KDD '22, page 252–262, New York, NY, USA. Association for Computing Machinery.

Damak, K., Khenissi, S., and Nasraoui, O. (2022). Debiasing the cloze task in sequential recommendation with bidirectional transformers. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, KDD '22, page 273–282, New York, NY, USA. Association for Computing Machinery.

Das, S., Donini, M., Gelman, J., Haas, K., Hardt, M., Katzman, J., Kenthapadi, K., Larroy, P., Yilmaz, P., and Zafar, M. B. (2021). Fairness measures for machine learning in finance. *The Journal of Financial Data Science*.

Dau, H. A., Bagnall, A., Kamgar, K., Yeh, C.-C. M., Zhu, Y., Gharghabi, S., Ratanamahatana, C. A., and Keogh, E. (2019). The UCR time series archive. *IEEE/CAA Journal of Automatica Sinica*, 6(6):1293–1305.

Dau, H. A., Bagnall, A. J., Kamgar, K., Yeh, C. M., Zhu, Y., Gharghabi, S., Ratanamahatana, C. A., and Keogh, E. J. (2018). The UCR time series archive. *CoRR*, abs/1810.07758.

Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine learning research*, 7(Jan):1–30.

Dennis J. Aigner, M. d. A. and Wiles, J. (2024). Statistical approaches for assessing disparate impact in fair housing cases. *Statistics and Public Policy*, 11(1):2263038.

DeSmet, C. and Cook, D. (2024). Hydragan: A cooperative agent model for multi-objective data generation. *ACM Trans. Intell. Syst. Technol.*, 15(3).

Dieterich, W., Mendoza, C., and Brennan, T. (2016). Compas risk scales: Demonstrating accuracy equity and predictive parity. *Northpointe Inc*, 7(4):1–36.

Dressel, J. and Farid, H. (2018). The accuracy, fairness, and limits of predicting recidivism. *Science Advances*, 4(1):eaao5580.

Eisenstein, J. (2019). *Introduction to Natural Language Processing*. MIT Press.

Esseen, C.-G. (1942). On the liapunoff limit of error in the theory of probability. *Arkiv för Matematik, Astronomi och Fysik*, A28(9):1–19.

Feldman, M., Friedler, S. A., Moeller, J., Scheidegger, C., and Venkatasubramanian, S. (2015). Certifying and removing disparate impact. In *Proceedings of the 21th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '15, page 259–268, New York, NY, USA. Association for Computing Machinery.

Feng, J., Gossmann, A., Pirracchio, R., Petrick, N., Pennello, G. A., and Sahiner, B. (2024). Is this model reliable for everyone? testing for strong calibration. In *International Conference on Artificial Intelligence and Statistics*, pages 181–189. PMLR.

Feng, Q., Du, M., Zou, N., and Hu, X. (2025). Fair machine learning in healthcare: A survey. *IEEE Transactions on Artificial Intelligence*, 6:493–507.

Fishburn, P. C. (1977). Condorcet social choice functions. *SIAM Journal on Applied Mathematics*, 33(3):469–489.

Freund, J. E., Miller, I., and Miller, M. (2014). *John E. Freund's Mathematical Statistics with Applications*. Pearson Education Limited, 8 edition. pp. 193.

Gao, J., Han, S., Zhu, H., Yang, S., Jiang, Y., Xu, J., and Zheng, B. (2023). Rec4ad: A free lunch to mitigate sample selection bias for ads ctr prediction in taobao. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, CIKM '23, page 4574–4580, New York, NY, USA. Association for Computing Machinery.

Garcia, S. and Herrera, F. (2008). An extension on" statistical comparisons of classifiers over multiple data sets" for all pairwise comparisons. *Journal of machine learning research*, 9(12).

Gastwirth, J. L. and Miao, W. (2009). Formal statistical analysis of the data in disparate impact cases provides sounder inferences than the U. S. government's 'four-fifths' rule: an examination of the statistical evidence in Ricci v. DeStefano. *Law, Probability and Risk*, 8(2):171–191.

Gibbard, A. (1973). Manipulation of voting schemes. *Econometrica*, 41(4):587–60.

Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., and Bengio, Y. (2014). Generative adversarial networks.

Górecki, T., Luczak, M., and Piasecki, P. (2024). An exhaustive comparison of distance measures in the classification of time series with 1NN method. *Journal of Computational Science*, 76:102235.

Goutte, C. and Gaussier, E. (2005). A probabilistic interpretation of precision, recall and f-score, with implication for evaluation. In *European conference on information retrieval*, pages 345–359. Springer.

Griggs v. Duke Power Co. (1971). 401 u.s. 424. Supreme Court of the United States.

Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., and Courville, A. C. (2017). Improved training of wasserstein gans. *Advances in neural information processing systems*, 30.

Gursoy, F. and Kakadiaris, I. A. (2022). Equal confusion fairness: Measuring group-based disparities in automated decision systems. In *2022 IEEE International Conference on Data Mining Workshops (ICDMW)*, pages 137–146.

gustav Esseen, C. (1956). A moment inequality with an application to the central limit theorem. *Scandinavian Actuarial Journal*, 1956:160–170.

Hardt, M., Price, E., and Srebro, N. (2016). Equality of opportunity in supervised learning. *CoRR*, abs/1610.02413.

He, J., Wang, W., Huang, M., Wang, S., and Guan, X. (2021). Bayesian inference under small sample sizes using general noninformative priors. *Mathematics*, 9(21).

Hebrail, G. and Berard, A. (2012). Household Power Consumption Dataset. https://archive.ics.uci.edu/ml/datasets/individual+household+electric+power+consumption. UCI Machine Learning Repository.

Hess, K. and Feuerriegel, S. (2024). Stabilized neural prediction of potential outcomes in continuous time. *arXiv preprint arXiv:2410.03514*.

Hong, J., Zhu, Z., Yu, S., Wang, Z., Dodge, H. H., and Zhou, J. (2021). Federated adversarial debiasing for fair and transferable representations. KDD '21, page 617–627, New York, NY, USA. Association for Computing Machinery.

Hubert, L. and Arabie, P. (1985). Comparing partitions. *Journal of classification*, 2(1):193–218.

Iman, R. L. and Davenport, J. M. (1980). Approximations of the critical region of the fbietkan statistic. *Communications in Statistics - Theory and Methods*, 9(6):571–595.

Johnson, N. L., Kemp, A. W., and Kotz, S. (2005). *Univariate Discrete Distributions*. Wiley Series in Probability and Statistics. John Wiley & Sons, Hoboken, NJ, 3rd edition.

Kleinberg, J., Mullainathan, S., and Raghavan, M. (2016). Inherent trade-offs in the fair determination of risk scores. *arXiv preprint arXiv:1609.05807*.

Kwon, S., Kim, S., Lee, S., Kim, J.-Y., An, S., and Kim, K. (2023). Addressing selection bias in computerized adaptive testing: A user-wise aggregate influence function approach. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, CIKM '23, page 4674–4680, New York, NY, USA. Association for Computing Machinery.

Lanczos, C. (1964). A precision approximation of the gamma function. *Journal of the Society for Industrial and Applied Mathematics Series B Numerical Analysis*, 1(1):86–96.

Larner, A. J. (2024). *The 2x2 Matrix: Contingency, Confusion and the Metrics of Binary Classification*. 2 edition.

Larson, J., Mattu, S., Kirchner, L., and Angwin, J. (2016). How we analyzed the compas recidivism algorithm. *ProPublica*. Accessed: 2024-10-08.

Le, T. and Deng, A. (2023). The price is right: Removing a/b test bias in a marketplace of expirable goods. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, CIKM '23, page 4681–4687, New York, NY, USA. Association for Computing Machinery.

Lemoine, N. P. (2019). Moving beyond noninformative priors: why and how to choose weakly informative priors in bayesian analyses. *Oikos*, 128(7):912–928.

List, C. and Pettit, P. (2002). Aggregating sets of judgments: An impossibility result. *Economics and Philosophy*, 18(1):89–110.

Liu, J., Chen, H., Shen, J., and Choo, K.-K. R. (2025). Faircompass: Operationalizing fairness in machine learning. *IEEE Transactions on Artificial Intelligence*, 6(2):281–291.

Liu, Q. and Paparrizos, J. (2024). The elephant in the room: Towards a reliable time-series anomaly detection benchmark. *Advances in Neural Information Processing Systems*, 37:108231–108261.

Matthews, B. W. (1975). Comparison of the predicted and observed secondary structure of t4 phage lysozyme. *Biochimica et Biophysica Acta (BBA)-Protein Structure*, 405(2):442–451.

May, K. O. (1952). A set of independent necessary and sufficient conditions for simple majority decision. *Econometrica*, 20(4):680–684.

McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., and Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys (CSUR)*, 54(6):1–35.

Microsoft Azure (2016). Azure Predictive Maintenance Dataset. `https://github.com/Azure/lstmbased-predictive-maintenance`. Telemetry and maintenance records from simulated aircraft engines.

Milgrom, P. (2024). Kenneth arrow's last theorem. *Journal of Mechanism and Institution Design*, 9(1):7–11.

Mirza, M. and Osindero, S. (2014). Conditional generative adversarial nets. *CoRR*, abs/1411.1784.

Moore, C., Ferguson, E., and Guerin, P. (2023). Pretrial Risk Assessment on the Ground: Algorithms, Judgments, Meaning, and Policy. *MIT Case Studies in Social and Ethical Responsibilities of Computing*, Summer 2023. https://mit-serc.pubpub.org/pub/czviu6qc.

Moulin, H. (1988). Condorcet's principle implies the no show paradox. *Journal of Economic Theory*, 45(1):53–64.

Nemenyi, P. B. (1963). *Distribution-free multiple comparisons.* Princeton University.

Paparrizos, J., Liu, C., Elmore, A. J., and Franklin, M. J. (2023). Querying time-series data: A comprehensive comparison of distance measures. *IEEE Data Eng. Bull.*, 46(3):69–88.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., and Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830.

Peizer, D. B. and Pratt, J. W. (1968). A normal approximation for binomial, f, beta, and other common, related tail probabilities, i. *Journal of the American Statistical Association*, 63(324):1416–1456.

ProPublica (2016). Compas analysis github repository. Accessed: 2024-10-08.

Ragab, M., Eldele, E., Tan, W. L., Foo, C.-S., Chen, Z., Wu, M., Kwoh, C.-K., and Li, X. (2023). Adatime: A benchmarking suite for domain adaptation on time series data. *ACM Trans. Knowl. Discov. Data*, 17(8).

Ricci v. DeStefano (2009). 557 u.s. 557. Supreme Court of the United States.

Robin, G. (1984). Grandes valeurs de la fonction somme des diviseurs et hypothése de riemann. *J. Math. Pures Appl.*, 63:187–213.

Rohbeck, M., De Brouwer, E., Bunne, C., Huetter, J.-C., Biton, A., Chen, K. Y., Regev, A., and Lopez, R. (2025). Modeling complex system dynamics with flow matching across time and conditions. In *The Thirteenth International Conference on Learning Representations*.

Rong, J., Sun, N., and Wang, D. (2019). A new evaluation criterion for allocation mechanisms with application to vehicle license allocations in china. *Journal of Mechanism and Institution Design*, 4(1):39–86.

Rudner, T. G., Zhang, Y. S., Wilson, A. G., and Kempe, J. (2024). Mind the gap: Improving robustness to subpopulation shifts with group-aware priors. In *International Conference on Artificial Intelligence and Statistics*, pages 127–135. PMLR.

Russo, F. and Tonia, F. (2023). Causal discovery and knowledge injection for contestable neural networks. In *Proceedings of the 26th European Conference on Artificial Intelligence (ECAI 2023)*, volume 372 of *Frontiers in Artificial Intelligence and Applications*. IOS Press.

Seabold, S. and Perktold, J. (2010). statsmodels: Econometric and statistical modeling with python. In *9th Python in Science Conference*.

Sen, A. and Pattanaik, P. K. (1969). Necessary and sufficient conditions for rational choice under majority decision. *Journal of Economic Theory*, 1(2):178–202.

Shah, N. (2023). Pushing the limits of fairness in algorithmic decision-making. In Elkind, E., editor, *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence, IJCAI-23*, pages 7051–7056. International Joint Conferences on Artificial Intelligence Organization. Early Career.

Shaham, S., Hajisafi, A., Quan, M. K., Nguyen, D. C., Krishnamachari, B., Peris, C., Ghinita, G., Shahabi, C., and Pathirana, P. N. (2025). Privacy and fairness in machine learning: A survey. *IEEE Transactions on Artificial Intelligence*, 6(7):1706–1726.

Shao, Z., Wang, F., Xu, Y., Wei, W., Yu, C., Zhang, Z., Yao, D., Sun, T., Jin, G., Cao, X., Cong, G., Jensen, C. S., and Cheng, X. (2025). Exploring progress in multivariate time series forecasting: Comprehensive benchmarking and heterogeneity analysis. *IEEE Transactions on Knowledge and Data Engineering*, 37(1):291–305.

Shevtsova, I. G. (2010). An improvement of convergence rate estimates in the lyapunov theorem. In *Doklady Mathematics*, volume 82.

Shifaz, A., Pelletier, C., Petitjean, F., and Webb, G. I. (2023). Elastic similarity and distance measures for multivariate time series. *Knowledge and Information Systems*, 65:2665–2698.

Sokolova, M. and Lapalme, G. (2009). A systematic analysis of performance measures for classification tasks. *Information Processing & Management*, 45(4):427–437.

Steinberg, E., Fries, J., Xu, Y., and Shah, N. (2023). Motor: A time-to-event foundation model for structured medical records, 2023. *URL https://arxiv. org/abs/2301.03150*.

U.S. Equal Employment Opportunity Commission (1978). Uniform guidelines on employee selection procedures. *Code of Federal Regulations, Title 29, Section 1607.4*. Available at: https://www.law.cornell.edu/cfr/text/29/1607.4.

Verma, S. and Rubin, J. (2018). Fairness definitions explained. In *2018 ieee/acm international workshop on software fairness (fairware)*. IEEE.

Wald, Y., Goldstein, M., Efroni, Y., van Amsterdam, W. A., and Ranganath, R. (2025). Time after time: Deep-q effect estimation for interventions on when and what to do. *arXiv preprint arXiv:2503.15890*.

Wang, X., Mueen, A., Ding, H., Trajcevski, G., Scheuermann, P., and Keogh, E. (2013). Experimental comparison of representation methods and distance measures for time series data. *Data Mining and Knowledge Discovery*, 26(2):275–309.

Wang, Z., Huang, N., Sun, F., Ren, P., Chen, Z., Luo, H., de Rijke, M., and Ren, Z. (2022). Debiasing learning for membership inference attacks against recommender systems. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, KDD '22, page 1959–1968, New York, NY, USA. Association for Computing Machinery.

Washington, A. L. (2019). Lessons from the COMPAS-ProPublica debate. *Colorado Technology Law Journal*, 17(1):131–164.

Watson-Daniels, J., Parkes, D. C., and Ustun, B. (2023). Predictive multiplicity in probabilistic classification. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 10306–10314.

Wilson, D. J. (2019). The harmonic mean p-value for combining dependent tests. *Proceedings of the National Academy of Sciences*, 116(4):1195–1200.

Wu, X., Qiu, X., Li, Z., Wang, Y., Hu, J., Guo, C., Xiong, H., and Yang, B. (2024). Catch: Channel-aware multivariate time series anomaly detection via frequency patching. *arXiv preprint arXiv:2410.12261*.

Ye, K., Walpole, R. E., Myers, R. H., and Myers, S. L. (2024). *Probability and Statistics for Engineers and Scientists*. Pearson, updated 9th edition. Section 6.5.

Young, H. P. (1975). Social choice scoring functions. *SIAM Journal on Applied Mathematics*, 28(4):824–838.

Zanger-Tishler, M., Nyarko, J., and Goel, S. (2023). Risk scores, label bias, and everything but the kitchen sink.

Zhang, J. M., Harman, M., Ma, L., and Liu, Y. (2020). Machine learning testing: Survey, landscapes and horizons. *IEEE Transactions on Software Engineering*.

Zhang, W., Wang, Z., Kim, J., Cheng, C., Oommen, T., Ravikumar, P., and Weiss, J. (2023). Individual fairness under uncertainty. 372.